A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, I: local matching and a global transfer under elliptic-support conditions

Draft 11

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November 8, 2026

Abstract

Let D be a central division algebra of prime degree r over a totally imaginary number field K containing μ_n, with n odd and $(n, r) = 1$. We compare the n-fold cover of $D_{\mathbb{A}}^{\times}$ constructed in [JC] with the Kazhdan–Patterson cover of $\mathrm{GL}(r, \mathbb{A}_K)$, realizing both as subgroups of one cover of $\mathrm{GL}(r, \mathbb{A}_L)$, where $L \subset D$ is a cyclic maximal subfield. *Theorem A:* at each ramified place of D, the covered elliptic tori on the two sides have the same commutator pairing and the same good elements, and they are identified on good elements by a canonical isomorphism. *Theorem B:* genuine test functions supported on the regular set have matching functions on the other side, in both directions. *Theorem C:* using a variant of the simple trace formula of Flicker–Kazhdan, we prove a locally finite identity of packets at the ramified places. From it we obtain a cuspidal transfer, with the same components away from the ramified places, for every genuine automorphic representation of the cover of $D_{\mathbb{A}}^{\times}$ that has a supercuspidal component at some split place, under a non-cancellation hypothesis that holds whenever the D-side packet is finite. Part II proves the matching of arbitrary functions and replaces this hypothesis by a weaker one; Part III studies the local character relations from the side of the division algebra.

Contents

1 Introduction **3**
1.1 Setting and aim . . . 3
1.2 Main results . . . 4
1.3 Goals of the later parts . . . 5
1.4 Conventions . . . 6

1.5 How results are labeled . . . 6
1.6 Organization . . . 8

2 Literature **9**

3 Notation and glossary **11**
3.1 Status labels . . . 11
3.2 Global data . . . 12
3.3 Local fields and absolute values . . . 12
3.4 Hilbert symbols . . . 13
3.5 Covers and the cocycle normalization . . . 13
3.6 Genuine objects, central characters and good elements . . . 14
3.7 Orbital integrals and matching . . . 15
3.8 Haar measures . . . 15
3.9 The division-side transform . . . 16
3.10 Named results and hypotheses . . . 16
3.11 Notation specific to Part I . . . 17

4 Local covers **18**
4.1 Why an ambient cover . . . 19
4.2 The ambient cover . . . 19
4.3 The restriction lemma . . . 20
4.4 Places where D splits . . . 22

5 Comparison of covered elliptic tori at the ramified places **23**
5.1 The commutator pairings . . . 23
5.2 Good elements and centers . . . 24
5.3 An example . . . 25
5.4 A canonical identification on good elements . . . 26
5.5 Consequences . . . 27

6 Orbital integrals and matching **28**
6.1 Orbital integrals and measures . . . 28
6.2 Definition of matching . . . 29
6.3 Transported orbital-integral functions . . . 29
6.4 The easy direction, in both directions . . . 30
6.5 What fails for general functions . . . 31
6.6 Orbital integrals on an elliptic torus . . . 32

7 Global covers and rational lifts **35**
7.1 The global covers . . . 35
7.2 Rational lifts correspond . . . 38
7.3 Measures . . . 39
7.4 Cusp forms on finite central covers . . . 40

8 The two trace formulas and their comparison **41**
8.1 The D side . . . 42
8.2 The GL side: a variant of [FK, §18] . . . 43
8.3 Comparison . . . 46

9 Spectral separation and the global transfer **47**
9.1 The place w_0 . . . 47
9.2 Separation . . . 48
9.3 Non-cancellation and the transfer . . . 51

10 What is left after Part I **54**

11 Conclusion: what the local theorem gives on its own **56**

12 Summary of results **56**

1 Introduction

This section states the main results, fixes the standing assumptions and conventions, explains how results are labeled, and describes what the later parts of the series do. It depends on nothing else in the paper. The global data $K, n, r, L, D, S, r', c, c_v$ are fixed in Assumption 1.1; all notation, including the status labels and the cocycle normalization, is collected in §3, and the results are summarized, with labels and pointers, in §12 at the end of the paper. Throughout, [JC] refers to the corrected version (`metaplectic_kazhdan_revised12.tex`) of [J], and a reference such as [JC, Lemma 6.1] is to the numbering there.

1.1 Setting and aim

We fix the global data, which are those of [JC, §1], and explain where this paper fits.

Standing assumptions 1.1. K is a totally imaginary number field containing μ_n, with $n \geq 1$ odd; L/K is cyclic of prime degree r with $(n, r) = 1$; D is a central division algebra over K of dimension r^2 containing L; and S is the set of places where D ramifies. Fix $r' \in \mathbb{Z}$ with $rr' \equiv 1 \pmod{n}$ and $c \in \mathbb{Z}$ with $n \mid r - 1 + 2c$, and put $c_v := r'c$, so that $n \mid r - 1 + 2rc_v$.

The paper [JC] lifts genuine automorphic representations of an n-fold cover of the torus $L_{\mathbb{A}}^{\times}$ to ϵ-invariant genuine automorphic representations of an n-fold cover $\overline{G}_D(\mathbb{A}_K)$ of $D_{\mathbb{A}}^{\times}$. Its Remark 10.3 outlines the missing step needed to reach the Kazhdan–Patterson cover $\overline{G}(\mathbb{A}_K)$ of $\mathrm{GL}(r, \mathbb{A}_K)$: a metaplectic Jacquet–Langlands transfer from $\overline{G}_D$ to $\overline{G}$. Part I (this paper) establishes the local matching of test functions that the transfer requires, and a global transfer for representations with suitable local components. The later parts of the series (§1.3) remove some of those conditions and study the local character relations.

Since r is prime, every $v \in S$ is finite (the archimedean places are complex), D_v is a division algebra, and $L_v := L \otimes_K K_v$ is a field. The local invariants of D are nonzero elements of $\frac{1}{r}\mathbb{Z}/\mathbb{Z}$ with sum 0, so $|S| \geq 2$, and $|S|$ is even if $r = 2$.

1.2 Main results

We state the three main results informally; precise statements are referenced.

Covers. Both $\overline{G}_D(\mathbb{A}_K)$ and $\overline{G}(\mathbb{A}_K)$ are realized as subgroups of one cover of $\mathrm{GL}(r, \mathbb{A}_L)$ (§4, §7), so that their centers, rational splittings and central characters are literally shared.

Theorem A (covered tori; Theorem 5.9). Let $v \in S$, let $E \subset D_v$ be a maximal subfield, and embed E in $M_r(K_v)$. The covered tori $\overline{E}_D \subset \overline{G}_D(K_v)$ and $\overline{E}_{\mathrm{GL}} \subset \overline{G}(K_v)$ have the same commutator pairing $(x, y) \mapsto ((Nx)^{1+2c_v}x^{-1}, y)_E$. Their good elements are the inverse image of $K_v^\times E^{\times n}$, of index $|n|_{K_v}^{-(r-1)}$. There is an isomorphism $\overline{E}_D \cong \overline{E}_{\mathrm{GL}}$ induced by conjugation in the ambient cover, and its restriction to the good elements is canonical: it is the same for every isomorphism that is the identity on the common center.

Theorem B (local matching; Theorem 6.5). With matching defined by $F(\tilde{\gamma}, \tilde{f}_D) = (-1)^{r-1}F(\jmath(\tilde{\gamma}), \tilde{f})$ at good regular elements, and vanishing of the GL-side orbital integrals at non-elliptic elements, every genuine $\tilde{f}_D$ on $\overline{G}_D(K_v)$ vanishing near the center has a matching $\tilde{f}$ supported on the good regular elliptic set of $\overline{G}(K_v)$. Conversely, every regular-supported $\tilde{f}$ with vanishing non-elliptic orbital integrals has a matching $\tilde{f}_D$. For general $\tilde{f}_D$ the method necessarily fails (Proposition 6.6), and germ expansions are needed.

Theorem C (global transfer; Proposition 9.6, Theorems 9.11 and 9.12). Let $\overline{\pi}_D$ be a genuine automorphic representation of $\overline{G}_D(\mathbb{A}_K)$ whose component at some finite place $w_0 \notin S$ is supercuspidal. Let $P_D \ni \overline{\pi}_D$ and P be the packets of D-side and cuspidal GL-side representations with the same components outside S. These packets satisfy an identity of traces at the places of S against all matching regular-supported test functions; the sums are locally finite. If the D-side packet is *not cancelled* on regular-supported test functions (hypothesis (H_D), which holds whenever P_D is finite), then there is a genuine cuspidal automorphic representation $\overline{\pi}$ of $\overline{G}(\mathbb{A}_K)$ with $\overline{\pi}_v \cong \overline{\pi}_{D,v}$ for all $v \notin S$. A converse holds under the analogous hypothesis (H) on the GL-side packet. Lemma 9.8 shows that (H_D) can fail only if, at every level, the coefficient function of the packet at S lies in the sum of the subspaces having a Plancherel factor at one place of S.

Theorem C rests on the simple trace formula of Flicker and Kazhdan [FK, §18]. We need it in a slightly different form, with the supercusp form at a place outside S; this variant is proved in Proposition 8.2. Since the multiplicity-one and local character theory needed to isolate single representations is not available here, Theorem C is a statement about packets.

1.3 Goals of the later parts

The goals below are needed to complete the metaplectic Jacquet–Langlands correspondence by removing the restrictions of Theorems B and C. Each item lists the inputs it needs, followed by its current status in the later parts of the series. References to Part II are to Draft 8 and those to Part III are to Draft 9.

(i) *Matching of arbitrary functions.* Extend Theorem B to all $\tilde{f}_D$, and to all $\tilde{f}$ with vanishing non-elliptic orbital integrals. This needs the germ expansions of [FK, §8] at central good elements, their restriction to the elliptic set, and a genuine function with the orbital integrals of a Steinberg pseudo-coefficient near the center, and an analog of [DKV, A.3] (§6.5, item (D1)). *Status:* proved in Part II (Theorem E).

(ii) *Local character relations.* Prove a metaplectic local Jacquet–Langlands correspondence: a bijection $\overline{\pi}_v \leftrightarrow \overline{\pi}_{D,v}$ between genuine square-integrable representations of $\overline{G}(K_v)$ and genuine irreducible representations of $\overline{G}_D(K_v)$, with $\Theta_{\overline{\pi}_{D,v}} = (-1)^{r-1}\Theta_{\overline{\pi}_v} \circ \jmath$ on good regular elements (item (D2)). This needs (i), the elliptic orthogonality relations of [FK, §§24–25], and multiplicity one and rigidity for $\overline{G}(\mathbb{A}_K)$ ([FK, §§27–28], as corrected in [Z]). *Status:* Part III constructs a unitary transform L_D between ordinary and genuine invariant class functions on $D_v^\times$ and its cover, and proves that it bijects irreducible characters at the places $v \in S$ of residue characteristic prime to n. When the square-integrable Flicker–Kazhdan correspondence is available in Zou's corrected form [Z, Proposition 3.6, Definition 3.7], Part III, Corollary 7.8, proves (D2) at such tame places; the published sufficient range $(n, N_r) = 1$ and the precise compatibility checks are recorded in Remark 10.1. Outside that range an appropriate metaplectic correspondence theorem is an additional input. At places dividing n the irreducible-character integrality needed for this composition remains open.

(iii) *An unconditional global correspondence.* Remove the auxiliary supercuspidal place in Theorem C, either via W.-W. Li's trace formula for covers [Li4] or via (i), and refine the packet identity of Theorem C to a correspondence of individual representations using (ii). *Status:* open; see Part II, §7(iii).

(iv) *Removing* (H_D) *and (H).* Prove finiteness of near-equivalence packets for $\overline{G}_D(\mathbb{A}_K)$ (for example, by bounding the level at S in terms of the components outside S), and linear independence of elliptic characters in GL-side packets, so that Theorem C becomes unconditional (Remark 9.13, item (D3)). *Status:* Part II replaces (H_D) by the weaker (H'_D) and reduces its removal to a bound on levels at S (Part II, Corollary 6.11); open in general.

(v) *Completing [JC, Remark 10.3].* Compose the lifting of [JC] with the transfer, to obtain genuine automorphic representations of the Kazhdan–Patterson cover of $\mathrm{GL}(r, \mathbb{A}_K)$ attached to genuine characters of the covered torus (item (D4)). Then compare the result with the Flicker–Kazhdan correspondence [FK] to $\mathrm{GL}(r, \mathbb{A}_K)$ and with Kazhdan's endoscopic lifting. *Status:* open.

1.4 Conventions

The conventions below are repeated, with the rest of the series' notation, in §3 (Hilbert symbols in §3.4, covers and the relabeling of μ_n in §3.5, genuine objects and good elements in §3.6).

All Hilbert symbols are n-th Hilbert symbols. For a finite extension F'/F of local fields containing μ_n, $a \in F^\times$ and $b \in F'^\times$, we use the norm compatibility $(a,b)_{F'} = (a, N_{F'/F}b)_F$ [Ser, Ch. XIV]; in particular $(a,b)_{F'} = (a,b)_F^{[F':F]}$ for $a, b \in F^\times$. For lifts $\tilde{x}, \tilde{y}$ of commuting elements, $[x,y] = \tilde{x}\tilde{y}\tilde{x}^{-1}\tilde{y}^{-1}$. For a cover $p : \overline{H} \to H$ and $H' \subset H$ we write $\overline{H'} = p^{-1}(H')$. Genuine functions and representations, good elements, and $C_c^\infty(\overline{H}, \tilde{\theta})_{\mathrm{gen}}$ are as in [JC, §§1–2]. As in [JC, §6.1], we relabel μ_n once, by $\zeta \mapsto \zeta^{r'}$, at every place and globally, and we say explicitly when a formula is stated before or after relabeling.

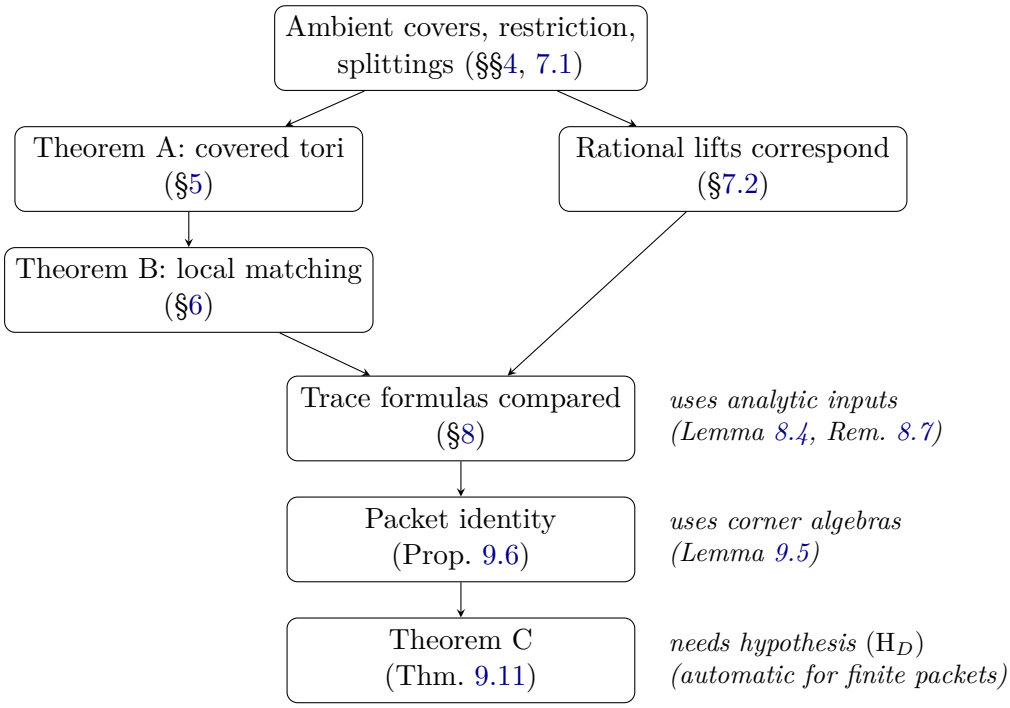
1.5 How results are labeled

The labels are those of the whole series (§3.1, which also defines the label [$\mathbf{P}^{\mathrm{c}}$] used in the companion paper; it does not occur here). Statements are marked by status:

- [**P**] *Proved here*: a complete argument is included, possibly using cited results marked [**E**].
- [**E**] *Established elsewhere*: a precisely cited theorem, used as a black box.
- [**O**] *Proof outlined*: the steps are identified, but at least one remains to be written out or checked.
- [**C**] *Conjectural*: requires new mathematical input (later parts of the series).

Statement	Where	Status
Both covers are subgroups of one ambient cover; the GL side is the Kazhdan–Patterson cover with parameter c_v	Prop. 4.3	**[P]**
Section and splitting conventions	Lemma 7.1	**[P]**
Canonical identification of the covers at $v \notin S$	Prop. 4.6	**[P]**
Covered elliptic tori at $v \in S$: pairings, good elements, centers; canonical identification (Theorem A)	§5	**[P]**
Matching for regular-supported functions (Theorem B)	Thm. 6.5, §6.6	**[P]**
Global covers; rational lifts correspond	§7	**[P]**
Geometric expansions, including class-counting constants	Props. 8.1, 8.2	**[P]**
Variant of the simple trace formula of [FK, §18]	Prop. 8.2	**[P]**
Comparison of the trace formulas	Thm. 8.6	**[P]**
Packet identity at S (locally finite)	Prop. 9.6	**[P]**
Characterization of cancellation	Lemma 9.8	**[P]**
Global transfer (Theorem C)	Thm. 9.11	**[P]** under (H_D); unconditional for finite packets
Converse transfer	Thm. 9.12	**[P]** under (H)
Finiteness of packets; removal of (H_D), (H)	§10	**[C]** (weakened in Part II)
Matching of arbitrary functions; local character relations	§10	**[C]** (matching: Part II; tame character relations with FK–Zou input: Part III)

The logical dependencies are as follows. Arrows mean "is used in", and the notes on the right list the additional inputs of each step.



Everything except the last box is unconditional, given the cited [**E**] inputs. The [**E**] inputs used in the global sections are listed in Remark 8.7.

A running example

Throughout the series we test the constructions on one case: $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4 - \zeta_3)_K$, which is ramified exactly at $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$, with $\mathfrak{p}_3 = (1 - \zeta_3)$ and $\mathfrak{p}_7 = (3 + \zeta_3)$. We take $L = K(i) = \mathbb{Q}(\zeta_{12})$, and $c = 1$, $r' = 2$, $c_v = 2$ in Assumption 1.1. At $\mathfrak{p}_3$, where $K_v = \mathbb{Q}_3(\zeta_3)$, the covered elliptic tori are non-abelian: the good elements have index 9, and genuine irreducibles of a covered torus have dimension 3 (Example 5.7). At $\mathfrak{p}_7$, where $K_v = \mathbb{Q}_7 \supset \mu_3$, they are abelian. The sign in Theorem 6.5 is -1. The note `notes/mjl_running_example.tex` in the project repository works this case out across Parts I–IV.

1.6 Organization

Section 2 reviews the literature for the whole series (Parts II–IV refer to it), and Section 3 collects the notation and glossary of the series (shared by all four Parts) together with the notation specific to Part I. Sections 4–6 contain the local theory; the facts about orbital integrals on elliptic tori that it needs are proved in §6.6. Sections 7–9 contain the global theory, and §7.4 supplies its analytic input for cusp forms on the cover. Section 10 lists what is left after this paper, Section 11 summarizes what the local theory gives on its own, and Section 12 lists the results with their labels.

2 Literature

This section reviews the literature on which the whole series draws, and places the series within it. Parts II–IV refer to this review and add only what is specific to them. It proves nothing and introduces no notation.

The Jacquet–Langlands correspondence. Jacquet and Langlands [JL] related automorphic representations of the multiplicative group of a quaternion algebra to those of GL(2) by comparing trace formulas. For a central division algebra of arbitrary degree the local correspondence is due to Deligne–Kazhdan–Vignéras [DKV] and Rogawski [Rog83], and Badulescu [Bad] proved the global correspondence for all inner forms of $\mathrm{GL}(n)$, with multiplicity one and strong multiplicity one. Flicker and Kazhdan [FK88] gave a simple trace formula for test functions with a single supercuspidal component and used it to extend both the simple-algebras correspondence and the metaplectic correspondence to automorphic forms with one supercuspidal component; Proposition 8.2 is a statement of the same kind for the cover $\overline{G}(\mathbb{A}_K)$, proved here by following [FK, §18], and we cite [FK88] for its approach only. The transfer of orbital integrals between $\mathrm{GL}(n)$ and its inner forms, with explicit matching functions, is studied by Cohen [Coh]; harmonic-analytic background is in Kottwitz [Kot] and Harish-Chandra [HC70]. Our definition of matching and the use of a simple trace formula follow the pattern of these works, with two new features: only good elements carry orbital integrals, and the two covers must first be identified on tori (§5). The sign convention is compared with Badulescu's in §6.2.

Covering groups. The n-fold covers of $\mathrm{SL}(2)$ and $\mathrm{GL}(2)$ over local and global fields go back to Kubota [Kub67], [Kub69]. Moore [Moo] and Matsumoto [Mat] classified the topological central extensions of the split simply connected groups by Steinberg symbols (see also Milnor [Mil]); this classification is the main input of Proposition 4.3. Kazhdan and Patterson [KP1] constructed the n-fold covers of $\mathrm{GL}(r)$ with a determinantal twisting parameter, developed their representation theory, including the exceptional representations, and formulated the generalized Shimura correspondence [KP2]. Explicit block-compatible cocycles are in Banks–Levy–Sepanski [BLS]. The modern framework is that of Brylinski and Deligne [BD], for which Weissman [Wei18] and Gan–Gao [GG18] formulated an L-group and a Langlands program; the historical introduction [GGW] surveys the subject. This series uses only the Kazhdan–Patterson covers and their restrictions to subgroups.

Metaplectic correspondences. For $n = 2$ the correspondence goes back to Shimura's work on modular forms of half-integral weight [Shi73]; its representation-theoretic form is due to Gelbart [Gel76] and Waldspurger [Wal80], [Wal91], through the Weil representation. Flicker [Fli80] treated the n-fold cover of $\mathrm{GL}(2)$ by the trace formula, and Flicker and Kazhdan [FK] established the metaplectic correspondence between genuine representations of the n-fold cover of $\mathrm{GL}(r)$ and representations of $\mathrm{GL}(r)$, using a simple trace formula [FK, §18]; our GL-side trace formula is theirs. Mezo generalized the local correspondence of [FK] and prepared a comparison of the invariant trace formulas of $\mathrm{GL}(r)$ and

its covers [Mez02], [Mez01], and constructed metaplectic tensor products for irreducible representations [Mez04]. Zou [Z] revisited the local metaplectic correspondence, restored and generalized parts of the local theory of [FK], and pointed out an error in [FK, §26.2]; Part III uses his corrected form. Savin [Sav88] proved a local Shimura correspondence at the level of Iwahori–Hecke algebras. W.-W. Li developed the Arthur–Selberg trace formula for covers of connected reductive groups [Li1], [Li2], [Li4]; it is one of the two routes to removing the auxiliary supercuspidal place (goal (iii) of §1.3). Kazhdan's lifting from tori [Kaz84] has a metaplectic analog [J], corrected in [JC], which is the source of the cover of $D_{\mathbb{A}}^{\times}$ used here. The present paper does not compose with the Flicker–Kazhdan correspondence; that would be the step after (D2).

Central extensions of division algebras. The topological central extensions of $\mathrm{SL}_1(D)$, and the metaplectic kernel for anisotropic groups, were determined by Prasad–Raghunathan [PRa88] and Prasad–Rapinchuk [PRp96]; see also Rapinchuk [Rap], [Rap88] and Sury [Sur]. These works concern the extensions themselves rather than automorphic forms on them.

The work closest to ours on the side of the division algebra is that of Klose, in his Göttingen dissertation (1983, written under S. J. Patterson) and the two papers drawn from it. He studied central extensions by μ_n of the multiplicative group of a quaternion division algebra. The local paper [Klo86a] concerns a p-adic quaternion division algebra; according to its indexing data, its subject is the commutator structure of $D^{\times}$, the structure of its group of one-units, and the classes in the second cohomology that describe its central extensions, including the metaplectic ones. The global paper [Klo86b] concerns a quaternion division algebra D over a number field. It constructs explicit cocycles, and in the description of Prasad and Rapinchuk [PRp96, Introduction] it obtains partial results on the topological central extensions of the adele group of $\mathrm{GL}_{1,D}$; their own computation of the metaplectic kernel is more precise. Hill [Hil, §§2.3, 3.8] attributes to Klose the facts on commutators and symmetric cocycles of abelian groups that were used there to determine the metaplectic extensions of $D^{\times}$; among them is a canonical isomorphism $H^2_{\mathrm{sym}}(\mu_m, \mu_m) \cong \mathbb{Z}/m$. In [J] it is noted that a result of Klose allows one to choose the quaternion algebra D so that the cover of $G_D(\mathbb{A}_K)$ exists without assuming $(n, r) = 1$, that is, also for even n. In this series n is odd, so for $r = 2$ that hypothesis holds automatically, and the cover of $D^{\times}$ is defined by restriction from the Kazhdan–Patterson cover of $\mathrm{GL}(r)$ over the splitting field L ([JC], §4). We have not been able to consult [Klo86a], [Klo86b] or the dissertation. The summary above rests on their indexing data (keywords and subject classification), on the descriptions in [PRp96], [Hil] and [J], and on the titles; the theorem statements of Klose's papers are not reproduced here for that reason. We have not compared Klose's extensions with the cover of [JC] for $r = 2$. Patel [Pat] shows that the 2-fold metaplectic cover of $\mathrm{GL}(2, E)$, for a quadratic extension E/F, splits over $\mathrm{GL}(2, F)$ and over the multiplicative group of the quaternion division algebra over F; this is the same mechanism of restricting a cover over an extension field to a subgroup, and it is the mechanism by which our ambient cover realizes both $\overline{G}_D$ and $\overline{G}$.

The present series. The present paper supplies the first steps of [JC, Remark 10.3]. We found no existing metaplectic Jacquet–Langlands correspondence for covers of inner forms of $\mathrm{GL}(r)$. In particular, among the papers that zbMATH lists as citing [FK] (Zbl 0616.10024), only [J] concerns a cover of $D^\times$; those mentioning division or simple algebras treat linear groups. Part II proves the matching of arbitrary functions, using germ expansions [FK, §8] and results on the linear groups (Shalika germs, Steinberg pseudo-coefficients), and improves Theorem C. Part III studies the division-algebra side through a unitary transform of class functions, using Adams operations at tame places and Kottwitz's framework for rational conjugacy. Part IV reconciles the cocycle conventions with [FK] and [Z], studies power maps and toral positivity at the places dividing n, and proves a conditional global comparison. The companion paper studies the remaining wild local problem in degree two, using the Heisenberg–Weil formalism and character theory of compact p-adic groups.

3 Notation and glossary

This section collects the notation and conventions of the series. Subsections 3.1–3.10 are shared by all four Parts and by the companion paper on the rank-two wild place (they are the file `glossary.tex` of the project repository); they include the status labels (§3.1) and the cocycle normalization (§3.5). Subsection 3.11 lists the notation specific to Part I, with pointers to the definitions. The section proves nothing and depends on nothing; it can be consulted at any point.

3.1 Status labels

Every statement in the series carries one of the following labels. A label is never upgraded without a proof, and every change of label is recorded in the revision log of the draft in which it is made.

[P] *Proved here*: a complete argument is included in the Part in which the label appears, possibly using results labeled **[E]**. A statement proved under an explicitly stated hypothesis is labeled **[P]** together with the hypothesis, as in "**[P]** under (H_D)", and the hypothesis itself is listed separately with its own label.

[E] *Established elsewhere*: a precisely cited theorem from the literature or from an earlier Part, used as a black box. "**[P]** (using **[E]**)" marks a proof that rests on such inputs; "**[E]**/**[P]**" marks a statement that is partly cited and partly proved.

[O] *Proof outlined*: the steps are identified, but at least one of them remains to be written out or checked.

[C] *Conjectural*: new mathematical input is required. This includes the goals deferred to later Parts and the hypotheses under which conditional statements are proved.

[$\mathbf{P}^{\mathrm{c}}$] *Proved modulo an exhaustive exact computation*: the proof consists of hand arguments together with a computation that is exhaustive over an explicitly described finite set and uses exact arithmetic only (no sampling, and no floating point without exact re-verification). Each such script is committed to the repository, cited by commit hash, and described in an appendix and in the scripts' README. At present this label is used only in the companion paper on the rank-two wild place.

3.2 Global data

K, n, μ_n K is a totally imaginary number field containing the group μ_n of n-th roots of unity, and $n \geq 1$ is odd. All archimedean places of K are complex.

r, D, L, S r is a prime with $(n, r) = 1$; D is a central division algebra over K of dimension r^2; $L \subset D$ is a cyclic maximal subfield, so $[L : K] = r$; S is the (finite) set of places at which D ramifies. Since r is prime, every $v \in S$ is finite and D_v is a division algebra; $|S| \geq 2$, and $|S|$ is even if $r = 2$ (Part I, §1.1).

r', c, c_v Integers with $rr' \equiv 1 \pmod{n}$ and $n \mid r - 1 + 2c$; then $c_v := r'c$ satisfies $n \mid r - 1 + 2rc_v$ (Part I, Assumption 1.1). Despite the subscript, c_v does not depend on the place v. The parameter c occurs *before* the relabeling of μ_n described in §3.5, and c_v *after* it.

$\mathbb{A}_K$, L_v $\mathbb{A}_K$ is the adele ring of K (written $\mathbb{A}$ when no confusion is possible), and $L_v := L \otimes_K K_v = \prod_{w|v} L_w$.

G_D The algebraic group $D^\times$ over K, so that $G_D(K_v) = D_v^\times$.

Running example $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4-\zeta_3)_K$, ramified exactly at $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$ with $\mathfrak{p}_3 = (1-\zeta_3)$ and $\mathfrak{p}_7 = (3+\zeta_3)$; $L = K(i) = \mathbb{Q}(\zeta_{12})$, $c = 1$, $r' = 2$, $c_v = 2$. The place $\mathfrak{p}_3$ divides n and $\mathfrak{p}_7$ does not. The note `notes/mjl_running_example.tex` works the example out across the series.

3.3 Local fields and absolute values

F, $\mathcal{O}_F$, ϖ, q_F, p F is a nonarchimedean local field of characteristic zero containing μ_n (usually $F = K_v$), with ring of integers $\mathcal{O}_F$, uniformizer ϖ, residue field of order q_F, and residual characteristic p.

$|\cdot|_F$, $|\cdot|_p$ $|\cdot|_F$ is the normalized absolute value, $|\varpi|_F = q_F^{-1}$; on $\mathbb{Q}_p$ it is written $|\cdot|_p$, with $|p|_p = p^{-1}$. In particular $|n|_F = 1$ if $p \nmid n$, and $|F^\times / F^{\times n}| = n^2 |n|_F^{-1}$ (Part I, proof of Proposition 5.5).

Places dividing n; tame and wild places A place v (or the field $F = K_v$) *divides* n if $p \mid n$. The places $v \in S$ with $p \nmid n$ are called *tame*, and those with $p \mid n$ *wild*. In the running example $\mathfrak{p}_7$ is tame and $\mathfrak{p}_3$ is wild ($|3|_{\mathfrak{p}_3} = 3^{-2}$).

3.4 Hilbert symbols

$(a,b)_F$ The n-th Hilbert symbol of F, with values in μ_n; Part IV and the companion paper also write $(a,b)_{n,F}$ or $(a,b)_n$. It is bimultiplicative, nondegenerate and antisymmetric, $(a,b)_F = (b,a)_F^{-1}$. At complex places it is trivial.

Norm compatibility For a finite extension F'/F, $a \in F^\times$ and $b \in F'^\times$, $(a,b)_{F'} = (a, N_{F'/F}b)_F$; in particular $(a,b)_{F'} = (a,b)_F^{[F':F]}$ for $a,b \in F^\times$ [Serre, *Local Fields*, Ch. XIV].

$(a,b)_{\mathbb{A}}$ For a number field $M \supseteq K$, $(a,b)_{\mathbb{A}_M} = \prod_w (a_w, b_w)_{M_w}$. Hilbert reciprocity: $(a,b)_{\mathbb{A}_M} = 1$ for $a,b \in M^\times$.

3.5 Covers and the cocycle normalization

Cover, p, $\overline{H'}$ An *n-fold cover* of a locally compact group H is a topological central extension $1 \to \mu_n \to \overline{H} \xrightarrow{p} H \to 1$. For a subgroup $H' \subset H$ we write $\overline{H'} := p^{-1}(H')$. Parts I and II write $\overline{H}$; Parts III, IV and the companion paper write $\widetilde{H}$, $\widetilde{Z}$.

Commutators For lifts $\tilde{x}, \tilde{y}$ of commuting elements, $[x,y] := \tilde{x}\tilde{y}\tilde{x}^{-1}\tilde{y}^{-1} \in \mu_n$; it does not depend on the lifts, and it is bimultiplicative on a torus [Flicker–Kazhdan, §3].

Kazhdan–Patterson cocycle σ_F The cocycle of [Kazhdan–Patterson, §0.1] and [Flicker–Kazhdan, §2] on $\mathrm{GL}(r,F)$ with twisting parameter 0. On the diagonal torus, $\sigma_F(\mathrm{diag}(a_i), \mathrm{diag}(b_i)) = \prod_{i<j}(a_i, b_j)_F$.

Parameter-c' cover $\overline{G}^{(c')}$ The cover of $\mathrm{GL}(r,F)$ with cocycle $\sigma_F(g,h)\,(\det g, \det h)_F^{c'}$. The comparison uses $c' = c_v$; the parameter is forced (Part I, Remark 5.3). On the diagonal torus the commutator of $\mathrm{diag}(t_i)$ and $\mathrm{diag}(u_j)$ is $\prod_{i,j}(t_i,u_j)_F^{A_{ij}}$ with $A = -I_r + (1+2c_v)J_r$, where J_r is the all-ones matrix (Part IV, §2.1).

Ambient cover $\overline{\mathcal{G}}_v$ The cover of $\mathcal{G}_v = \mathrm{GL}(r, L_v) = \prod_{w|v} \mathrm{GL}(r, L_w)$ with cocycle $\prod_{w|v} \sigma_{L_w}(g_w, h_w)\,(\det g_w, \det h_w)_{L_w}^{c_v}$, the copies of μ_n being identified (Part I, §4.2 and equation (1)). Both covers compared in the series are subgroups of it, so their common central subgroup $p^{-1}(K_v^\times)$ is literally the same subgroup on the two sides.

Restriction and relabeling Pulled back along $\iota_D : D_v^\times \hookrightarrow \mathcal{G}_v$, the ambient cocycle is the local component of the cocycle $\beta = \sigma_L \cdot (\mathrm{Nrd}, \mathrm{Nrd})^c$ of the corrected 1988 paper [JC, equation (4)]. Pulled back along the diagonal embedding of $\mathrm{GL}(r, K_v)$, it is isomorphic, *as an extension*, to the extension with cocycle $\sigma_F^r(\det, \det)_F^c$ (Part I, Proposition 4.3). The group μ_n is relabeled *once*, at every place and globally, by $\zeta \mapsto \zeta^{r'}$; after relabeling the second extension is $\overline{G}^{(c_v)}$. Formulas state whether they hold before or after relabeling. The isomorphism is one of extensions, not an equality of cocycles, and it is unique up to $\tilde{g} \mapsto \chi(\det p(\tilde{g}))\,\tilde{g}$ with $\chi : F^\times \to \mu_n$

continuous. Central characters are therefore never transported through it (Part I, Remark 4.5).

Scalar centrality Because $n \mid r - 1 + 2rc_v$, the whole inverse image $\overline{Z} = p^{-1}(F^\times)$ of the scalars is central (Part I, Lemma 4.2). A cover with this property is called *scalar-central*.

Global normalization Globally one uses the normalized cocycle $\tau_w(g,h) = \sigma_w(g,h)\,\kappa_w(g)\kappa_w(h)\kappa_w(gh)^{-1}$ of [Flicker–Kazhdan, §2], where κ_w is trivial for $w \mid n\infty$. The compact splitting is $\kappa_w^*(k) = (k,1)_\tau$ on $\mathrm{GL}(r,\mathcal{O}_{L_w})$ for $w \nmid n\infty$, and the rational splitting is $\iota(x) = s_\tau(x)\,\kappa_L(x)^{-1}$; in the unnormalized models the rational splitting is $x \mapsto (x,1)_\sigma$ at every place (Part I, Lemma 7.1). The global ambient cover $\overline{\mathcal{G}}(\mathbb{A}_L)$ has cocycle $\tau_L \cdot (\det,\det)_{\mathbb{A}_L}^{c_v}$.

Star map For h in the base group and any lift $\tilde{h}$, $h^* := \tilde{h}^{\,n}$. It does not depend on the lift, it is conjugation equivariant, $(ah)^* = a^*h^*$ for scalars a, and it is multiplicative on every torus because n is odd (Part III, Definition 2.1; Part IV, §2.2). For $\zeta \in \mu_n(F)$ one has $\zeta^* = 1$.

3.6 Genuine objects, central characters and good elements

Genuine representation A representation of $\overline{H}$ on which μ_n acts by the tautological character $\zeta \mapsto \zeta$.

$\tilde{\theta}$, θ A genuine character $\tilde{\theta}$ of $\overline{Z}$ (unitary from Part I, §7.1 on; globally, a character of $\overline{Z}(\mathbb{A}_K)$ trivial on $\iota(K^\times)$). The same $\tilde{\theta}$ serves both covers, since $\overline{Z}$ is a common subgroup. The ordinary character attached to it is $\theta(a) := \tilde{\theta}(a^*)$; it is trivial on μ_n, and θ does not determine $\tilde{\theta}$ (Part IV, §2.2). Part III requires only that $\tilde{\theta}|_{\mu_n}$ be faithful.

Genuine test functions $C_c^\infty(\overline{H},\tilde{\theta})_{\mathrm{gen}}$ is the space of smooth functions with support compact modulo $\overline{Z}$ and $f(z\tilde{h}) = \tilde{\theta}(z)^{-1}f(\tilde{h})$ for $z \in \overline{Z}$; in particular $f(\zeta\tilde{h}) = \zeta^{-1}f(\tilde{h})$. Characters and invariant class functions of genuine representations transform by $\tilde{\theta}$ itself (Part III, §2).

Regular, elliptic An element of $\mathrm{GL}(r,F)$ is *regular* if its eigenvalues are distinct, and *elliptic regular* if moreover its characteristic polynomial is irreducible. Since r is prime, the regular elements of $D_v^\times$ are the noncentral ones, and they are all elliptic. An element of a cover is regular (elliptic) if its image is.

Good element A regular $\tilde{\gamma}$ is *good* if it commutes with $p^{-1}(T_\gamma)$, where T_γ is the centralizer of $\gamma = p(\tilde{\gamma})$; equivalently, $\tilde{\gamma}$ is central in the covered torus $\overline{T_\gamma}$. Genuine invariant functions vanish at regular elements that are not good. For a maximal subfield E of D_v (or an elliptic torus $E^\times$ of $\mathrm{GL}(r,F)$), the good elements of $\overline{E^\times}$ are

$p^{-1}(F^\times E^{\times n}) = \overline{Z} \cdot \{t^*\}$, of index $|n|_F^{-(r-1)}$, and every genuine irreducible representation of $\overline{E^\times}$ has dimension $|n|_F^{-(r-1)/2}$ (Part I, Proposition 5.5). Covered elliptic tori are abelian exactly at the places not dividing n.

3.7 Orbital integrals and matching

Δ For regular semisimple γ, $\Delta(\gamma) = |D(\gamma)|_F^{1/2} |\det \gamma|_F^{-(r-1)/2}$, where $D(\gamma)$ is the discriminant of the (reduced) characteristic polynomial and det = Nrd on the division-algebra side. Thus $\Delta(\gamma)^2$ is the Weyl discriminant $|\det(1 - \mathrm{Ad}(\gamma^{-1}))|_{\mathfrak{g}/\mathfrak{t}}|_F$ (Part I, Definition 6.1 and Lemma 6.12).

Φ, $F = \Delta\Phi$ $\Phi(\tilde{\gamma}, \tilde{f})$ is the orbital integral over $T_\gamma \backslash G$ of a genuine test function at a good regular $\tilde{\gamma}$, and $F(\tilde{\gamma}, \tilde{f}) = \Delta(\gamma)\Phi(\tilde{\gamma}, \tilde{f})$ is the normalized orbital integral, set to 0 at regular elements that are not good (Part I, Definition 6.1). Part III writes $J(t, f) = \Delta(t)\, O(t, f)$ for the same function, with its own measure conventions.

Canonical identification $\jmath$ At $v \in S$ and for a maximal subfield $E \subset D_v$ with an embedding $j : E \hookrightarrow M_r(F)$, every isomorphism of covered tori that is the identity on $\overline{Z}$ has the same restriction $\jmath_{E,j}$ to the good elements; these restrictions induce a bijection $\jmath$ from the good regular classes of the cover of $D_v^\times$ to the good regular elliptic classes of the cover of $\mathrm{GL}(r, F)$ (Part I, Theorem 5.9 and Corollary 5.12).

Matching Genuine test functions $\tilde{f}_D$ and $\tilde{f}$ *match* if (M1) $F(\tilde{\gamma}, \tilde{f}_D) = (-1)^{r-1} F(\jmath(\tilde{\gamma}), \tilde{f})$ at every good regular $\tilde{\gamma}$, and (M2) $F(\tilde{\delta}, \tilde{f}) = 0$ at every good regular non-elliptic $\tilde{\delta}$ (Part I, Definition 6.3). The sign sits in the matching of functions, so that the expected character relation is $\Theta_{\overline{\pi}_D} = (-1)^{r-1} \Theta_{\overline{\pi}} \circ \jmath$.

Supercusp form A genuine test function on the cover of $\mathrm{GL}(r, F)$ whose integrals along the unipotent radicals of all proper parabolic subgroups vanish, for example a matrix coefficient of a supercuspidal representation [Flicker–Kazhdan, §13].

Germ The asymptotic behavior of orbital integrals near a singular element, expressed through Shalika germs [Flicker–Kazhdan, §8]; used near the center in Part II.

3.8 Haar measures

Part I At $v \in S$ the Haar measures on $D_v^\times / F^\times$ and $\mathrm{GL}(r, F)/F^\times$ are arbitrary but fixed, and matching is defined with respect to them; torus measures are transported by conjugation and do not affect matching. At $v \notin S$ the measures correspond under the identifications ψ_v (Part I, §§6.1, 7.3).

Part II At $v \in S$, $\mathrm{vol}(D_v^\times / F^\times) = d(\mathrm{St})^{-1}$, where $d(\mathrm{St})$ is the formal degree of the Steinberg representation for the chosen measure on $\mathrm{GL}(r, F)/F^\times$ (Part II, Definition 2.1).

Part III For the transform L_D, the quotients H/Z and T/Z carry their Haar *probability* measures (Part III, §4).

Formal degree For a representation σ that is square integrable modulo the center, the constant $d(\sigma)$ in the Schur orthogonality relations; it depends on the Haar measure.

3.9 The division-side transform

H, $\widetilde{H}$, L_D For a local division algebra D of degree r over F, $H = D^\times$ and $\widetilde{H}$ is its cover from Part I. The transform L_D sends θ-equivariant invariant L^2 functions φ on H to genuine $\tilde{\theta}$-equivariant invariant functions on $\widetilde{H}$ by $(L_D\varphi)(z\,t^*) = b_D\,\tilde{\theta}(z)\,\Delta(t)\Delta(t^n)^{-1}\varphi(t)$ at good regular elements, and 0 at the others; it is unitary (Part III, Definition 4.1 and Theorem 4.3).

b_D, c_D $b_D = |n|_F^{-(r-1)/2}$ and $c_D = |n|_F^{-(r^2-1)/2}$ (Part III, §§4–5). Both equal 1 at a tame place; in the running example $b_D = 3$ and $c_D = 27$ at $\mathfrak{p}_3$.

Integrality of L_D The question whether L_D carries irreducible characters to irreducible genuine characters. It holds at tame places (Part III) and is open in general at wild places (Part IV; companion paper).

3.10 Named results and hypotheses

Theorems A, B, C Part I: the comparison of covered elliptic tori, the matching of regular-supported test functions, and the global transfer under (H_D) (with its converse under (H)).

Theorems D, E, C′ Part II: the equivariant section near the center and its compatibility with $\jmath$; the matching of arbitrary genuine test functions; and the global transfer under the weaker hypothesis (H'_D) (with its converse under (H′)).

(H_D), **(H)**, (H'_D), (H′) Non-cancellation hypotheses: the character of a near-equivalence packet at the places of S is not identically zero on the relevant matching test functions (Part I, Definition 9.9 and Theorem 9.12; Part II, Definition 6.4 and §6.3). (H_D) holds when the D-side packet is finite (Part I, Proposition 9.10); sufficient conditions for (H) are in Part I, Remark 9.13; and (H_D) ⇒ (H'_D), (H) ⇒ (H′) (Part II, §6.3).

(D1)–(D4) The steps left open by Part I: matching of arbitrary functions, local character relations, finiteness of near-equivalence packets, and composition with [JC] (Part I, §10). Their status in the later Parts is recorded in Part I, §1.3.

Targets A, B, C The companion paper's three local problems at the wild place of the running example and beyond (companion paper, §1).

3.11 Notation specific to Part I

ι_D, ι_{GL}; $\overline{G}_D(K_v)$, $\overline{G}_v$, $\overline{Z}_v$
: The embeddings of $D_v^\times$ and $\mathrm{GL}(r, K_v)$ in $\mathcal{G}_v = \mathrm{GL}(r, L_v)$, and the inverse images of their images and of $K_v^\times$ in the ambient cover $\overline{\mathcal{G}}_v$ (§4.2, Definition 4.1).

α_v, ψ_v
: For $v \notin S$: an isomorphism $D_v \cong M_r(K_v)$, and the induced isomorphism of covers $\overline{G}_D(K_v) \to \overline{G}_v$ given by conjugation in $\overline{\mathcal{G}}_v$ (Proposition 4.6). $\psi^V = \otimes_{v \notin V} \psi_v$ (§9.2).

E, j, N; $\overline{E}_D$, $\overline{E}_{\mathrm{GL}}$
: At $v \in S$: a maximal subfield $E \subset D_v$, an embedding $j : E \hookrightarrow M_r(F)$, the norm $N = N_{E/F}$, and the covered tori $p^{-1}(\iota_D(E^\times))$ and $p^{-1}(\iota_{\mathrm{GL}} j(E^\times))$ (§5).

$\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$
: Isomorphisms of covered tori over $\iota_{\mathrm{GL}} j \iota_D^{-1}$ that are the identity on μ_n and on $\overline{Z}_v$; a torsor under the characters $E^\times \to \mu_n$ trivial on $F^\times$ (Theorem 5.9).

$\mathcal{C}^g(\overline{G}_D(K_v))$, $\mathcal{C}^g_{\mathrm{ell}}(\overline{G}_v)$
: The good regular, respectively good regular elliptic, conjugacy classes (§5.5).

E_i, j_i, $\jmath_i$; dg_D, dg, dt_i
: Representatives of the degree-r extensions of F with fixed embeddings, the corresponding identifications, and the local Haar measures (§6.1).

Admissible function; $W_{i,\bullet}$
: A locally constant, $\tilde{\theta}_v^{-1}$-equivariant, $N_{G_\bullet}(T_{i,\bullet})$-invariant function on good regular elements of a covered torus with support compact modulo $\overline{Z}_v$; $W_{i,\bullet} = N_{G_\bullet}(T_{i,\bullet})/T_{i,\bullet}$ (§6.3).

$\overline{\mathcal{G}}(\mathbb{A}_L)$, $\overline{G}_D(\mathbb{A}_K)$, $\overline{G}(\mathbb{A}_K)$, $\overline{Z}(\mathbb{A}_K)$
: The global ambient cover and its subgroups (§7.1).

ι, κ_v^*, κ_w, τ_w, s_N
: The rational splitting, the compact splittings, the Flicker–Kazhdan function κ_w, the normalized cocycle (Lemma 7.1), and the unipotent splittings (Proposition 7.5(e)).

S_D
: A finite set of places containing S, the archimedean places, the places above n, and the places where L or $\tilde{\theta}$ ramifies or the integrality conditions of Proposition 4.6(c) fail (§7.3).

W_γ, $c(\gamma)$, $\mathcal{O}_H$
: For a regular elliptic rational γ: the automorphisms of $K(\gamma)/K$ fixing $K^\times\gamma$, the class-counting constant, and the set of good regular elliptic rational classes (§8).

w_0, u; **admissible pair**
: A finite place $w_0 \notin S$ carrying the supercusp form, a place $u \in S$ with elliptic support, and the pairs of test functions of Definition 8.5.

(A1), (A2); (R1)–(R5)
The analytic inputs of the proof of Proposition 8.2, and the properties of $\overline{G}(\mathbb{A}_K)$ verified in §7.4.

σ, $\tilde{f}_\sigma$, $d(\sigma)$
A genuine supercuspidal representation of $\overline{G}_{w_0}$, its normalized matrix coefficient, and its formal degree (Lemma 9.1).

V, ρ
$V = S \cup \{w_0\}$, and ρ an admissible genuine irreducible representation of $\overline{G}(\mathbb{A}^V)$ (§9.2).

Corner Hecke algebra; e_v, $e_{\mathfrak{t}}$, A_v, A_∞, A, $e\tau$
For an idempotent (or idempotent multiplier) e, the algebra $e\mathcal{H}e$; the central-character idempotents, the archimedean K_∞-type projector, the corner algebras and the module $e\tau$ of Lemma 9.5. Irreducible τ with $e\tau \neq 0$ correspond to simple $e\mathcal{H}e$-modules.

Near-equivalence packets $P_D(\rho)$, $P(\rho)$
The automorphic (resp. cuspidal) representations with components ρ away from V and σ at w_0, counted with multiplicity; they need not be finite (Proposition 9.6). Ω_D, Ω are the finite sets of representations that can contribute at a fixed level (proof of Proposition 9.6).

$\mathcal{T}_v$, a, Λ_a
The genuine irreducibles of $\overline{G}_D(K_v)$ with fixed vectors at a given level, a coefficient function on $\prod_{v\in S}\mathcal{T}_v$, and the corresponding trace functional (Lemma 9.8).

(H_D), **(H); elliptic detectability**
The condition that the character at S of the D-side (resp. GL-side) packet is not identically zero on matching regular-supported test functions (Definition 9.9, Theorem 9.12).

(D1)–(D4)
The deferred steps of §10.

Ψ, $\tilde{\Psi}$, $\overline{G}_T$, T^{reg}, $\overline{T}^{g,\mathrm{reg}}$
The conjugation maps, the open set of good elements conjugate into T^{reg}, the regular elements of an elliptic torus T, and the good regular elements of $\overline{T}$ (§6.6).

4 Local covers

This section realizes the cover of $D_v^\times$ from [JC] and the Kazhdan–Patterson cover of $\mathrm{GL}(r, K_v)$ as subgroups of one cover of the larger group $\mathrm{GL}(r, L \otimes_K K_v)$ (§4.1–4.2). It then shows that the GL-side subgroup is the Kazhdan–Patterson cover with parameter c_v, as an extension (§4.3), and identifies the two covers canonically at the places where D splits (§4.4). It depends only on the Kazhdan–Patterson cocycle [KP1, §0.1], [FK, §2] and the Matsumoto–Moore classification [Mat], [Moo], and uses from [JC] only the definition

of the cover of $D_v^\times$. It introduces the ambient cover $\overline{\mathcal{G}}_v$, the embeddings ι_D and ι_{GL}, the local covers $\overline{G}_D(K_v)$ and $\overline{G}_v$, their common central subgroup $\overline{Z}_v$, and the identifications ψ_v at $v \notin S$.

4.1 Why an ambient cover

We explain the idea before the computations.

Equality of commutator pairings does not, by itself, give a preferred isomorphism between two central extensions. Any two isomorphisms differ by a character with values in μ_n, and such characters can be nontrivial on the center. A careless choice would therefore make the comparison of central characters, and hence of genuine functions, ambiguous. Our approach is to realize both covers as *subgroups* of a single cover of $\mathrm{GL}(r, L \otimes_K K_v)$. Conjugation in this larger group then produces isomorphisms between the relevant subgroups. These isomorphisms are in general not unique, but their restrictions to the centers of covered tori are canonical (Theorem 5.9). Because both covers sit in one group, the common central subgroup $p^{-1}(K_v^\times)$ is literally the same subgroup on the two sides. This uses no choice of cocycle, so the question of whether two cocycles agree exactly or only up to a coboundary does not arise for it.

4.2 The ambient cover

We define the ambient cover at each place and the two subgroups; the notation $\overline{\mathcal{G}}_v$, ι_D, ι_{GL}, $\overline{G}_D(K_v)$, $\overline{G}_v$, $\overline{Z}_v$ introduced here is used throughout.

Fix once and for all an isomorphism $D \otimes_K L \cong M_r(L)$. For a place v of K put $L_v = L \otimes_K K_v = \prod_{w|v} L_w$ and $\mathcal{G}_v := \mathrm{GL}(r, L_v) = \prod_{w|v} \mathrm{GL}(r, L_w)$. Let $\overline{\mathcal{G}}_v$ be the cover of $\mathcal{G}_v$ obtained from the product of the Kazhdan–Patterson covers of the $\mathrm{GL}(r, L_w)$ with twisting parameter c_v, by identifying their copies of μ_n (i.e. the quotient of the product by $\{(\zeta_w) : \prod_w \zeta_w = 1\}$). A cocycle for $\overline{\mathcal{G}}_v$ is

$$\prod_{w|v} \sigma_{L_w}(g_w, h_w)\, (\det g_w, \det h_w)_{L_w}^{c_v}, \tag{1}$$

where σ_{L_w} is the Kazhdan–Patterson cocycle with parameter 0 [KP1, §0.1], [FK, §2]. Let

$$\iota_D : D_v^\times \hookrightarrow \mathcal{G}_v, \qquad \iota_{\mathrm{GL}} : \mathrm{GL}(r, K_v) \hookrightarrow \mathcal{G}_v$$

be the embedding induced by $D \otimes_K L \cong M_r(L)$, and the diagonal inclusion. Note that $\det \circ \iota_D = \mathrm{Nrd}$, and ι_D and ι_{GL} agree on the scalars $K_v^\times$.

Definition 4.1. $\overline{G}_D(K_v) := p^{-1}(\iota_D(D_v^\times)) \subset \overline{\mathcal{G}}_v$ and $\overline{G}_v := p^{-1}(\iota_{\mathrm{GL}}(\mathrm{GL}(r, K_v))) \subset \overline{\mathcal{G}}_v$. Let $\overline{Z}_v := p^{-1}(K_v^\times)$; it is a subgroup of both.

$$\begin{array}{ccccc} \overline{G}_D(K_v) & \hookrightarrow & \overline{\mathcal{G}}_v & \hookleftarrow & \overline{G}_v \\ \downarrow & & \downarrow & & \downarrow \\ D_v^\times & \xhookrightarrow{\iota_D} & \mathrm{GL}(r, L_v) & \xhookleftarrow{\iota_{\mathrm{GL}}} & \mathrm{GL}(r, K_v) \end{array} \qquad \overline{Z}_v = p^{-1}(K_v^\times) \subset \overline{G}_D(K_v) \cap \overline{G}_v. \tag{2}$$

Lemma 4.2. ***[P]****The inverse image in $\overline{\mathcal{G}}_v$ of the center $L_v^\times$ of $\mathcal{G}_v$ is central in $\overline{\mathcal{G}}_v$. In particular $\overline{Z}_v$ is central in $\overline{\mathcal{G}}_v$, in $\overline{G}_D(K_v)$ and in $\overline{G}_v$.*

Proof. By [FK, §2] (see [JC, Lemma 2.1]), the center of the parameter-c_v cover of $\mathrm{GL}(r, L_w)$ is the inverse image of $\{z : z^{r-1+2rc_v} \in L_w^{\times n}\}$, which is all of $L_w^\times$ since $n \mid r - 1 + 2rc_v$. This is an **[E]** input. Centrality in the product with identified μ_n's follows factor by factor. □

4.3 The restriction lemma

We now identify the two subgroups of Definition 4.1. The D side is the cover of [JC] by construction. The GL side is isomorphic, as a central extension, to the Kazhdan–Patterson cover with parameter c_v. This replaces the cocycle-level claim in the proof of [JC, Lemma 6.1(c)] by a statement about extensions, with a proof that does not depend on the internal form of the Kazhdan–Patterson cocycle.

Proposition 4.3. ***[P]*** *(using **[E]** inputs) Let v be a finite place.*

(a) The pull-back of the cocycle (1) along ι_D is the local component at v of the cocycle $\beta = \sigma_L \cdot (\mathrm{Nrd}, \mathrm{Nrd})^c$ of [JC, equation (4)]. Hence $\overline{G}_D(K_v)$ is the local cover of [JC, §6.1].

(b) There is an isomorphism of central extensions of $\mathrm{GL}(r, K_v)$ by μ_n (the identity on μ_n and over $\mathrm{GL}(r, K_v)$)

$$\overline{G}_v \;\cong\; \text{the extension with cocycle } \sigma_{K_v}^r\,(\det,\det)_{K_v}^c.$$

After relabeling by $\zeta \mapsto \zeta^{r'}$, the right side is the Kazhdan–Patterson cover $\overline{G}^{(c_v)}$ of $\mathrm{GL}(r, K_v)$, normalized as in [JC, Lemma 2.1].

(c) Any two isomorphisms as in (b) differ by an automorphism $\tilde{g} \mapsto \chi(\det p(\tilde{g}))\tilde{g}$ with $\chi : K_v^\times \to \mu_n$ a continuous character. Every result about $\overline{G}^{(c_v)}$ that we import from [FK] or [KP1] is stated in terms of the group $\overline{G}^{(c_v)}$, the projection p and μ_n alone (commutators, good elements, orbital integrals, characters, supercusp forms, germ expansions). It therefore holds for $\overline{G}_v$ via any such isomorphism. The only exception is the use of the splittings κ_v^* over maximal compact subgroups; these are defined directly on $\overline{G}_v$ in Proposition 7.5(d), and the local results of [JC] hold for any splitting [JC, Remarks after Lemma 6.1].

Proof. (a) For $g, h \in D_v^\times$ we have $\det \iota_D(g)_w = \mathrm{Nrd}\, g \in K_v^\times$ for each $w \mid v$. By norm compatibility, $\prod_{w|v}(\mathrm{Nrd}\, g, \mathrm{Nrd}\, h)_{L_w}^{c_v} = (\mathrm{Nrd}\, g, \mathrm{Nrd}\, h)_{K_v}^{rc_v} = (\mathrm{Nrd}\, g, \mathrm{Nrd}\, h)_{K_v}^{c}$, since $\sum_w [L_w : K_v] = r$, $rc_v = rr'c \equiv c \pmod n$, and the values lie in μ_n. The remaining factor is the restriction of the parameter-0 cocycle of $\mathrm{GL}(r, L_v)$, which is the local component of σ_L in [JC, equation (4)]. (Globally one uses the normalized cocycle τ_L of [FK, §2], which differs from the product of the σ_{L_w} by the coboundary of κ. This changes the cocycle model of $\overline{\mathcal{G}}_v$ but not the group $\overline{\mathcal{G}}_v$, nor the subgroups of Definition 4.1.)

(b) The twisting factor restricts to $(\det,\det)^c_{K_v}$ exactly as in (a). So it suffices to show that, for each $w \mid v$ with $d_w := [L_w : K_v]$, the pull-back X_1 of the parameter-0 cover of $\mathrm{GL}(r, L_w)$ to $\mathrm{GL}(r, K_v)$ is isomorphic to the extension X_2 with cocycle $\sigma_{K_v}^{d_w}$, i.e. the push-out of the parameter-0 cover of $\mathrm{GL}(r, K_v)$ along $\zeta \mapsto \zeta^{d_w}$. Taking the Baer sum over $w \mid v$ then gives the cocycle $\sigma^r_{K_v}$. Write $F = K_v$ and $F' = L_w$.

Step 1: the cocycles agree on the diagonal torus. By [FK, §2], the Kazhdan–Patterson cocycle restricted to the diagonal torus is $\sigma_{F'}(a, a') = \prod_{i<j}(a_i, a'_j)_{F'}$. For $a, a' \in A(F)$, norm compatibility gives $\sigma_{F'}(a, a') = \prod_{i<j}(a_i, a'_j)_F^{d_w} = \sigma_F(a, a')^{d_w}$. So the cocycles of X_1 and X_2 (in these models) agree *literally* on $A(F) \times A(F)$.

Step 2: the restrictions to $\mathrm{SL}(r, F)$ *are isomorphic.* By Matsumoto and Moore [Mat], [Moo], topological central extensions of $\mathrm{SL}(r, F)$ by μ_n are classified by continuous Steinberg symbols $c : F^\times \times F^\times \to \mu_n$. Every such symbol is $(\cdot,\cdot)_F^k$ for some $k \in \mathbb{Z}/n$ (Moore), and $\mathrm{SL}(r, F)$ is perfect, so an isomorphism between two extensions, if it exists, is unique [**E**]. In the extension with symbol c, the commutator of lifts of $a, a' \in A(F) \cap \mathrm{SL}(r, F)$ is $\prod_i c(a_i, a'_i)^\epsilon$ for a universal sign ϵ [Mat, §5], [Mil, §§8–9] [**E**]. Let $X_1|_{\mathrm{SL}}$ and $X_2|_{\mathrm{SL}}$ have symbols $(\cdot,\cdot)^{k_1}$ and $(\cdot,\cdot)^{k_2}$. Their torus commutators agree by Step 1. Take $a = \mathrm{diag}(x, x^{-1}, 1, \dots, 1)$ and $a' = \mathrm{diag}(y, y^{-1}, 1, \dots, 1)$; this gives $(x, y)^{2\epsilon k_1} = (x, y)^{2\epsilon k_2}$ for all x, y. As the Hilbert symbol is surjective onto μ_n and n is odd, $k_1 = k_2$. Hence the Baer difference $X := X_1 - X_2$ splits over $\mathrm{SL}(r, F)$, by a unique homomorphism $s : \mathrm{SL}(r, F) \to X$.

Step 3: X splits over $\mathrm{GL}(r, F)$. Let $A_1 = \{\mathrm{diag}(a, 1, \dots, 1)\} \cong F^\times$, so that $\mathrm{GL}(r, F) = \mathrm{SL}(r, F) \rtimes A_1$. By Step 1 the cocycle of X (the quotient of the two cocycles) is identically 1 on $A(F) \times A(F)$, so $t(a) = (a, 1)$ is a homomorphism $A_1 \to X$. For $a \in A_1$, $h \mapsto t(a)s(a^{-1}ha)t(a)^{-1}$ is again a splitting of X over $\mathrm{SL}(r, F)$, so it equals s by uniqueness. Hence $S(ha) := s(h)t(a)$ ($h \in \mathrm{SL}(r, F)$, $a \in A_1$) is a homomorphism: $S(ha\, h'a') = s(h)\, s(ah'a^{-1})\, t(aa') = s(h)t(a)s(h')t(a)^{-1}t(a)t(a') = S(ha)S(h'a')$. So X is split, i.e. $X_1 \cong X_2$.

For the relabeling, raising $\sigma^r_{K_v}(\det,\det)^c_{K_v}$ to the power r' gives $\sigma_{K_v}(\det,\det)^{c_v}_{K_v}$, the Kazhdan–Patterson cocycle with parameter c_v.

(c) Two isomorphisms of extensions over the identity differ by a continuous homomorphism $\mathrm{GL}(r, K_v) \to \mu_n$. Since $\mathrm{SL}(r, K_v)$ is perfect, such a homomorphism factors through det. The rest is a remark on how the cited results are used. □

Remark 4.4. (1) Proposition 4.3(b) is the form in which [JC, Lemma 6.1(c)] should be stated: an isomorphism of extensions, not an equality of cocycles. The proof there, which appeals to the Bruhat decomposition, is replaced by Steps 1–3 above. An equality at the level of explicit cocycles would follow from the formula of [BLS], where σ is a product of Hilbert symbols of ratios of minors (which lie in K_v when $g, h \in \mathrm{GL}(r, K_v)$); we do not need it.

(2) For v split in L, (b) is literal: the diagonal restriction of a product of r copies of a cocycle is its r-th power.

(3) At archimedean v all Hilbert symbols are trivial, and all covers in sight are split.

Remark 4.5 (Central characters and the choice of isomorphism)**.** An isomorphism $\phi : \overline{G}_v \to \overline{G}^{(c_v)}$ as in Proposition 4.3(b) is determined up to twisting by $\chi \circ \det$. On the inverse

image of a scalar z this twist multiplies by $\chi(z^r)$. So ϕ transports a genuine character $\tilde{\theta}_v$ of $\overline{Z}_v$ to a central character of $\overline{G}^{(c_v)}$ that depends on ϕ. We avoid this ambiguity as follows.

(1) Central characters live on $\overline{Z}_v \subset \overline{\mathcal{G}}_v$ and are *never* transported through ϕ. The same $\tilde{\theta}_v$ serves $\overline{G}_D(K_v)$ and $\overline{G}_v$ (Lemma 4.2).

(2) The results imported through ϕ are statements about all genuine functions or representations, for all central characters at once, or about objects that do not involve central characters: commutators and good elements [FK, §3], convergence of orbital integrals [FK, §6], supercusp forms and Selberg's principle [FK, §13], Harish-Chandra's character theory, and the analytic steps of [FK, §18]. Such a statement for $\overline{G}^{(c_v)}$ with central character $\tilde{\theta}_v \circ \phi^{-1}$ is equivalent, via ϕ, to the same statement for $\overline{G}_v$ with central character $\tilde{\theta}_v$. So these results hold for $\overline{G}_v$ independently of the choice of ϕ.

(3) Results tied to a normalized identification are *not* used in Part I. These include the unramified theory relative to the splitting of [KP1], the Flicker–Kazhdan correspondence relating specific central characters of $\overline{G}^{(c_v)}$ and $\mathrm{GL}(r, K_v)$, and exceptional representations. For the later parts we note the following. For almost all v, two splittings over $\mathrm{GL}(r, \mathcal{O}_v)$ differ by $\chi_0 \circ \det$ with χ_0 a character of $\mathcal{O}_v^\times$. Extending χ_0 to $K_v^\times$, one can choose ϕ to carry κ_v^* of Proposition 7.5(d) to the Kazhdan–Patterson splitting. Such ϕ are then unique up to $\chi \circ \det$ with χ unramified, and the remaining ambiguity on $\overline{Z}_v$ is $\chi(\varpi^r) \in \mu_n$, which must be fixed separately when the Flicker–Kazhdan correspondence is applied.

4.4 Places where D splits

At $v \notin S$ the groups $D_v^\times$ and $\mathrm{GL}(r, K_v)$ are isomorphic. We show that the corresponding covers are identified by conjugation in $\overline{\mathcal{G}}_v$, canonically once the isomorphism of the base groups is fixed, and compatibly with the splittings over maximal compact subgroups at almost all places. This is step (2) of [JC, Remark 10.3].

Proposition 4.6. ***[P]****Let $v \notin S$ and fix a K_v-algebra isomorphism $\alpha_v : D_v \xrightarrow{\sim} M_r(K_v)$.*

(a) There is $g_v \in \mathcal{G}_v$ with $\mathrm{Ad}(g_v) \circ \iota_D = \iota_{\mathrm{GL}} \circ \alpha_v$ on $D_v^\times$. It is unique up to the center $L_v^\times$ of $\mathcal{G}_v$.

(b) $\psi_v := \mathrm{Ad}(\tilde{g}_v)|_{\overline{G}_D(K_v)} : \overline{G}_D(K_v) \to \overline{G}_v$ (any lift $\tilde{g}_v$) is an isomorphism of covers over α_v. It is the identity on μ_n and on $\overline{Z}_v$, and depends only on α_v. Replacing α_v by $\mathrm{Ad}(h) \circ \alpha_v$ with $h \in \mathrm{GL}(r, K_v)$ replaces ψ_v by $\mathrm{Ad}(\tilde{h}) \circ \psi_v$.

(c) Suppose v is finite, $v \nmid n$, $\alpha_v(\mathcal{O}_{D,v}) = M_r(\mathcal{O}_v)$ for the global order $\mathcal{O}_D$ used in [JC, Lemma 6.1(d)], and $\iota_D(\mathcal{O}_{D,v}) \otimes \mathcal{O}_{L_v} = M_r(\mathcal{O}_{L_v})$ (true for almost all v). Then g_v may be taken in $\mathrm{GL}(r, \mathcal{O}_{L_v})$, and ψ_v carries the splitting $\kappa_v^*(G_D(\mathcal{O}_v))$ of [JC, Lemma 6.1(d)] to the splitting $\kappa_v^*(\mathrm{GL}(r, \mathcal{O}_v))$ of Proposition 7.5(d).

Proof. (a) Both ι_D and $\iota_{\mathrm{GL}} \circ \alpha_v$ extend L_v-linearly to L_v-algebra isomorphisms $D_v \otimes_{K_v} L_v \to M_r(L_v)$, so they differ by an L_v-algebra automorphism of $M_r(L_v) = \prod_w M_r(L_w)$. Such an automorphism preserves the idempotents of L_v, hence each factor, and on each factor it is inner (Skolem–Noether). Uniqueness up to the centralizer of $\iota_{\mathrm{GL}}(M_r(K_v)) \otimes L_v = M_r(L_v)$, which is $L_v^\times$, is clear.

(b) $\mathrm{Ad}(\tilde{g}_v)$ is an automorphism of $\overline{\mathcal{G}}_v$, trivial on μ_n, over $\mathrm{Ad}(g_v)$, so it maps $p^{-1}\iota_D(D_v^\times)$ onto $p^{-1}\iota_{\mathrm{GL}}(\mathrm{GL}(r, K_v))$. Changing $\tilde{g}_v$ within $p^{-1}(g_v L_v^\times)$ does not change $\mathrm{Ad}(\tilde{g}_v)$, since $p^{-1}(L_v^\times)$ is central (Lemma 4.2). The same lemma shows ψ_v is the identity on $\overline{Z}_v$. The last assertion is clear.

(c) Under the hypotheses, $\mathrm{Ad}(g_v)$ restricts to an automorphism of the Azumaya algebra $M_r(\mathcal{O}_{L_v})$ over $\mathcal{O}_{L_v} = \prod_w \mathcal{O}_{L_w}$, a product of local rings. Such an automorphism is inner by an element of $\mathrm{GL}(r, \mathcal{O}_{L_v})$ (Skolem–Noether for Azumaya algebras over local rings, whose Picard groups are trivial). So $g_v \in L_v^\times \mathrm{GL}(r, \mathcal{O}_{L_v})$, and we may take $g_v \in \mathrm{GL}(r, \mathcal{O}_{L_v})$. The splittings in question are the restrictions of the splitting $\kappa^*_{\mathcal{G}_v}$ of $\overline{\mathcal{G}}_v$ over $\mathrm{GL}(r, \mathcal{O}_{L_v})$ (Proposition 7.5(d) and [JC, Lemma 6.1(d)]). Since $\kappa^*_{\mathcal{G}_v}$ is a homomorphism, $\mathrm{Ad}(\kappa^*_{\mathcal{G}_v}(g_v))$ carries $\kappa^*_{\mathcal{G}_v}(\iota_D(k))$ to $\kappa^*_{\mathcal{G}_v}(g_v \iota_D(k) g_v^{-1}) = \kappa^*_{\mathcal{G}_v}(\iota_{\mathrm{GL}} \alpha_v(k))$. □

5 Comparison of covered elliptic tori at the ramified places

This section shows that, at a place $v \in S$, the covered torus $\overline{E}_D \subset \overline{G}_D(K_v)$ attached to a maximal subfield $E \subset D_v$ and the covered torus $\overline{E}_{\mathrm{GL}} \subset \overline{G}_v$ attached to an embedding $E \hookrightarrow M_r(K_v)$ have the same commutator pairing, the same good elements and the same centers. They are identified on good elements by a canonical isomorphism. We compute the pairing (§5.1) and the good elements (§5.2), give an example (§5.3), construct the identification (§5.4), and record its consequences (§5.5). It depends on §4 (Lemma 4.2 and Proposition 4.3) and on the commutator formula of [FK, §3], [KP1, Prop. 0.1.5]. It introduces the covered tori $\overline{E}_D$ and $\overline{E}_{\mathrm{GL}}$, the set $\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$, the canonical identification $\jmath_{E,j}$ and the induced bijection $\jmath$ of good regular elliptic classes.

Throughout this section, $v \in S$, $F := K_v$, w is the unique place of L over v, so that $\mathcal{G}_v = \mathrm{GL}(r, L_w)$, and L_w/F is cyclic of degree r. Let $E \subset D_v$ be a maximal subfield, so $[E : F] = r$, and let $j : E \hookrightarrow M_r(F)$ be an F-algebra embedding. (One exists, e.g. the regular representation; any two are $\mathrm{GL}(r, F)$-conjugate.) We write $N = N_{E/F}$, $\overline{E}_D := p^{-1}(\iota_D(E^\times)) \subset \overline{G}_D(K_v)$ and $\overline{E}_{\mathrm{GL}} := p^{-1}(\iota_{\mathrm{GL}} j(E^\times)) \subset \overline{G}_v$.

5.1 The commutator pairings

We compute the commutator pairing of the covered torus on each side, inside $\overline{\mathcal{G}}_v$, and find the same answer.

For $x, y \in E^\times$ write $[x, y]_D$ and $[x, y]_{\mathrm{GL}}$ for the commutators of lifts of $\iota_D(x), \iota_D(y)$ and of $\iota_{\mathrm{GL}} j(x), \iota_{\mathrm{GL}} j(y)$. They are well defined and bimultiplicative [FK, §3].

Proposition 5.1. *[**P**]For all $x, y \in E^\times$,*

$$[x,y]_D = [x,y]_{\mathrm{GL}} = (Nx, Ny)_F^{r+2c}\,(x,y)_E^{-r} \qquad \textit{(before relabeling).} \tag{3}$$

After relabeling by $\zeta \mapsto \zeta^{r'}$ this becomes

$$[x,y] = (Nx, Ny)_F^{1+2c_v}\,(x,y)_E^{-1} = \Big((Nx)^{1+2c_v}x^{-1},\, y\Big)_E, \tag{4}$$

which is [JC, equation (2)] for the elliptic torus $j(E^\times)$ of $\overline{G}^{(c_v)}$.

Proof. Commutators are invariants of the extension, so we may compute in any model. We use the formula of [FK, §3] and [KP1, Prop. 0.1.5] [**E**]: for a maximal torus $T' = \prod_i F_i'^\times$ of $\mathrm{GL}(r, F')$ and the parameter-0 cover, $[x,y] = (\det x, \det y)_{F'} \prod_i (x_i, y_i)_{F_i'}^{-1}$. Twisting by $(\det,\det)^m$ multiplies commutators by $(\det x, \det y)^{2m}$.

The D side. $\iota_D(x), \iota_D(y)$ lie in the maximal torus $\iota_D(E \otimes_F L_w)^\times$ of $\mathcal{G}_v$. Write $E \otimes_F L_w = \prod_i E_i$ with fields $E_i \supseteq E$; then $\sum_i [E_i : L_w] = r$, and $[E_i : E] = [E_i : L_w]$ since $[E : F] = [L_w : F]$. We have $\det \iota_D(x) = Nx \in F^\times$, so

$$(\det x, \det y)_{L_w} = (Nx, Ny)_F^r, \qquad \prod_i (x,y)_{E_i} = \prod_i (x, N_{E_i/E}\, y)_E = (x,y)_E^r.$$

The twist by $(\det,\det)_{L_w}^{c_v}$ contributes $(Nx, Ny)_F^{2rc_v} = (Nx, Ny)_F^{2c}$.

The GL *side.* By Proposition 4.3(b) the extension $\overline{G}_v$ has cocycle $\sigma_F^r(\det,\det)_F^c$ up to isomorphism. For the parameter-0 cover and the elliptic torus $j(E^\times)$, [FK, §3] gives $(Nx, Ny)_F (x,y)_E^{-1}$. Raising to the r-th power and multiplying by $(Nx, Ny)_F^{2c}$ gives the same expression.

Relabeling. Raise (3) to the power r'. For the last equality in (4), $(Nx, Ny)_F = (Nx, y)_E$ by norm compatibility, since $Nx \in F^\times$. □

Remark 5.2. The equality $[x,y]_D = [x,y]_{\mathrm{GL}}$ uses the formula of [FK, §3] only through its multiplicative shape. If the formula were $(\det x, \det y)^a \prod_i (x_i, y_i)^b$, both sides of (3) would become $(Nx, Ny)_F^{ra+2c}(x,y)_E^{rb}$.

Remark 5.3 (The twisting parameter is forced)**.** If the GL-side cover had parameter c' instead of c_v, its pairing on $j(E^\times)$ would be $(Nx, Ny)_F^{1+2c'}(x,y)_E^{-1}$. This agrees with (4) if and only if $(Nx, Ny)_F^{2(c'-c_v)} = 1$ for all $x, y \in E^\times$. Since $N(E^\times) \supseteq F^{\times r}$ and $(F^{\times r}, F^{\times r})_F = \mu_n$, and n is odd, this holds if and only if $c' \equiv c_v \pmod n$. So matching requires exactly $\overline{G}^{(c_v)}$, compared with the D side through the same relabeling of μ_n.

5.2 Good elements and centers

From the explicit pairing we determine the good elements on both sides, the index of the center, and hence the dimension of genuine irreducible representations of the covered torus.

Lemma 5.4. *[**P**]$F^\times \cap E^{\times n} = F^{\times n}$. Hence $F^\times/F^{\times n} \to E^\times/E^{\times n}$ is injective.*

Proof. If $z = y^n$ with $y \in E^\times$, then $z^r = N(z) = N(y)^n \in F^{\times n}$. Also $z^n \in F^{\times n}$, and $(r, n) = 1$, so $z \in F^{\times n}$. □

Proposition 5.5. ***[P]****Let* $\bullet \in \{D, \mathrm{GL}\}$.

(a) An element $\tilde{x} \in \overline{E}_\bullet$ *is good (central in* $\overline{E}_\bullet$*) if and only if* $p(\tilde{x}) \in F^\times E^{\times n}$*. No regularity assumption is needed.*

(b) $[\overline{E}_\bullet : Z(\overline{E}_\bullet)] = [E^\times : F^\times E^{\times n}] = |n|_F^{-(r-1)}$*. In particular* $\overline{E}_\bullet$ *is abelian if and only if* $v \nmid n$*.*

(c) Every genuine irreducible smooth representation of $\overline{E}_\bullet$ *has dimension* $|n|_F^{-(r-1)/2}$ *(an integer: if* $p \mid n$ *and* $p^k \| n$*, then* $\mu_{p^k} \subset F$ *forces* $p - 1 \mid [F : \mathbb{Q}_p]$*, so* $|n|_F^{-1} = p^{k[F:\mathbb{Q}_p]}$ *is a square), and is determined by its central character on* $Z(\overline{E}_\bullet)$ *[JC, Prop. 2.4].*

(d) If $x \in E^\times \smallsetminus F^\times$*, then* $F(x^n) = E$*. In particular, if* x *is regular, so is* x^n*.*

Proof. (a) By (4), $\tilde{x}$ is central if and only if $((Nx)^{1+2c_v} x^{-1}, y)_E = 1$ for all $y \in E^\times$. By nondegeneracy of the Hilbert symbol, this holds if and only if $x \equiv (Nx)^{1+2c_v} \pmod{E^{\times n}}$. If so, $x \in F^\times E^{\times n}$. Conversely, if $x = zy^n$ with $z \in F^\times$, then $(Nx)^{1+2c_v} \equiv z^{r(1+2c_v)} = z \cdot z^{r-1+2rc_v} \equiv z \equiv x \pmod{E^{\times n}}$.

(b) Since $\mu_n \subset F$, we have $|F^\times/F^{\times n}| = n^2|n|_F^{-1}$ and $|E^\times/E^{\times n}| = n^2|n|_E^{-1} = n^2|n|_F^{-r}$ [Neu, Ch. II, §5] **[E]**. Now use Lemma 5.4.

(c) Stone–von Neumann: the dimension is the square root of the index.

(d) Put $F' = F(x^n) \subseteq E$. Since $\mu_n \subset F'$, the degree of the Kummer extension $F'(x)/F'$ divides n. It also divides r, so it is 1. As r is prime and $x \notin F$, $F(x) = E$, so $F' = E$. □

Remark 5.6. Part (b) generalizes [JC, Lemma 2.5] from unramified E to all maximal subfields when $v \nmid n$. Part (d) shows that for regular elliptic elements the condition "x^* regular" in [FK, §18] is automatic.

5.3 An example

We illustrate the index computation with the smallest case in which a place above n makes the covered torus non-abelian.

Example 5.7. Take $r = 2$, $n = 3$, and $F = K_v = \mathbb{Q}_3(\zeta_3)$, so D_v is the quaternion division algebra over F. We may take $c = 1$ (so $3 \mid r - 1 + 2c = 3$), $r' = 2$ and $c_v = 2$, and indeed $3 \mid r - 1 + 2rc_v = 9$. For any quadratic extension E/F, (4) reads $[x, y] = (Nx, Ny)_F^5 (x, y)_E^{-1} = (Nx, Ny)_F^2 (x, y)_E^{-1}$. The field F is totally ramified of degree 2 over $\mathbb{Q}_3$, so $|3|_F = 3^{-2}$ and

$$[\overline{E} : Z(\overline{E})] = |3|_F^{-1} = 9.$$

So every genuine irreducible representation of $\overline{E}_D$ or $\overline{E}_{\mathrm{GL}}$ has dimension 3, and the good elements form a subgroup of index 9 in $E^\times$. (As a check, $|F^\times/F^{\times 3}| = 3 \cdot 3 \cdot 9 = 81$.) By contrast, at a place $v \in S$ of residual characteristic $\neq 3$, every covered elliptic torus is abelian.

5.4 A canonical identification on good elements

Equal commutator pairings leave open whether the two covered tori are isomorphic, and if so how canonically. We show that they are isomorphic, by conjugation inside $\overline{\mathcal{G}}_v$, and that every isomorphism that is the identity on the common central subgroup $\overline{Z}_v$ has the same restriction to good elements. This defines the identification $\jmath_{E,j}$ used in the rest of the paper.

Definition 5.8. Let $\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$ be the set of isomorphisms of topological groups $\varphi : \overline{E}_D \to \overline{E}_{\mathrm{GL}}$ with $p \circ \varphi = (\iota_{\mathrm{GL}} j \iota_D^{-1}) \circ p$ that are the identity on μ_n and on $\overline{Z}_v$.

Theorem 5.9. ***[P]***

(a) There is $g \in \mathcal{G}_v$ with $g\,\iota_D(e)\,g^{-1} = \iota_{\mathrm{GL}}(j(e))$ for all $e \in E$. For any lift $\tilde{g}$, $\mathrm{Ad}(\tilde{g})$ restricts to an element $\varphi_g \in \mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$.

(b) Any two elements of $\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$ agree on $Z(\overline{E}_D)$. The common restriction

$$\jmath_{E,j} : Z(\overline{E}_D) \xrightarrow{\sim} Z(\overline{E}_{\mathrm{GL}})$$

is an isomorphism and a homeomorphism. It is the identity on $\overline{Z}_v$, satisfies $p \circ \jmath_{E,j} = (\iota_{\mathrm{GL}} j \iota_D^{-1}) \circ p$, commutes with n-th powers, and preserves regularity.

(c) $\mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$ is a torsor under the group of continuous characters $E^\times \to \mu_n$ trivial on $F^\times$; all of these are trivial on $F^\times E^{\times n}$.

Proof. (a) Via ι_D and $\iota_{\mathrm{GL}} \circ j$, the space L_w^r is a module over the étale L_w-algebra $A := E \otimes_F L_w = \prod_i E_i$, faithful in both cases. A faithful A-module of L_w-dimension $r = \dim_{L_w} A$ is free of rank one, since each E_i must act nontrivially and dimensions add up. So the two module structures are conjugate by some $g \in \mathcal{G}_v$. Then $\mathrm{Ad}(\tilde{g})$ is an automorphism of $\overline{\mathcal{G}}_v$ over $\mathrm{Ad}(g)$, trivial on μ_n, mapping $\overline{E}_D$ onto $\overline{E}_{\mathrm{GL}}$. It is the identity on $\overline{Z}_v$ by Lemma 4.2, and it is continuous with continuous inverse.

(b) If $\varphi, \varphi' \in \mathrm{Iso}_{\overline{Z}}$, then $\varphi' \varphi^{-1}$ is an automorphism of $\overline{E}_{\mathrm{GL}}$ over the identity, trivial on μ_n, so it is $\tilde{x} \mapsto \chi(x)\tilde{x}$ for a continuous character $\chi : E^\times \to \mu_n$. Then $\chi|_{F^\times} = 1$, and χ is trivial on $E^{\times n}$ since it has values in μ_n. So $\varphi = \varphi'$ on $p^{-1}(F^\times E^{\times n}) = Z(\overline{E}_D)$ (Proposition 5.5(a)). The other properties hold for any φ, and φ_g is a homeomorphism by (a).

(c) This is the computation in (b), together with the fact that $\tilde{x} \mapsto \chi(x)\tilde{x}$ is an automorphism for every such χ. □

$$\begin{array}{ccc} Z(\overline{E}_D) & \xrightarrow[\sim]{\jmath_{E,j}} & Z(\overline{E}_{\mathrm{GL}}) \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p} \\ F^\times E^{\times n} & \xrightarrow{=} & F^\times E^{\times n} \end{array} \qquad \text{(identity on } \overline{Z}_v\text{, on the bottom identified via } \iota_D,\ \iota_{\mathrm{GL}} \circ j\text{).} \tag{5}$$

Remark 5.10. Off the good elements, φ depends on choices through a character of $E^\times/F^\times E^{\times n}$, a group that is nontrivial exactly when $v \mid n$. This does no harm, since genuine orbital integrals vanish at non-good elements (§6) and genuine representations of covered tori are determined by their central characters.

5.5 Consequences

We record that the identification is natural for conjugation, independent of j, and induces bijections of good regular elliptic classes and of genuine representations of covered tori. The class sets $\mathcal{C}^g(\overline{G}_D(K_v))$, $\mathcal{C}^g_{\mathrm{ell}}(\overline{G}_v)$ and the bijection $\jmath$ defined here enter the definition of matching in §6.

Proposition 5.11 (Naturality)**.** ***[P]****Let $y \in D_v^\times$ and $h \in \mathrm{GL}(r,F)$, and put $E' = yEy^{-1}$ and $j' = \mathrm{Ad}(h)\circ j \circ \mathrm{Ad}(y)^{-1} : E' \hookrightarrow M_r(F)$. Then on $Z(\overline{E}_D)$*

$$\jmath_{E',j'} \circ \mathrm{Ad}(\tilde{y}) = \mathrm{Ad}(\tilde{h}) \circ \jmath_{E,j}.$$

In particular $\jmath_{E,j}$ depends on j only through its $\mathrm{GL}(r,F)$-conjugacy class, which is unique. If y normalizes E and h normalizes $j(E)$, inducing the same automorphism of E, then $\jmath_{E,j} \circ \mathrm{Ad}(\tilde{y}) = \mathrm{Ad}(\tilde{h}) \circ \jmath_{E,j}$.

Proof. Put $j_1 := \mathrm{Ad}(h)\circ j = j' \circ \mathrm{Ad}(y)$. Let $\varphi \in \mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$ and $\varphi' \in \mathrm{Iso}_{\overline{Z}}(\overline{E}'_D, \overline{E}'_{\mathrm{GL}})$, relative to j and j'. Then $\mathrm{Ad}(\tilde{h})\circ\varphi$ and $\varphi' \circ \mathrm{Ad}(\tilde{y})$ are isomorphisms $\overline{E}_D \to \overline{E}'_{\mathrm{GL}}$ over j_1. Both are the identity on μ_n and on the central subgroup $\overline{Z}_v$ (Lemma 4.2). By Theorem 5.9(b) for j_1, they agree on $Z(\overline{E}_D)$. □

A regular element $\tilde{\gamma}$ of $\overline{G}_D(K_v)$ or $\overline{G}_v$ is *good* if it is central in $\overline{T_\gamma}$, where T_γ is the centralizer of $\gamma = p(\tilde{\gamma})$. Let $\mathcal{C}^g(\overline{G}_D(K_v))$ be the set of conjugacy classes of good regular elements of $\overline{G}_D(K_v)$ (all regular elements of $D_v^\times$ are elliptic), and $\mathcal{C}^g_{\mathrm{ell}}(\overline{G}_v)$ the set of conjugacy classes of good regular elliptic elements of $\overline{G}_v$.

Corollary 5.12. ***[P]****The maps $\jmath_{E,j}$ induce a well-defined bijection*

$$\jmath : \mathcal{C}^g(\overline{G}_D(K_v)) \xrightarrow{\sim} \mathcal{C}^g_{\mathrm{ell}}(\overline{G}_v), \qquad [\tilde{\gamma}] \mapsto [\jmath_{F(\gamma),j}(\tilde{\gamma})].$$

It preserves (reduced) characteristic polynomials and commutes with multiplication by $\overline{Z}_v$.

Proof. Well defined: by Proposition 5.11. *Surjective:* if $\tilde{\gamma}'$ is good regular elliptic, $E' = F(p(\tilde{\gamma}'))$ is a field of degree r, which embeds in D_v [W, Ch. XIII]. Choose $E \subset D_v$ and j with $j(E) = E'$; then $\tilde{\gamma}' \in Z(\overline{E}_{\mathrm{GL}})$ is $\jmath_{E,j}$ of a unique element. *Injective:* if $\jmath(\tilde{\gamma}_1)$ and $\jmath(\tilde{\gamma}_2)$ are conjugate, then $p(\tilde{\gamma}_1)$ and $p(\tilde{\gamma}_2)$ have the same characteristic polynomial, hence are conjugate in $D_v^\times$. By naturality we may assume $p(\tilde{\gamma}_1) = p(\tilde{\gamma}_2)$, so $\tilde{\gamma}_2 = \zeta\tilde{\gamma}_1$ with $\zeta \in \mu_n$ and $\zeta\jmath(\tilde{\gamma}_1)$ is conjugate to $\jmath(\tilde{\gamma}_1)$. If $x^{-1}\tilde{\delta}x = \zeta\tilde{\delta}$ with $\tilde{\delta}$ regular and good, then $p(x)$ centralizes $p(\tilde{\delta})$, so $x \in \overline{T_\delta}$ commutes with $\tilde{\delta}$, and $\zeta = 1$. □

Corollary 5.13. ***[P]****For $\varphi \in \mathrm{Iso}_{\overline{Z}}(\overline{E}_D, \overline{E}_{\mathrm{GL}})$, $\chi \mapsto \chi\circ\varphi^{-1}$ induces a bijection between isomorphism classes of genuine irreducible smooth representations of $\overline{E}_D$ and $\overline{E}_{\mathrm{GL}}$, independent of φ. On good elements $\mathrm{tr}\,\chi(\tilde{t}) = \mathrm{tr}(\chi\circ\varphi^{-1})(\jmath_{E,j}(\tilde{t}))$, and both characters vanish off the good elements.*

Proof. By [JC, Prop. 2.4] the isomorphism class is determined by the central character ζ_χ on $Z(\overline{E}_\bullet)$, and the central character of $\chi \circ \varphi^{-1}$ is $\zeta_\chi \circ \jmath_{E,j}^{-1}$, independent of φ by Theorem 5.9(b). □

6 Orbital integrals and matching

This section defines normalized orbital integrals of genuine functions and the notion of matching (§§6.1–6.2). It then proves that matching functions exist, in both directions, for functions supported on the regular set (§§6.3–6.4). Finally it explains exactly what fails for general functions (§6.5). The last subsection, §6.6, proves the facts about orbital integrals on elliptic tori that the section uses. It depends on §5. It introduces the normalized orbital integrals Φ and $F = \Delta\Phi$, the matching conditions (M1)–(M2), the representatives E_i, j_i, $\jmath_i$ of elliptic tori, and admissible functions on covered tori. Throughout, $v \in S$ and $F = K_v$, and $\tilde{\theta}_v$ is a genuine character of $\overline{Z}_v$; by Lemma 4.2 the same character serves both sides.

6.1 Orbital integrals and measures

We fix measures, define the normalized orbital integrals, and record their basic properties. The representatives E_i, j_i, $\jmath_i$ and the measures dg_D, dg, dt_i fixed here are used in the rest of the paper.

Representatives of tori. The field extensions of F of degree r fall into finitely many isomorphism classes; fix representatives $E_1, \dots, E_m$. For each i fix an embedding $E_i \subset D_v$ (it exists [W, Ch. XIII]) and an embedding $j_i : E_i \hookrightarrow M_r(F)$. Every maximal subfield of D_v is $D_v^\times$-conjugate to exactly one E_i, and every elliptic maximal torus of $\mathrm{GL}(r,F)$ is conjugate to exactly one $j_i(E_i^\times)$ (Skolem–Noether). Write $\jmath_i := \jmath_{E_i,j_i}$.

Measures. Fix Haar measures dg_D on $D_v^\times/F^\times$ and dg on $\mathrm{GL}(r,F)/F^\times$, and for each i a Haar measure dt_i on $E_i^\times/F^\times$. Use dt_i on both $\iota_D(E_i^\times)/F^\times$ and $j_i(E_i^\times)/F^\times$, and transport it by conjugation to every torus conjugate to these. Quotient measures on $T\backslash G$ are dg/dt. The matching relation below does not depend on the choice of the dt_i: rescaling dt_i by λ multiplies the orbital integrals on both sides at elements of tori of type i by λ^{-1}. It does depend on dg_D and dg, which are fixed.

Definition 6.1. For $\bullet \in \{D, \mathrm{GL}\}$, $G_\bullet \in \{D_v^\times, \mathrm{GL}(r,F)\}$, a genuine $\tilde{f} \in C_c^\infty(\overline{G_\bullet}, \tilde{\theta}_v)_{\mathrm{gen}}$ and a good regular semisimple $\tilde{\gamma}$ with centralizer T_γ, put

$$\Phi(\tilde{\gamma}, \tilde{f}) := \int_{T_\gamma \backslash G_\bullet} \tilde{f}(\tilde{x}^{-1}\tilde{\gamma}\tilde{x})\, d\dot{x}, \qquad F(\tilde{\gamma}, \tilde{f}) := \Delta(\gamma)\, \Phi(\tilde{\gamma}, \tilde{f}),$$

with $\Delta(\gamma) = |D(\gamma)|_F^{1/2} |\det \gamma|_F^{-(r-1)/2}$ as in [FK, §6], where $D(\gamma)$ is the discriminant of the (reduced) characteristic polynomial and det = Nrd on the D side. For non-good regular $\tilde{\gamma}$ put $F(\tilde{\gamma}, \tilde{f}) := 0$.

Lemma 6.2. ***[P]*** *(convergence:* ***[E]****)*

(a) *The integrand of* Φ *is independent of the lift* $\tilde{x}$ *and left* T_γ*-invariant (because* $\tilde{\gamma}$ *is good). The integral converges absolutely. On the* D *side* $T_\gamma \backslash D_v^\times$ *is compact, since* $D_v^\times / F^\times$ *is. On the* GL *side the orbit of a regular semisimple element is closed, and convergence is [FK, §6] (following Rao)* ***[E]****.*

(b) $F(\cdot, \tilde{f})$ *is constant on conjugacy classes, and* $F(z\tilde{\gamma}, \tilde{f}) = \tilde{\theta}_v(z)^{-1} F(\tilde{\gamma}, \tilde{f})$ *for* $z \in \overline{Z}_v$*; in particular it is genuine.*

(c) *If* $\tilde{\gamma}$ *is regular elliptic and not good, then the formal integral* $\int_{T_\gamma \backslash G_\bullet}$ *(over* $\overline{T_\gamma} \backslash \overline{G_\bullet}$*) vanishes; this is [JC, Prop. 2.7] with* $\epsilon = 1$*. Thus the convention* $F = 0$ *at non-good elements loses nothing.*

(d) *For* $\gamma \in E_i^\times$ *regular,* $\Delta(\gamma)$ *is the same on the two sides. If* E_i/F *is cyclic, then* Δ *equals* $|t_0^2|_F^{1/2}$ *times the factor* Δ *of [JC, §4.1] with* $\epsilon = 1$*, for any* t_0 *as there, so that* $F(\cdot, \tilde{f}) = |t_0^2|_F^{1/2}\, \tilde{f}^T$ *in the notation of [JC, §4.1].*

Proof. (a) and (b) are immediate from the definitions, Lemma 4.2 and conjugation invariance of the measures. (c) is [JC, Prop. 2.7]. (d) The characteristic polynomials agree. For [JC]'s factor (defined only for cyclic E_i): $\tilde{\Delta}(t)^2 = \pm D(t)$ and $t_0^2 \in F^\times$, so $|\tilde{\Delta}(t) t_0^{-1}|_F = |D(t)|_F^{1/2} |t_0^2|_F^{-1/2}$. □

6.2 Definition of matching

We define matching with the sign $(-1)^{r-1}$ and with vanishing at good regular non-elliptic elements, using the bijection of Corollary 5.12.

Definition 6.3. Functions $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$ and $\tilde{f} \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ *match* if

(M1) $F(\tilde{\gamma}, \tilde{f}_D) = (-1)^{r-1} F(\jmath(\tilde{\gamma}), \tilde{f})$ for every good regular $\tilde{\gamma} \in \overline{G}_D(K_v)$, and

(M2) $F(\tilde{\delta}, \tilde{f}) = 0$ for every good regular non-elliptic $\tilde{\delta} \in \overline{G}_v$.

By Corollary 5.12 and Lemma 6.2(b), (M1) is a condition on conjugacy classes, and it is compatible with the common central character $\tilde{\theta}_v$. With the sign placed here, the expected character relation $\Theta_{\overline{\pi}_D} = (-1)^{r-1} \Theta_{\overline{\pi}} \circ \jmath$ on good regular elements would give $\operatorname{tr} \overline{\pi}_D(\tilde{f}_D) = \operatorname{tr} \overline{\pi}(\tilde{f})$ for matching functions. Badulescu [Bad, §4.6] puts no sign in the matching of functions and keeps the sign $(-1)^{nd-n}$ in the character relation; the two conventions are equivalent.

6.3 Transported orbital-integral functions

We show that the orbital integrals of a regular-supported function on one side, transported to the other side by the identifications $\jmath_i$, have exactly the properties that characterize orbital integrals of regular-supported functions there. The proofs, for arbitrary (not necessarily cyclic) maximal subfields, are in §6.6. The notion of an admissible function defined here is used again in §6.6.

For $\bullet \in \{D, \mathrm{GL}\}$ and each i, let $\overline{E}_{i,\bullet}$ be the covered torus of type i (Definition in §5, with $E = E_i$, $j = j_i$), $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ its good regular elements, and $W_{i,\bullet} := N_{G_\bullet}(T_{i,\bullet})/T_{i,\bullet}$, which acts on $Z(\overline{E}_{i,\bullet})$ by conjugation and injects into $\mathrm{Aut}(E_i/F)$. Call a function h on $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ *admissible* if

(i) h is locally constant;

(ii) $h(z\tilde{t}) = \tilde{\theta}_v(z)^{-1}h(\tilde{t})$ for $z \in \overline{Z}_v$ (in particular h is genuine);

(iii) the support of h is compact modulo $\overline{Z}_v$ and contained in $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$, i.e. its closure in $Z(\overline{E}_{i,\bullet})$ avoids the non-regular elements;

(iv) $h(\tilde{y}\tilde{t}\tilde{y}^{-1}) = h(\tilde{t})$ for $y \in N_{G_\bullet}(T_{i,\bullet})$.

Lemma 6.4. ***[P]***

(a) If $\tilde{f} \in C_c^\infty(\overline{G_\bullet}, \tilde{\theta}_v)_{\mathrm{gen}}$ is supported in the regular set, then for each i the restriction of $F(\cdot, \tilde{f})$ to $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ is admissible.

(b) $h \mapsto h \circ \jmath_i^{-1}$ is a bijection from admissible functions on $Z(\overline{E}_{i,D})^{\mathrm{reg}}$ to admissible functions on $Z(\overline{E}_{i,\mathrm{GL}})^{\mathrm{reg}}$.

(c) Conversely, every admissible h on $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ is of the form $F(\cdot, \tilde{f}_i)|_{Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}}$ for some $\tilde{f}_i \in C_c^\infty(\overline{G_\bullet}, \tilde{\theta}_v)_{\mathrm{gen}}$ supported on the set of $G_\bullet$-conjugates of $\operatorname{supp} h$. *In particular $\tilde{f}_i$ is supported in the good regular set of type i, and $F(\tilde{\delta}, \tilde{f}_i) = 0$ for every regular $\tilde{\delta}$ not conjugate into $Z(\overline{E}_{i,\bullet})$.*

Proof. (a) This is Lemma 6.10, applied to $G_\bullet$ and the elliptic torus $T_{i,\bullet}$. For cyclic E_i it is also [JC, Lemma 4.5(a)] with $\epsilon = 1$, up to the constant of Lemma 6.2(d).

(b) $\jmath_i$ is a homeomorphism (Theorem 5.9(b)), so (i) is preserved. It is the identity on $\overline{Z}_v$ (so (ii) and compactness modulo $\overline{Z}_v$ in (iii) are preserved) and preserves regularity in both directions (so the rest of (iii) is preserved). For (iv): let $y \in N_{D_v^\times}(E_i)$ induce $\sigma \in \mathrm{Aut}(E_i/F)$. By Skolem–Noether there is $h \in N_{\mathrm{GL}(r,F)}(j_i(E_i))$ inducing $j_i\sigma j_i^{-1}$, and conversely. By Proposition 5.11, $\jmath_i \circ \mathrm{Ad}(\tilde{y}) = \mathrm{Ad}(\tilde{h}) \circ \jmath_i$, so invariance under $\mathrm{Ad}(\tilde{y})$ on one side is equivalent to invariance under $\mathrm{Ad}(\tilde{h})$ on the other. (Elements of $T_{i,\bullet}$ itself act trivially on good elements.) The same argument with $\jmath_i^{-1}$ gives the inverse.

(c) This is Lemma 6.11, applied to $G_\bullet$ and $T_{i,\bullet}$. A regular $\tilde{\delta}$ is conjugate into $Z(\overline{E}_{i,\bullet})$ only if its centralizer is conjugate to $T_{i,\bullet}$. □

6.4 The easy direction, in both directions

We combine the previous lemmas into the existence theorem for matching functions supported on the regular set, and check the hypotheses of the simple trace formula of [FK, §18].

Theorem 6.5. ***[P]***

(a) *Let $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$ vanish on a neighborhood of $\overline{Z}_v$. Then there is $\tilde{f} \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$, supported on the good regular elliptic set, matching $\tilde{f}_D$.*

(b) *Let $\tilde{f} \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ be supported in the regular set, with $F(\tilde{\delta}, \tilde{f}) = 0$ for all good regular non-elliptic $\tilde{\delta}$. Then there is $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$ vanishing on a neighborhood of $\overline{Z}_v$ and matching $\tilde{f}$.*

(c) *Every $\tilde{f}$ as in (a) satisfies, at the place v, both local hypotheses of [FK, §18, Theorem]: $F(x, \tilde{f}) = 0$ for all regular non-elliptic x, and $\tilde{f}$ vanishes on $\{x : x^* \text{ singular}\}$, where $x^* = \tilde{x}^n$.*

Proof. (a) Since r is prime, the non-regular elements of $D_v^\times$ are the central ones. So $\tilde{f}_D$, which vanishes near $\overline{Z}_v$ and has support compact modulo $\overline{Z}_v$, is supported in the regular set. For each i, Lemma 6.4(a) shows that $h_i^D := F(\cdot, \tilde{f}_D)|_{Z(\overline{E}_{i,D})^{\mathrm{reg}}}$ is admissible. By (b) of that lemma, $h_i := (-1)^{r-1} h_i^D \circ \jmath_i^{-1}$ is admissible on $Z(\overline{E}_{i,\mathrm{GL}})^{\mathrm{reg}}$, and by (c) there is $\tilde{f}_i$ on $\overline{G}_v$ with $F(\cdot, \tilde{f}_i) = h_i$ on $Z(\overline{E}_{i,\mathrm{GL}})^{\mathrm{reg}}$ and $F(\tilde{\delta}, \tilde{f}_i) = 0$ for regular $\tilde{\delta}$ not conjugate into $Z(\overline{E}_{i,\mathrm{GL}})$. Put $\tilde{f} := \sum_i \tilde{f}_i$, which is supported on the good regular elliptic set.

(M2): a good regular non-elliptic $\tilde{\delta}$ is not conjugate into any elliptic torus, so $F(\tilde{\delta}, \tilde{f}) = 0$.

(M1): let $\tilde{\gamma} \in \overline{G}_D(K_v)$ be good regular. It is conjugate to some $\tilde{t} \in Z(\overline{E}_{i,D})^{\mathrm{reg}}$ for a unique i, and $\jmath(\tilde{\gamma})$ is conjugate to $\jmath_i(\tilde{t})$ (Corollary 5.12). The tori of types $k \neq i$ do not contribute at $\jmath_i(\tilde{t})$. Hence

$$F(\jmath(\tilde{\gamma}), \tilde{f}) = F(\jmath_i(\tilde{t}), \tilde{f}_i) = h_i(\jmath_i(\tilde{t})) = (-1)^{r-1} F(\tilde{t}, \tilde{f}_D) = (-1)^{r-1} F(\tilde{\gamma}, \tilde{f}_D).$$

(b) For each i, the restriction of $F(\cdot, \tilde{f})$ to $Z(\overline{E}_{i,\mathrm{GL}})^{\mathrm{reg}}$ is admissible by Lemma 6.4(a). Transport it by $\jmath_i^{-1}$, multiply by $(-1)^{r-1}$, realize it on $\overline{G}_D(K_v)$ by Lemma 6.4(c), and sum over i. (M1) follows as in (a), and (M2) is the hypothesis. The support of $\tilde{f}_D$ is closed, compact modulo $\overline{Z}_v$, and contained in the regular set, so it avoids a neighborhood of $\overline{Z}_v$.

(c) The first hypothesis is (M2) together with the vanishing at non-good elements. For the second, every $x \in \operatorname{supp} \tilde{f}$ is regular elliptic, so $p(x)^n$ is regular by Proposition 5.5(d) (applied in the torus $F(p(x))^\times$), and $p(x^*) = p(x)^n$. □

6.5 What fails for general functions

We explain why Theorem 6.5(a) cannot be extended to all $\tilde{f}_D$ by the same method, by computing the orbital integrals of $\tilde{f}_D$ near the center exactly.

Proposition 6.6. ***[P]****Let $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$. There are an open compact subgroup $U \subset D_v^\times$, normal in $D_v^\times$, and a splitting $\kappa_U : U \to \overline{G}_D(K_v)$ with the following properties.*

(a) *$\mathrm{Ad}(\tilde{x}) \circ \kappa_U = \kappa_U \circ \mathrm{Ad}(x)$ on U for every $x \in D_v^\times$, and $\kappa_U(t)$ is good for every regular $t \in U$.*

(b) *For every $z \in \overline{Z}_v$, every maximal subfield E and every regular $t \in E^\times \cap U$,*

$$F(z\,\kappa_U(t), \tilde{f}_D) = \Delta(t)\ \mathrm{vol}(T \backslash D_v^\times)\ \tilde{f}_D(z), \qquad T = E^\times.$$

Proof. Let $U_k = 1 + \mathfrak{P}_D^k$, where $\mathfrak{P}_D$ is the maximal ideal of the maximal order of D_v. These are normal subgroups of $D_v^\times$ and form a basis of neighborhoods of 1. The cover splits over $U_{k'}$ for k' large, by restriction from $\overline{\mathcal{G}}_v$ [FK, §2]; fix such k' and a splitting κ over $U_{k'}$. For $k \geq k'$ large enough, every $t \in U_k$ is the n-th power of an element $s \in U_{k'}$ that is a limit of polynomials in t. If $p \nmid n$, take s in the pro-p group $\overline{\langle t \rangle}$, on which n-th powering is bijective. If $p \mid n$, take $s = \sum_i \binom{1/n}{i}(t-1)^i$, which converges for k large. Put $U = U_k$ and $\kappa_U = \kappa|_U$.

(a) For $x \in D_v^\times$, $\mathrm{Ad}(\tilde{x}) \circ \kappa \circ \mathrm{Ad}(x)^{-1}$ is another splitting over $U_{k'}$, so it differs from κ by a character $U_{k'} \to \mu_n$. Such a character is trivial on n-th powers, hence on U. If $t \in U \cap E^\times$ is regular, then $s \in F[t] \subseteq E$ (as E is closed), so $t \in E^{\times n}$ and $\kappa_U(t)$ is good (Proposition 5.5(a)).

(b) Since $\tilde{f}_D$ is smooth with compact support modulo $\overline{Z}_v$, after enlarging k we have $\tilde{f}_D(z\,\kappa_U(u)) = \tilde{f}_D(z)$ for all $z \in \overline{Z}_v$ and $u \in U$. Then, by (a) and centrality of z, $\tilde{f}_D(\tilde{x}^{-1} z \kappa_U(t) \tilde{x}) = \tilde{f}_D(z\,\kappa_U(x^{-1}tx)) = \tilde{f}_D(z)$ for all x. Integrate over the compact quotient $T\backslash D_v^\times$. □

Remark 6.7. Proposition 6.6 shows that $F(\cdot, \tilde{f}_D)$ is *not* compactly supported in the regular set unless $\tilde{f}_D$ vanishes on $\overline{Z}_v$: near each $z \in \overline{Z}_v$ it is a constant times $\Delta(t)$, which tends to 0 but is nonzero. A matching $\tilde{f}$ on $\overline{G}_v$ must then have orbital integrals with this behavior near z on every elliptic torus, and vanishing orbital integrals on all non-elliptic tori. Such an $\tilde{f}$ cannot be supported in the regular set. Producing it requires the germ expansion of [FK, §8] at the central good element z, the restriction of the germs to the elliptic set, and a function realizing exactly the "identity" germ on the elliptic set (a genuine function with the orbital integrals of a Steinberg pseudo-coefficient near the center; Part II, Remark 5.6, explains why it is not a pseudo-coefficient of a single genuine representation). This is item (D1) of §10.

6.6 Orbital integrals on an elliptic torus

This subsection proves, directly and for an arbitrary elliptic maximal torus (cyclic or not), the facts about orbital integrals of regular-supported functions used in §6: their properties (Lemma 6.10), their realization (Lemma 6.11), and the fact that traces of regular-supported functions depend only on orbital integrals (Lemma 6.12). It makes the local matching independent of [JC, Lemma 4.5], which is stated there only for cyclic extensions. The only input is Harish-Chandra's submersion principle and a properness statement (Lemma 6.8). The subsection introduces the maps Ψ and $\tilde{\Psi}$ and the open set $\overline{G}_T$, which are used only here.

Setting. F is a p-adic field containing μ_n, and G is $D_F^\times$ or $\mathrm{GL}(r, F)$, with D_F a central division algebra of degree r. The group $\overline{G}$ is an n-fold cover in which $\overline{Z} = p^{-1}(Z)$, $Z = F^\times$, is central, and $\tilde{\theta}$ is a genuine character of $\overline{Z}$. $T = E^\times \subset G$ is an elliptic maximal torus, with E/F separable of degree r, and $W = N_G(T)/T$ (a finite group). T^{reg} is the set of regular elements of T (here $T^{\mathrm{reg}} = E^\times \smallsetminus F^\times$ when r is prime), and $\overline{T}^{g,\mathrm{reg}}$ is the set of good regular elements of $\overline{T}$. Measures dg, dt on G/Z, T/Z and $d\dot{x}$ on $T\backslash G$ are as in §6.1. Note

that $Z(\overline{T})$ is an open and closed subgroup of $\overline{T}$, since its image $F^\times E^{\times n}$ (Proposition 5.5) contains the open subgroup $E^{\times n}$.

Lemma 6.8. ***[E]***

(a) The map $\Psi : T\backslash G \times T^{\mathrm{reg}} \to G$, $(\dot{x}, t) \mapsto x^{-1}tx$, is a local analytic isomorphism onto the open set G_T of elements conjugate into T^{reg}. W acts freely on the source by $w \cdot (\dot{x}, t) = (T\tilde{w}x, wtw^{-1})$, and the fibers of Ψ are exactly the W-orbits.

(b) If $C \subset T^{\mathrm{reg}}$ is compact modulo Z and $\Omega \subset G$ is compact modulo Z, then $\{\dot{x} \in T\backslash G : x^{-1}Cx \cap \Omega \neq \emptyset\}$ is compact.

Proof. (a) For regular t, the differential of Ψ is surjective, because $\mathrm{Ad}(t^{-1}) - 1$ is invertible on a complement of $\operatorname{Lie} T$. Dimensions agree, so the inverse function theorem for analytic maps over F applies (Harish-Chandra's submersion principle [HC70], [Kot, §§7, 13]). If $x^{-1}tx = y^{-1}t'y$, then $w := yx^{-1}$ conjugates t to t'; since t is regular, w normalizes T, which gives the fibers. Freeness holds because $w \in T$ if $Twx = Tx$. (b) This is [HC70, Part I], [Kot, §13]. On the D side it is trivial, since $T\backslash G$ is compact. □

Lemma 6.9. ***[P]****The map $\tilde{\Psi} : T\backslash G \times \overline{T}^{g,\mathrm{reg}} \to \overline{G}$, $(\dot{x}, \tilde{t}) \mapsto \tilde{x}^{-1}\tilde{t}\tilde{x}$, is well defined. It commutes with multiplication by $\overline{Z}$ (in particular by μ_n) on the second factor, and it is a covering map of degree $|W|$ onto the open set $\overline{G}_T := p^{-1}(G_T) \cap \{\text{good elements}\}$. Its fibers are the W-orbits for the action $w \cdot (\dot{x}, \tilde{t}) = (T\tilde{w}x, \tilde{w}\tilde{t}\tilde{w}^{-1})$.*

Proof. Conjugation by $\tilde{x}$ does not depend on the lift. Replacing x by sx with $s \in T$ does not change $\tilde{x}^{-1}\tilde{t}\tilde{x}$, because $\tilde{t}$ is good. Similarly $\tilde{w}\tilde{t}\tilde{w}^{-1}$ does not depend on the lift $\tilde{w}$ of $w \in N_G(T)$ or on the representative of w modulo T. The fiber statement follows from Lemma 6.8(a) by lifting: if $\tilde{x}^{-1}\tilde{t}\tilde{x} = \tilde{y}^{-1}\tilde{t}'\tilde{y}$, then $w = yx^{-1}$ normalizes T and $\tilde{t}' = \tilde{w}\tilde{t}\tilde{w}^{-1}$. The induced map from the quotient by W is therefore injective. It is a local homeomorphism, because Ψ is and because $p : \overline{G} \to G$ and $p : \overline{T} \to T$ are finite coverings that are trivial over small open sets (the cover splits over small open compact subgroups). So it is a homeomorphism onto its open image. Since W is finite and acts freely, $\tilde{\Psi}$ is a covering of degree $|W|$ onto its image. The image consists of good elements, because conjugates of good elements are good, and it is all of $\overline{G}_T$, because every good regular element conjugate into T is conjugate to an element of $\overline{T}^{g,\mathrm{reg}}$. □

Lemma 6.10. ***[P]****Let $\tilde{f} \in C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen}}$ be supported in the regular set. Then $h(\tilde{t}) := F(\tilde{t}, \tilde{f})$, for $\tilde{t} \in \overline{T}^{g,\mathrm{reg}}$, is admissible in the sense of §6.3: (i) locally constant; (ii) $h(z\tilde{t}) = \tilde{\theta}(z)^{-1}h(\tilde{t})$; (iii) with support compact modulo $\overline{Z}$ inside $\overline{T}^{g,\mathrm{reg}}$; (iv) invariant under $N_G(T)$.*

Proof. (ii) holds because $\tilde{f}$ transforms by $\tilde{\theta}^{-1}$ and $\Delta(zt) = \Delta(t)$.

(iv) For $m \in N_G(T)$, substitute $x \mapsto mx$. The map $Tx \mapsto Tmx$ is a well-defined, measure-preserving bijection of $T\backslash G$, since $\mathrm{Ad}(m)$ preserves dt and W is finite. Also $\Delta(mtm^{-1}) = \Delta(t)$.

(iii) The function $\delta(g) := |D(g)|_F \, |\det g|_F^{-(r-1)}$ is continuous, conjugation invariant and Z-invariant, and it vanishes exactly at the non-regular elements. Since $p(\operatorname{supp} \tilde{f})$ is

compact modulo Z and regular, $\delta \geq c > 0$ on it. If $h(\tilde{t}) \neq 0$, some conjugate of t lies in $p(\operatorname{supp} \tilde{f})$, so $\delta(t) \geq c$. The set $B = \{t \in T : \delta(t) \geq c\}$ is closed, Z-stable and regular, and T/Z is compact, so B is compact modulo Z. Hence $\operatorname{supp} h \subseteq p^{-1}(B) \cap Z(\overline{T})$, which is compact modulo $\overline{Z}$ and contained in $\overline{T}^{g,\mathrm{reg}}$.

(i) Fix $\tilde{t}_0 \in \overline{T}^{g,\mathrm{reg}}$. Let U_T be an open compact subgroup of T with $t_0 U_T \subset T^{\mathrm{reg}}$, over which the cover splits by some κ. Every element of $\overline{T}$ near $\tilde{t}_0$ is of the form $\tilde{t}_0\kappa(u)$ with $u \in U_T$ (these are good, $Z(\overline{T})$ being open). By Lemma 6.8(b) applied to $C = Zt_0U_T$ and $\Omega = p(\operatorname{supp} \tilde{f})$, the set X of $\dot{x}$ with $x^{-1}Cx \cap \Omega \neq \emptyset$ is compact. The function $(\dot{x}, u) \mapsto \tilde{f}(\tilde{\Psi}(\dot{x}, \tilde{t}_0\kappa(u)))$ is locally constant on $X \times U_T$ and vanishes for $\dot{x} \notin X$. By compactness of X there is an open subgroup $U' \subset U_T$ such that its value at $(\dot{x}, u)$ equals its value at $(\dot{x}, 1)$ for all $\dot{x}$ and all $u \in U'$. Integrating over $T\backslash G$ gives $\Phi(\tilde{t}_0\kappa(u), \tilde{f}) = \Phi(\tilde{t}_0, \tilde{f})$ for $u \in U'$, and Δ is locally constant on T^{reg}. □

Lemma 6.11. ***[P]****Let h be admissible on $\overline{T}^{g,\mathrm{reg}}$. Choose $\beta \in C_c^\infty(T\backslash G)$ with $\int \beta\, d\dot{x} = 1$, and define $\tilde{f}$ on $\overline{G}$ by*

$$\tilde{f}(\tilde{y}) := \frac{1}{|W|} \sum_{(\dot{x},\tilde{t}) \in \tilde{\Psi}^{-1}(\tilde{y})} \beta(\dot{x})\, h(\tilde{t})\, \Delta(t)^{-1} \quad (\tilde{y} \in \overline{G}_T), \qquad \tilde{f}(\tilde{y}) := 0 \quad (\tilde{y} \notin \overline{G}_T).$$

Then $\tilde{f} \in C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen}}$, its support is contained in $\tilde{\Psi}(\operatorname{supp} \beta \times \operatorname{supp} h)$ (good regular elements conjugate into $\overline{T}$), $F(\cdot, \tilde{f}) = h$ on $\overline{T}^{g,\mathrm{reg}}$, and $F(\tilde{\delta}, \tilde{f}) = 0$ for every regular $\tilde{\delta}$ not conjugate into $\overline{T}$.

Proof. Genuine, with central character $\tilde{\theta}^{-1}$. We have $\tilde{\Psi}^{-1}(z\tilde{y}) = \{(\dot{x}, z\tilde{t}) : (\dot{x}, \tilde{t}) \in \tilde{\Psi}^{-1}(\tilde{y})\}$, $\Delta(zt) = \Delta(t)$, and (ii).

Smooth, with compact support modulo $\overline{Z}$. The set $K_f := \tilde{\Psi}(\operatorname{supp} \beta \times \operatorname{supp} h)$ is compact modulo $\overline{Z}$ (by (iii) and continuity), hence closed, and it is contained in the open set $\overline{G}_T$. On $\overline{G}_T$, $\tilde{f}$ is locally constant, because near any point $\tilde{\Psi}$ has $|W|$ continuous local inverses (Lemma 6.9) and β, h, Δ are locally constant. Outside the closed set K_f, $\tilde{f}$ vanishes. The two open sets $\overline{G}_T$ and $\overline{G} \smallsetminus K_f$ cover $\overline{G}$, since $K_f \subseteq \overline{G}_T$, and $\tilde{f}$ is locally constant on each. So $\tilde{f}$ is locally constant on $\overline{G}$; there is no boundary issue, because the support K_f stays inside the open set $\overline{G}_T$ and away from its boundary.

Orbital integrals. For $\tilde{t}_0 \in \overline{T}^{g,\mathrm{reg}}$ and $x \in G$, the fiber of $\tilde{\Psi}$ over $\tilde{x}^{-1}\tilde{t}_0\tilde{x}$ is $\{(Twx, \tilde{w}\tilde{t}_0\tilde{w}^{-1}) : w \in W\}$. By (iv) and W-invariance of Δ,

$$\tilde{f}(\tilde{x}^{-1}\tilde{t}_0\tilde{x}) = \frac{h(\tilde{t}_0)}{|W|\Delta(t_0)} \sum_{w \in W} \beta(Twx).$$

Integrating over $T\backslash G$, and using that $Tx \mapsto Twx$ preserves $d\dot{x}$, gives $\Phi(\tilde{t}_0, \tilde{f}) = h(\tilde{t}_0)\Delta(t_0)^{-1}$, i.e. $F(\tilde{t}_0, \tilde{f}) = h(\tilde{t}_0)$. If $\tilde{\delta}$ is regular and not conjugate into $\overline{T}$, its conjugacy class does not meet $\overline{G}_T \supseteq \operatorname{supp} \tilde{f}$, so $\Phi(\tilde{\delta}, \tilde{f}) = 0$. □

Lemma 6.12. ***[P]*** *(using **[E]**) Let $\overline{\pi}$ be an irreducible admissible genuine representation of $\overline{G}$ with central character $\tilde{\theta}$, and let $\tilde{f}_1, \tilde{f}_2 \in C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen}}$ be supported in the regular set. If $F(\tilde{\gamma}, \tilde{f}_1) = F(\tilde{\gamma}, \tilde{f}_2)$ for every good regular $\tilde{\gamma}$, then $\operatorname{tr} \overline{\pi}(\tilde{f}_1) = \operatorname{tr} \overline{\pi}(\tilde{f}_2)$.*

Proof. The character $\Theta_{\overline{\pi}}$ is a locally constant function on the regular set with $\operatorname{tr}\overline{\pi}(\tilde{f}) = \int_{\overline{G}/\overline{Z}} \tilde{f}\,\Theta_{\overline{\pi}}$ for regular-supported $\tilde{f}$ ([FK, §13], after Harish-Chandra) [**E**]. The product $\tilde{f}\Theta_{\overline{\pi}}$ is invariant under $\overline{Z}$ (hence under μ_n), so it is a function on G/Z. The Weyl integration formula on G/Z [HC70] [**E**] gives

$$\operatorname{tr}\overline{\pi}(\tilde{f}) = \sum_T \frac{1}{|W_T|} \int_{T/Z} \Delta(t)^2\,\Theta_{\overline{\pi}}(\tilde{t})\,\Phi(\tilde{t},\tilde{f})\,dt,$$

summed over representatives T of the (finitely many) conjugacy classes of maximal tori, with $W_T = N_G(T)/T$. The measures are dg on G/Z, a Haar measure dt on T/Z for each T, and $d\dot{x} = dg/dt$ on $T\backslash G$ in Φ; the formula holds with these choices for any dt, since rescaling dt rescales $d\dot{x}$ inversely [HC70], [Kot, §7]. Here $\Delta(t)^2 = |D(t)|_F |\det t|_F^{-(r-1)}$ is the Weyl discriminant $|\det(1-\mathrm{Ad}(t^{-1}))|_{\mathfrak{g}/\mathfrak{t}}|_F$. We used $\Theta_{\overline{\pi}}(\tilde{x}^{-1}\tilde{t}\tilde{x}) = \Theta_{\overline{\pi}}(\tilde{t})$. At a non-good regular $\tilde{t}$ the integrand vanishes, because $\Theta_{\overline{\pi}}(\tilde{t}) = 0$ (conjugating by $\overline{T}$ multiplies $\tilde{t}$ by some $\zeta \neq 1$). So the trace depends only on the values of $F(\cdot,\tilde{f}) = \Delta\Phi$ at good regular elements. □

7 Global covers and rational lifts

This section constructs the global GL-side cover inside the same ambient adelic cover as the D-side cover of [JC] (§7.1). It then proves that, under the local identifications of §§4–6, the rational lift of an elliptic element on the D side corresponds to the rational lift of the corresponding element on the GL side, so that the elliptic terms of the two trace formulas agree term by term (§7.2). The measure conventions are fixed in §7.3, and §7.4 supplies the sup-norm bound for cusp forms on the cover that the trace formula of §8 uses. The section depends on §§4–6 and on the global normalization of the Kazhdan–Patterson cocycle [FK, §2], [KP1, §0.1]. It introduces the global covers $\overline{\mathcal{G}}(\mathbb{A}_L)$, $\overline{G}_D(\mathbb{A}_K)$, $\overline{G}(\mathbb{A}_K)$ and $\overline{Z}(\mathbb{A}_K)$, the splittings ι and κ_v^*, the global central character $\tilde{\theta}$ (unitary from here on), and the finite set of places S_D.

7.1 The global covers

We define the global ambient cover and the two global covers as subgroups, and list the properties needed for the trace formulas. The splitting conventions of Lemma 7.1 and the unitarity of $\tilde{\theta}$ are used in all later sections.

Let $\overline{\mathcal{G}}(\mathbb{A}_L)$ be the cover of $\mathrm{GL}(r,\mathbb{A}_L)$ with cocycle $\tau_L \cdot (\det,\det)_{\mathbb{A}_L}^{c_v}$, where $\tau_L = \prod_w \tau_{L,w}$ is the normalized Kazhdan–Patterson cocycle of [FK, §2] (parameter 0) and $(a,b)_{\mathbb{A}_L} = \prod_w (a_w,b_w)_{L_w}$; here $c_v = r'c$ is independent of v. It is the restricted product of the local covers $\overline{\mathcal{G}}_v$ with respect to the splittings $\kappa_{\mathcal{G}_v}^*$ over $\mathrm{GL}(r,\mathcal{O}_{L_v})$ for $v \nmid n\infty$, modulo $\{(\zeta_v) : \prod \zeta_v = 1\}$. Let $\iota : \mathrm{GL}(r,L) \to \overline{\mathcal{G}}(\mathbb{A}_L)$ be the rational splitting described in Lemma 7.1.

We first make the section and splitting conventions explicit, since the normalized and unnormalized cocycle models describe the same splittings by different formulas.

Lemma 7.1 (Splitting conventions). ***[P]*** *(using* ***[E]****) For a place w of L let $\sigma_w = \sigma_{L_w}$ be the Kazhdan–Patterson cocycle (parameter 0) of $\mathrm{GL}(r, L_w)$, and $s_\sigma(g) = (g,1)_\sigma$ the corresponding section. For $w \nmid n\infty$ let $\kappa_w : \mathrm{GL}(r, L_w) \to \mu_n$ be the function of [FK, §2]: on $K_w = \mathrm{GL}(r, \mathcal{O}_{L_w})$ it satisfies*

$$\sigma_w(k,k') = \kappa_w(kk')\,\kappa_w(k)^{-1}\kappa_w(k')^{-1} \qquad (k,k' \in K_w), \tag{6}$$

and it is extended to $\mathrm{GL}(r, L_w)$ as in [FK, §2] (with $\kappa_w = 1$ for $w \mid n\infty$). Put

$$\tau_w(g,h) = \sigma_w(g,h)\,\kappa_w(g)\,\kappa_w(h)\,\kappa_w(gh)^{-1}, \tag{7}$$

let $s_\tau(g) = (g,1)_\tau$ be the section of the τ-model, and let $\phi_w : (g,\zeta)_\sigma \mapsto (g, \zeta\kappa_w(g)^{-1})_\tau$. Then:

(a) ϕ_w is an isomorphism of extensions over the identity, trivial on μ_n.

(b) Compact splitting. *For $w \nmid n\infty$, $\tau_w \equiv 1$ on $K_w \times K_w$, so $\kappa_w^* := s_\tau|_{K_w}$, i.e. $\kappa_w^*(k) = (k,1)_\tau$, is a homomorphism. In the σ-model it is $k \mapsto (k, \kappa_w(k))_\sigma$, i.e. $\phi_w^{-1} \circ \kappa_w^* = s_\sigma \cdot \kappa_w$.*

(c) Rational splitting. *For $x \in \mathrm{GL}(r,L)$, $\kappa_w(x) = 1$ for almost all w [KP1, Prop. 0.1.3], so $\kappa_L(x) := \prod_w \kappa_w(x)$ is defined. The map*

$$\iota(x) = s_\tau(x)\,\kappa_L(x)^{-1} = \Big((x, \kappa_w(x)^{-1})_\tau\Big)_w$$

is a homomorphism $\mathrm{GL}(r,L) \to \overline{\mathcal{G}}(\mathbb{A}_L)$ splitting p. Componentwise, $\iota(x)_w = \phi_w(s_\sigma(x))$, i.e. in the σ-models the rational splitting is $x \mapsto (x,1)_\sigma$ at every place.

(d) For $x \in \mathrm{GL}(r,L)$, $\iota(x)_w = \kappa_w^*(x)$ if and only if $x \in K_w$ and $\kappa_w(x) = 1$, which holds for almost all w.

(e) The twisting factor $(\det,\det)_{\mathbb{A}_L}^{c_v}$ does not affect (a)–(d): it is a common factor of both models, it is trivial on $\mathrm{GL}(r,L)\times\mathrm{GL}(r,L)$ by Hilbert reciprocity, and it is trivial on $K_w \times K_w$ for $w \nmid n$.

Proof. (a) Using (7),

$$\begin{aligned}\phi_w(g,\zeta)\,\phi_w(h,\eta) &= (gh,\ \zeta\eta\,\kappa_w(g)^{-1}\kappa_w(h)^{-1}\tau_w(g,h)) \\ &= (gh,\ \zeta\eta\,\sigma_w(g,h)\,\kappa_w(gh)^{-1}) = \phi_w\Big((g,\zeta)(h,\eta)\Big).\end{aligned}$$

(b) By (6) and (7), $\tau_w(k,k') = 1$, so $s_\tau(k)s_\tau(k') = \tau_w(k,k')s_\tau(kk') = s_\tau(kk')$. Also $\phi_w((k,\kappa_w(k))_\sigma) = (k,1)_\tau$. (c) The global τ-cocycle is $\tau = \prod_w \tau_w$, a finite product for each pair [FK, §2]. For $x, x' \in \mathrm{GL}(r,L)$, the values $\sigma_w(x,x')$ are products of Hilbert symbols of elements of $L^\times$ [KP1, §0.1], [BLS], so they equal 1 for almost all w and $\prod_w \sigma_w(x,x') = 1$ by Hilbert reciprocity. Hence

$$\iota(x)\iota(x') = \tau(x,x')\kappa_L(x)^{-1}\kappa_L(x')^{-1}s_\tau(xx') = \Big(\prod_w \sigma_w(x,x')\Big)\kappa_L(xx')^{-1}s_\tau(xx') = \iota(xx').$$

This is the statement of [FK, §2] that $x \mapsto s(x)/\kappa(x)$ is a homomorphism on $G(F)$, with $s = s_\tau$. The componentwise formula is $\phi_w((x,1)_\sigma) = (x, \kappa_w(x)^{-1})_\tau$. (d) Compare (b) and (c). (e) is clear. □

Remark 7.2 (Conventions)**.** Throughout, ι and κ_v^* are the splittings of Lemma 7.1: $\iota(x) = s(x)/\kappa_L(x)$ and $\kappa_w^*(k) = s(k)$ in the normalized model. These are the conventions of [JC, Lemma 6.1(a),(d)].

Remark 7.3 (No circularity with [JC])**.** [JC] cites Proposition 4.3 of the present paper for the local identification of covers. That proposition is purely local. Its proof uses only the explicit restriction of the Kazhdan–Patterson cocycle to the diagonal torus [FK, §2], the Matsumoto–Moore classification [Mat], [Moo], and the commutator formula [Mil]. It uses no result of [JC]. Conversely, the present paper uses from [JC] only the definition of the global cover of $D_{\mathbb{A}}^{\times}$ and the local lemmas cited by number (all from [JC, §§2–4]). It uses none of [JC]'s global theorems.

Unitarity. From now on $\tilde{\theta} : \overline{Z}(\mathbb{A}_K) \to \mathbb{C}^{\times}$ is a genuine *unitary* character trivial on $\iota(K^{\times})$, and its local components $\tilde{\theta}_v$ are unitary. This is needed for the L^2 spaces, adjoints, formal degrees and Schur orthogonality used in §§8–9.

Definition 7.4. $\overline{G}_D(\mathbb{A}_K) := p^{-1}(\iota_D(G_D(\mathbb{A}_K)))$ and $\overline{G}(\mathbb{A}_K) := p^{-1}(\iota_{\mathrm{GL}}(\mathrm{GL}(r, \mathbb{A}_K)))$, subgroups of $\overline{\mathcal{G}}(\mathbb{A}_L)$, and $\overline{Z}(\mathbb{A}_K) := p^{-1}(\mathbb{A}_K^{\times})$.

Proposition 7.5. ***[P]***

(a) The cocycle of $\overline{\mathcal{G}}(\mathbb{A}_L)$ restricts on $G_D(\mathbb{A}_K)$ to $\beta_\tau := \tau_L \cdot (\mathrm{Nrd}, \mathrm{Nrd})^c$. This is the cocycle β of [JC, equation (4)] when σ_L there is read as the normalized cocycle τ_L, which is the reading under which [JC]'s global cover is defined and split over $\mathrm{GL}(r, L)$. The rational splitting $\iota|_{G_D(K)}$ is that of [JC, Lemma 6.1(a),(d)], and the compact splittings κ_v^* are those of [JC, Lemma 6.1(d)] (Remark 7.2).

(b) ι splits $\overline{G}(\mathbb{A}_K)$ over $\mathrm{GL}(r, K)$, and $\overline{G}_D(\mathbb{A}_K)$ over $G_D(K)$.

(c) $\overline{Z}(\mathbb{A}_K)$ is a common subgroup of $\overline{G}_D(\mathbb{A}_K)$ and $\overline{G}(\mathbb{A}_K)$, central in both, and $\iota(K^{\times})$ is a common subgroup. Hence a genuine unitary character $\tilde{\theta}$ of $\overline{Z}(\mathbb{A}_K)$ trivial on $\iota(K^{\times})$ serves both sides.

(d) For almost all v, $\kappa_{\mathcal{G}_v}^*$ restricts to splittings κ_v^* of $\overline{G}_D(K_v)$ over $G_D(\mathcal{O}_v)$ and of $\overline{G}_v$ over $\mathrm{GL}(r, \mathcal{O}_v)$; these agree with [JC, Lemma 6.1(d)] on the D side and are carried to each other by ψ_v (Proposition 4.6(c)). For γ rational and integral outside a finite set, $\iota(\gamma)_v = \kappa_v^*(\gamma)$ for almost all v. The covers are the restricted products of their local components with respect to these splittings.

(e) Let N be the unipotent radical of a parabolic subgroup of $\mathrm{GL}(r)$ defined over K. There is a unique splitting s_N of $\overline{G}(\mathbb{A}_K)$ over $N(\mathbb{A}_K)$, and $s_N|_{N(K)} = \iota|_{N(K)}$.

(f) The local components of $\overline{G}(\mathbb{A}_K)$ are the covers $\overline{G}_v$ of Definition 4.1, isomorphic to $\overline{G}^{(c_v)}$ after relabeling (Proposition 4.3(b)).

Proof. (a) As in Proposition 4.3(a), $(\det, \det)_{\mathbb{A}_L}^{c_v}$ restricts on $G_D(\mathbb{A}_K)$ to $\prod_v (\mathrm{Nrd}, \mathrm{Nrd})_{K_v}^{rc_v} = (\mathrm{Nrd}, \mathrm{Nrd})^c$, so the restricted cocycle is β_τ. In [JC], σ_L is described as the Kazhdan–Patterson cocycle of $\mathrm{GL}(r, \mathbb{A}_L)$ "which splits over $\mathrm{GL}(r, L)$", with rational splitting $x \mapsto$

$s(x)/\kappa_L(x)$. A global product of local cocycles is defined only after the normalization (7) [FK, §2]. So σ_L in [JC] is the normalized cocycle τ_L, and its rational splitting is ι of Lemma 7.1(c). The compact splittings are those of Lemma 7.1(b) (Remark 7.2). Central characters are defined on our $\overline{Z}(\mathbb{A}_K)$ and are not transported.

(b) τ_L is split over $\mathrm{GL}(r,L)$ by ι (Lemma 7.1(c)), and $(\det g, \det h)_{\mathbb{A}_L} = 1$ for $g, h \in \mathrm{GL}(r,L)$ by Hilbert reciprocity for L. So ι splits $\overline{\mathcal{G}}(\mathbb{A}_L)$ over $\mathrm{GL}(r,L)$, and hence over its subgroups $\iota_{\mathrm{GL}}(\mathrm{GL}(r,K))$ and $\iota_D(G_D(K))$.

(c) ι_D and ι_{GL} agree on $\mathbb{A}_K^\times$, and on $K^\times$ the splitting is ι in both cases. Centrality is Lemma 4.2 at each place.

(d) For almost all v (namely $v \nmid n\infty$, v unramified in L, and the integrality conditions of Proposition 4.6(c)), $\iota_D(G_D(\mathcal{O}_v))$ and $\iota_{\mathrm{GL}}(\mathrm{GL}(r,\mathcal{O}_v))$ lie in $\mathrm{GL}(r,\mathcal{O}_{L_v})$. There $\kappa^*_{\mathcal{G}_v} := \prod_{w|v} \kappa^*_w$, with $\kappa^*_w(k) = (k,1)_\tau$, is a homomorphism by Lemma 7.1(b),(e). This agrees with [JC, Lemma 6.1(d)] (Remark 7.2). Compatibility with ψ_v is Proposition 4.6(c). The statement about $\iota(\gamma)$ is Lemma 7.1(d) in $\overline{\mathcal{G}}(\mathbb{A}_L)$, hence in the subgroups.

(e) Locally, the section s of [FK, §2] splits the cover over $N(K_v)$. A splitting is unique, because $N(K_v)$ is divisible (it is unipotent in characteristic 0), so it has no nontrivial homomorphisms to μ_n. At almost all v the local splitting agrees with κ^*_v on $N(\mathcal{O}_v)$, again by uniqueness: $N(\mathcal{O}_v)$ is pro-p with $p \nmid n$. Hence the local splittings define a splitting of the restricted product over $N(\mathbb{A}_K)$. It is unique, since $N(\mathbb{A}_K)$ is divisible (for X in the Lie algebra with integral components almost everywhere, X/n is again such). The restriction to $N(K)$ is a splitting of $\overline{G}(\mathbb{A}_K)$ over the divisible group $N(K)$, as is $\iota|_{N(K)}$, so they agree.

(f) is the definition of the local components together with Proposition 4.3. □

7.2 Rational lifts correspond

We prove that the local identifications of §§4–6 carry the rational lift of a rational elliptic element of $G_D(K)$ to the rational lift of the corresponding element of $\mathrm{GL}(r,K)$, so that the elliptic terms of the two trace formulas agree.

For $\gamma \in G_D(K)$ and genuine factorizable $\tilde{f}_D = \otimes_v \tilde{f}_{D,v}$, write $F(\iota(\gamma), \tilde{f}_D) := \prod_v F(x_v, \tilde{f}_{D,v})$, where (x_v) is any representative of $\iota(\gamma)$ in the restricted product. This is well defined because the product of the local phases is 1 [JC, Lemma 6.1, Remark (3)]. Similarly on the GL side.

Theorem 7.6. ***[P]****Let $E \subset D$ be a maximal subfield, $j : E \hookrightarrow M_r(K)$ a K-embedding, and $\gamma \in E^\times$ regular and good at every place on the D side. Fix α_v and ψ_v for $v \notin S$ as in Proposition 4.6. Let $\tilde{f}_D = \otimes \tilde{f}_{D,v}$ and $\tilde{f} = \otimes \tilde{f}_v$ be genuine and factorizable with $\tilde{f}_{D,v} = \tilde{f}_v \circ \psi_v$ for $v \notin S$ (with $dg_{D,v} = \psi_v^* dg_v$) and $\tilde{f}_{D,v}$, $\tilde{f}_v$ matching for $v \in S$. Then $j(\gamma)$ is good at every place on the* GL *side, and*

$$F(\iota(\gamma), \tilde{f}_D) = F(\iota(j(\gamma)), \tilde{f}).$$

Proof. A rational conjugating element. As in Theorem 5.9(a), L^r is a free module of rank one over the étale L-algebra $E \otimes_K L$ via both ι_D and $\iota_{\mathrm{GL}} \circ j$, so there is $g \in \mathrm{GL}(r,L)$ with $\mathrm{Ad}(g) \circ \iota_D = \iota_{\mathrm{GL}} \circ j$ on E. Since ι is a homomorphism on $\mathrm{GL}(r,L)$, $\mathrm{Ad}(\iota(g))$ maps $\iota(\iota_D \gamma)$

to $\iota(\iota_{\mathrm{GL}} j\gamma)$. Conjugation in the restricted product is componentwise, so if (x_v) represents $\iota(\gamma)$ on the D side, then $(\mathrm{Ad}(\tilde{g}_v)x_v)$ represents $\iota(j\gamma)$ on the GL side, where $\tilde{g}_v$ are any lifts of the components of g. Hence

$$F(\iota(j\gamma), \tilde{f}) = \prod_v F(\mathrm{Ad}(\tilde{g}_v)x_v, \tilde{f}_v).$$

Places $v \in S$. Here $E_v = E \otimes_K K_v$ is a maximal subfield of D_v, and g conjugates $\iota_D|_{E_v}$ to $\iota_{\mathrm{GL}} \circ j$. So $\mathrm{Ad}(\tilde{g}_v)|_{\overline{E}_{D,v}} \in \mathrm{Iso}_{\overline{Z}}$, and by Theorem 5.9(b) it equals $\jmath_{E_v,j}$ on the good element x_v. Hence $\mathrm{Ad}(\tilde{g}_v)x_v$ is good, and by (M1) $F(\mathrm{Ad}(\tilde{g}_v)x_v, \tilde{f}_v) = (-1)^{r-1}F(x_v, \tilde{f}_{D,v})$.

Places $v \notin S$. The two embeddings $\alpha_v|_{E_v}$ and j of the étale K_v-algebra E_v into $M_r(K_v)$ are conjugate by some $h_v \in \mathrm{GL}(r, K_v)$ (again by the free-module argument). Then $\mathrm{Ad}(\tilde{h}_v) \circ \psi_v$ and $\mathrm{Ad}(\tilde{g}_v)$, restricted to $\overline{E}_{D,v} = p^{-1}(\iota_D(E_v^\times))$, are isomorphisms onto $p^{-1}(\iota_{\mathrm{GL}} j(E_v^\times))$ lying over j. Both are the identity on μ_n and on the central $\overline{Z}_v$. As in Theorem 5.9(b), they differ by a character of $E_v^\times$ with values in μ_n, trivial on $K_v^\times$. Such a character is trivial on $K_v^\times E_v^{\times n}$, which contains the good regular elements of $\overline{E}_{D,v}$ [FK, §3, Proposition] **[E]**. So they agree at x_v. Hence $\mathrm{Ad}(\tilde{g}_v)x_v$ is good and

$$F(\mathrm{Ad}(\tilde{g}_v)x_v, \tilde{f}_v) = F(\mathrm{Ad}(\tilde{h}_v)\psi_v(x_v), \tilde{f}_v) = F(\psi_v(x_v), \tilde{f}_v) = F(x_v, \tilde{f}_{D,v}),$$

using conjugation invariance and the fact that ψ_v is an isomorphism of groups carrying $dg_{D,v}$ to dg_v and torus measures to torus measures.

Conclusion. Multiplying over v gives $F(\iota(j\gamma), \tilde{f}) = (-1)^{(r-1)|S|}F(\iota(\gamma), \tilde{f}_D)$, and $(-1)^{(r-1)|S|} = 1$, since $r-1$ is even for odd r and $|S|$ is even for $r = 2$. □

Corollary 7.7. ***[P]****The map $\gamma \mapsto j(\gamma)$ induces a bijection between $G_D(K)$-conjugacy classes (modulo $Z(K)$) of regular γ that are good at every place, and $\mathrm{GL}(r, K)$-conjugacy classes (modulo $Z(K)$) of regular elliptic γ' that are good at every place and whose field $K(\gamma')$ embeds in D. For γ' regular elliptic whose field does not embed in D, $F(\iota(\gamma'), \tilde{f}) = 0$ whenever $\tilde{f}_v$ satisfies (M2) at all $v \in S$.*

Proof. Rational regular classes are determined by characteristic polynomials (Skolem–Noether on both sides), and a degree-r field embeds in D if and only if it is a field locally at every $v \in S$ (Albert–Brauer–Hasse–Noether). Goodness corresponds place by place by Theorem 7.6. If $K(\gamma')$ does not embed in D, then γ' is non-elliptic at some $v \in S$, and the factor at v vanishes by (M2) (or because γ' is not good there). □

7.3 Measures

We fix the global measures and explain why no further compatibility between the measures on the two groups is needed. The finite set S_D of places defined here is used in §§8–9.

Fix a finite set $S_D \supseteq S$ of places containing the archimedean places, the places above n, the places ramified in L or where $\tilde{\theta}$ is ramified, and the places where the integrality conditions of Proposition 4.6(c) fail. For $v \notin S_D$ normalize dg_v on $\mathrm{GL}(r, K_v)/K_v^\times$ by $\mathrm{vol}(\mathrm{GL}(r, \mathcal{O}_v)K_v^\times/K_v^\times) = 1$, and put $dg_{D,v} := \psi_v^* dg_v$; it gives $G_D(\mathcal{O}_v)K_v^\times/K_v^\times$ volume 1 by Proposition 4.6(c). For $v \in S_D \smallsetminus S$ take any dg_v and $dg_{D,v} := \psi_v^* dg_v$. For $v \in S$ take

any dg_v and $dg_{D,v}$; matching at v is defined with respect to these. Let $dg = \prod dg_v$ and $dg_D = \prod dg_{D,v}$. For a rational maximal subfield E use one product measure $dt = \prod dt_v$ on $\mathbb{A}_E^\times/\mathbb{A}_K^\times$, normalized at almost all places by $\mathrm{vol}(\mathcal{O}_{E_v}^\times K_v^\times/K_v^\times) = 1$, on *both* sides (via ι_D and $\iota_{\mathrm{GL}} \circ j$).

With these choices, the elliptic term of γ on either side is

$$|W_\gamma|^{-1}\,\mathrm{vol}(\mathbb{A}_K^\times E^\times\backslash\mathbb{A}_E^\times; dt)\,F(\iota(\gamma),\cdot),$$

the local orbital integrals being taken with respect to dg_v/dt_v or $dg_{D,v}/dt_v$. The volume factor is literally the same number on the two sides, since it is the same group with the same measure. The spectral sides are $\sum m(\overline{\pi})\,\mathrm{tr}\,\overline{\pi}(\tilde{f}; dg)$ and $\sum m(\overline{\pi}_D)\,\mathrm{tr}\,\overline{\pi}_D(\tilde{f}_D; dg_D)$, with the same dg and dg_D used in the orbital integrals. So the comparison requires only that matching at $v \in S$ be defined with respect to the same dg_v and $dg_{D,v}$ that are used in the trace formulas. A canonical normalization (for example, transfer of measures between inner forms, or Euler–Poincaré measures) is needed only to state local character relations with the correct constants (§10). That the elliptic term has the stated form, with the class-counting constant $1/|W_\gamma|$, is proved in Propositions 8.1 and 8.2.

7.4 Cusp forms on finite central covers

This subsection supplies the analytic input of Lemma 8.4(a): the uniform sup-norm bound for $r(\phi)$ on cusp forms, for the cover $\overline{G}(\mathbb{A}_K)$. For $\mathrm{GL}(r,\mathbb{A})$ the bound is classical [GGPS, Ch. 3], [God], [MW, I.2]. Moeglin and Waldspurger formulate their theory so as to include finite central coverings, and their theory of Eisenstein series is used for covering groups in this generality, for example by Gao [Gao] for Mp_4. Rather than rely on that alone, we list the properties (R1)–(R5) of the group that the classical proof uses, and we verify each of them for $\overline{G}(\mathbb{A}_K)$. The subsection depends on §7 (Proposition 7.5(e)) and on Lemma 8.4.

Lemma 7.8. ***[P]*** *(using* ***[E]****) Let ϕ be a genuine test function on $\overline{G}(\mathbb{A}_K)$ transforming by $\tilde{\theta}^{-1}$ under $\overline{Z}(\mathbb{A}_K)$ (K_∞-finite at the archimedean places). There is C_ϕ with $\sup_x |(r(\phi)\psi)(x)| \le C_\phi\|\psi\|_2$ for all $\psi \in L^2_{\mathrm{cusp}}$.*

Structure of the classical proof. Fix a Siegel set $\mathfrak{S}$ for $\mathrm{GL}(r,\mathbb{A})$ relative to the upper triangular Borel subgroup, so that $\mathrm{GL}(r,\mathbb{A}) = \mathrm{GL}(r,K)Z(\mathbb{A})\mathfrak{S}$. For $x \in \mathfrak{S}$, write $(r(\phi)\psi)(x) = \int_X K_\phi(x,y)\psi(y)\,dy$. Since ψ is cuspidal, the constant-term kernels $K_{\phi,P}(x,y) = \int_{N_P(K)\backslash N_P(\mathbb{A})} K_\phi(s(n)x,y)\,dn$ (P standard, proper) integrate to 0 against ψ. One may therefore replace K_ϕ by the modified kernel $K_\phi^\natural := \sum_P (-1)^{\dim A_P/A_G} K_{\phi,P}$, the sum over all standard P including G. One proves that $\sup_{x\in\mathfrak{S}} \|K_\phi^\natural(x,\cdot)\|_{L^2(X)} < \infty$, whence the bound by Cauchy–Schwarz. The estimate of $K_\phi^\natural$ uses Poisson summation along the abelian quotients of the unipotent radicals, with bounds on the non-constant Fourier coefficients obtained by integration by parts at the archimedean places and from the level at the finite places, together with the contraction of N_P by the torus part of $\mathfrak{S}$ [GGPS, Ch. 3, §2], [MW, I.2]. The proof uses exactly the following properties (R1)–(R5) of the group and the test function.

(R1) *Reduction theory and finite volume of X.* For the cover, the quotient

$$X = \iota(\mathrm{GL}(r,K))\overline{Z}(\mathbb{A}_K)\backslash\overline{G}(\mathbb{A}_K)$$

is homeomorphic to $Z(\mathbb{A})\mathrm{GL}(r,K)\backslash\mathrm{GL}(r,\mathbb{A})$, with matching measures (Lemma 8.4, proof). For genuine $\tilde{\theta}$-equivariant ψ, the function $|\psi|$ is a function on X. So Siegel sets, finiteness of volume and the inner product are those of the linear group.

(R2) *Unipotent splittings.* For every standard unipotent radical N there is a unique splitting s_N of $N(\mathbb{A})$, which agrees with ι on $N(K)$ (Proposition 7.5(e)). Hence constant terms and Fourier coefficients along $N(K)\backslash N(\mathbb{A})$ of genuine automorphic functions are defined, and cuspidality has its usual meaning.

(R3) *Equivariance of the splittings.* For $m \in \mathrm{GL}(r,\mathbb{A})$ normalizing N and any lift $\tilde{m}$, $\mathrm{Ad}(\tilde{m}) \circ s_N = s_N \circ \mathrm{Ad}(m)$, since both sides are splittings of $N(\mathbb{A})$ and the splitting is unique. This is what is used when the torus part of $\mathfrak{S}$ is conjugated past N.

(R4) *Archimedean places.* K is totally imaginary, and at complex places all Hilbert symbols are trivial, so the cover is canonically split over $\mathrm{GL}(r,K_v)$. Hence derivatives and integration by parts along $N(K_\infty)$ are those of the linear group.

(R5) *Test functions.* For $\tilde{a},\tilde{b} \in \overline{G}(\mathbb{A})$, the function $n \mapsto \phi(\tilde{a}\, s_N(n)\, \tilde{b})$ on $N(\mathbb{A})$ is a Schwartz–Bruhat function. Its seminorms (support, level, and finitely many archimedean derivatives) are bounded, uniformly for $\tilde{a},\tilde{b}$ in compact sets, by finitely many seminorms of ϕ. Indeed, s_N is a continuous homomorphism and the cover is split over small open subgroups and at the archimedean places. The classical proof uses no other property of ϕ; in particular, it never multiplies values of ϕ at different points.

Proof of Lemma 7.8. By (R1)–(R5), each step of the classical proof applies verbatim to $\overline{G}(\mathbb{A}_K)$. We list where each property enters.

- The reduction to $x \in \mathfrak{S}$ and the final Cauchy–Schwarz step use (R1).
- The vanishing of $\int K_{\phi,P}(x,y)\psi(y)\,dy$ for cuspidal ψ, and Poisson summation, use (R2).
- The Fourier-coefficient bounds use (R3)–(R5). These bounds depend only on the seminorms in (R5) and on the position of x in $\mathfrak{S}$.

The resulting bound $\sup_{x\in\mathfrak{S}} \|K_\phi^\natural(x,\cdot)\|_{L^2(X)} \le C_\phi$ gives the lemma. □

8 The two trace formulas and their comparison

This section computes the geometric side of the trace formula on the D side, including the class-counting constants (§8.1). It then proves the variant of the simple trace formula

of [FK, §18] that we need on the GL side, with the supercusp form at a place outside S (§8.2), and shows that the two trace formulas agree for matching test functions (§8.3). Measures are as in §7.3, and $\tilde{\theta}$ is a genuine unitary character of $\overline{Z}(\mathbb{A}_K)$ trivial on $\iota(K^\times)$ (Proposition 7.5(c)). The section depends on §7, on Theorem 6.5, and on the analytic inputs listed in Remark 8.7 (with §7.4). It introduces W_γ, $c(\gamma)$ and the class sets $\mathcal{O}_H$ below, the auxiliary places w_0 and u, and admissible pairs of test functions (Definition 8.5).

For $H \in \{G_D, \mathrm{GL}(r)\}$ and a regular elliptic $\gamma \in H(K)$, put $E = K(\gamma)$ (a field of degree r), so that $H_\gamma = E^\times$, and

$$W_\gamma := \{\sigma \in \mathrm{Aut}(E/K) : \sigma(\gamma) \in K^\times \gamma\}, \qquad c(\gamma) := \frac{\mathrm{vol}(\mathbb{A}_K^\times E^\times \backslash \mathbb{A}_E^\times; dt)}{|W_\gamma|}.$$

Write $\mathcal{O}_H$ for the set of $H(K)$-conjugacy classes in $Z(K)\backslash H(K)$ of regular elliptic elements that are good at every place. (On the D side every regular element is elliptic.)

8.1 The D side

We compute the trace of the right regular representation on the compact quotient. This is the untwisted case of [JC, Prop. 6.6], except that all maximal subfields of D now contribute, not only L.

Proposition 8.1. ***[P]****Let $\tilde{f}_D = \otimes_v \tilde{f}_{D,v}$ be smooth, genuine, compactly supported modulo $\overline{Z}(\mathbb{A}_K)$, with $\tilde{f}_D(z\tilde{g}) = \tilde{\theta}(z)^{-1}\tilde{f}_D(\tilde{g})$, and let $\tilde{R}_D$ be the right regular representation on $L^2(\iota(G_D(K))\backslash \overline{G}_D(\mathbb{A}_K), \tilde{\theta})$. Then $\tilde{R}_D(\tilde{f}_D)$ is of trace class, and*

$$\sum_{\overline{\pi}_D} m(\overline{\pi}_D)\,\mathrm{tr}\,\overline{\pi}_D(\tilde{f}_D) = \mathrm{vol}(Z(\mathbb{A})G_D(K)\backslash G_D(\mathbb{A}))\,\tilde{f}_D(1) + \sum_{[\gamma]\in\mathcal{O}_{G_D}} c(\gamma)\,F(\iota(\gamma), \tilde{f}_D).$$

The left side converges absolutely, and the sum on the right is finite. If $\tilde{f}_{D,u}$ vanishes near $\overline{Z}_u$ for some $u \in S$, the first term on the right vanishes.

Proof. The quotient $Z(\mathbb{A})G_D(K)\backslash G_D(\mathbb{A})$ is compact, and the cover is finite, so $\tilde{R}_D(\tilde{f}_D)$ is an integral operator with smooth kernel on a compact space. It is therefore of trace class, and the spectral expansion converges absolutely **[E]** (Gelfand–Graev–Piatetski-Shapiro). Its kernel is $K(x,y) = \sum_{\gamma\in Z(K)\backslash G_D(K)} \tilde{f}_D(x^{-1}\iota(\gamma)y)$. This is well defined because $\tilde{f}_D(\iota(z)\,\cdot) = \tilde{\theta}(\iota(z))^{-1}\tilde{f}_D = \tilde{f}_D$ for $z \in K^\times$. The trace is $\int K(x,x)\,dx$ over $Z(\mathbb{A})G_D(K)\backslash G_D(\mathbb{A})$. As in [FK, §18], only finitely many classes contribute, since characteristic polynomials of rational elements form a discrete set.

Group the γ into classes $[\gamma]$ for the conjugation action of $G_D(K)$ on $Z(K)\backslash G_D(K)$. The stabilizer of $Z(K)\gamma$ is $H_\gamma^Z(K) := \{\delta : \delta\gamma\delta^{-1} \in K^\times\gamma\}$. Since ι is a homomorphism on $G_D(K)$, the contribution of $[\gamma]$ is

$$\int_{Z(\mathbb{A})H_\gamma^Z(K)\backslash G_D(\mathbb{A})} \tilde{f}_D(x^{-1}\iota(\gamma)x)\,dx$$
$$= \frac{1}{[H_\gamma^Z(K) : Z(K)G_\gamma(K)]} \int_{Z(\mathbb{A})G_\gamma(K)\backslash G_D(\mathbb{A})} \tilde{f}_D(x^{-1}\iota(\gamma)x)\,dx.$$

If γ is central, then $\gamma \in Z(K)$, the class is that of 1, and the contribution is $\mathrm{vol}\cdot\tilde{f}_D(1)$. Otherwise γ is regular, since r is prime, so $G_\gamma = E^\times$ with $E = K(\gamma)$ a field. Any $\delta \in H^Z_\gamma(K)$ normalizes E, and $H^Z_\gamma(K)/G_\gamma(K) \cong W_\gamma$ by Skolem–Noether. The index is therefore $|W_\gamma|$, because $Z(K) \subseteq G_\gamma(K)$. Factoring the integral through $G_\gamma(\mathbb{A})\backslash G_D(\mathbb{A})$ gives

$$\int_{G_\gamma(\mathbb{A})\backslash G_D(\mathbb{A})} \tilde{f}_D(x^{-1}\iota(\gamma)x) \int_{Z(\mathbb{A})G_\gamma(K)\backslash G_\gamma(\mathbb{A})} c_\gamma(s)^{-1}\, ds\, dx,$$

where $c_\gamma(s) \in \mu_n$ is defined by $\tilde{s}^{-1}\iota(\gamma)\tilde{s} = c_\gamma(s)\iota(\gamma)$. This c_γ is a character of $G_\gamma(\mathbb{A})$, trivial on $Z(\mathbb{A})$ (Lemma 4.2) and on $G_\gamma(K)$ (as ι is a homomorphism). The inner integral is therefore $\mathrm{vol}(\mathbb{A}_K^\times E^\times\backslash\mathbb{A}_E^\times)$ if $c_\gamma \equiv 1$, and 0 otherwise. As $G_\gamma(\mathbb{A})$ is the restricted product of the $G_\gamma(K_v)$, $c_\gamma \equiv 1$ if and only if $\iota(\gamma)$ is good at every place. In that case the outer integral is the product of the local orbital integrals, i.e. $\Phi(\iota(\gamma), \tilde{f}_D)$, which equals $F(\iota(\gamma), \tilde{f}_D)$ by the product formula $\prod_v \Delta_v(\gamma) = 1$. Interchanging the finite sum with the integral is legitimate, by compactness. The last assertion holds because $\tilde{f}_D(1) = \prod_v \tilde{f}_{D,v}(1)$ and $\tilde{f}_{D,u}(1) = 0$. □

8.2 The GL side: a variant of [FK, §18]

The simple trace formula [FK, §18, Theorem] requires the supercusp form at one of the two places where non-elliptic orbital integrals vanish. We want it at a place $w_0 \notin S$, where matching is the identity. We state the variant we need and prove it by following the proof in [FK, §18], noting which hypothesis each step uses. In fact we need only *one* place with elliptic support, which then plays the roles of all three of the places u, u', u'' in [FK].

Proposition 8.2. ***[P]*** *(using **[E]**) Let $\tilde{f} = \otimes_v \tilde{f}_v$ be smooth, genuine, compactly supported modulo $\overline{Z}(\mathbb{A}_K)$, transforming by $\tilde{\theta}^{-1}$ under $\overline{Z}(\mathbb{A}_K)$, with $\tilde{f}_v$ the unit element of the genuine spherical Hecke algebra (relative to κ_v^*) for almost all v. Suppose:*

(i) at some place w_0, $\tilde{f}_{w_0}$ is a supercusp form [FK, §13], i.e. $\int_{N(K_{w_0})} \tilde{f}_{w_0}(\tilde{x}\, s_N(n)\, \tilde{y})\, dn = 0$ for every unipotent radical N of a proper parabolic subgroup and all $\tilde{x}, \tilde{y}$;

(ii) at some finite place u, $\tilde{f}_u$ is supported on the regular elliptic set of $\overline{G}_u$.

Then $r(\tilde{f})$ maps $L^2(\iota(\mathrm{GL}(r,K))\backslash\overline{G}(\mathbb{A}_K), \tilde{\theta})$ into the cuspidal subspace, is of trace class, and

$$\sum_{\overline{\pi}\ \text{cuspidal}} m(\overline{\pi})\, \mathrm{tr}\, \overline{\pi}(\tilde{f}) = \sum_{[\gamma]\in\mathcal{O}_{\mathrm{GL}}} c(\gamma)\, F(\iota(\gamma), \tilde{f}),$$

with absolute convergence on the left and finitely many nonzero terms on the right. The places w_0 and u may be arbitrary (in particular, they may be distinct, and w_0 need not be one of the places of (ii)).

Proof. We follow [FK, §18] step by step.

Step 1 (spectral side; uses only (i)). For φ in the L^2 space and a unipotent radical N, the constant term of $r(\tilde{f})\varphi$ along N is

$$\int_{N(K)\backslash N(\mathbb{A})} (r(\tilde{f})\varphi)(s_N(n)\tilde{x})\, dn = \int_{N(K)\backslash\overline{G}(\mathbb{A})} \varphi(\tilde{y}) \int_{N(\mathbb{A})} \tilde{f}(\tilde{x}^{-1} s_N(n)\tilde{y})\, dn\, d\tilde{y}.$$

This uses that φ is left invariant under $\iota(N(K)) = s_N(N(K))$ (Proposition 7.5(e)) and that s_N splits $N(\mathbb{A})$. The inner integral factors over places, and its factor at w_0 vanishes by (i). So $r(\tilde{f})$ maps L^2 into the cuspidal subspace L^2_{cusp}, and so does $r(\tilde{f}^*)$, since $\tilde{f}^*(\tilde{g}) = \overline{\tilde{f}(\tilde{g}^{-1})}$ again satisfies (i). Let P be the orthogonal projection onto L^2_{cusp}; it commutes with $r(\phi)$ for every test function ϕ. Then $r(\tilde{f}) = Pr(\tilde{f})$, and taking adjoints in $r(\tilde{f}^*) = Pr(\tilde{f}^*)$ gives $r(\tilde{f}) = r(\tilde{f})P$.

We now justify the analytic conclusions explicitly. The inputs **[E]** are:

(A1) for every test function ϕ, $r(\phi)P$ is Hilbert–Schmidt, and for $\phi = \phi_1 * \phi_2$ its kernel $K^{\mathrm{cusp}}_{\phi}(x,y) = \sum_i (r(\phi_1)\varphi_i)(x)\,\overline{(r(\phi_2^*)\varphi_i)(y)}$ (where $\{\varphi_i\}$ is an orthonormal basis of L^2_{cusp}) converges absolutely and locally uniformly to a continuous function. This is Lemma 8.4 below;

(A2) the Dixmier–Malliavin theorem [DM]: every test function is a finite sum $\sum_j \phi_j * \phi'_j$ of convolutions of test functions.

Write $\tilde{f} = \sum_j \phi_j * \phi'_j$ by (A2). Then $r(\tilde{f}) = r(\tilde{f})P = \sum_j (r(\phi_j)P)(r(\phi'_j)P)$ is a finite sum of products of two Hilbert–Schmidt operators (Lemma 8.4(b)), hence of trace class. Its trace is

$$\operatorname{tr} r(\tilde{f}) = \sum_j \iint K^{\mathrm{cusp}}_{\phi_j}(x,y) K^{\mathrm{cusp}}_{\phi'_j}(y,x)\,dy\,dx = \int K^{\mathrm{cusp}}_{\tilde{f}}(x,x)\,dx,$$

with absolute convergence by Cauchy–Schwarz (both kernels are square integrable), the integrals being over $Z(\mathbb{A})\mathrm{GL}(r,K)\backslash\mathrm{GL}(r,\mathbb{A})$. Finally, the operator $r(\tilde{f})(1-P) = 0$ has kernel $K_{\tilde{f}} - K^{\mathrm{cusp}}_{\tilde{f}}$, where $K_{\tilde{f}}(x,y) = \sum_{\gamma \in Z(K)\backslash \mathrm{GL}(r,K)} \tilde{f}(x^{-1}\iota(\gamma)y)$ is the full kernel (a locally finite sum of smooth functions). Both kernels are continuous, so $K_{\tilde{f}} = K^{\mathrm{cusp}}_{\tilde{f}}$ everywhere, and in particular on the diagonal. Hence $\operatorname{tr} r(\tilde{f}) = \int K_{\tilde{f}}(x,x)\,dx$, and the spectral side is $\sum m(\overline{\pi}) \operatorname{tr} \overline{\pi}(\tilde{f})$ over cuspidal $\overline{\pi}$, converging absolutely. This is the content of the first sentences of the proof in [FK, §18]. It is the only step of [FK]'s proof that uses the supercusp form, and the argument does not use the place at which the supercusp form sits. (The absolute convergence of $\int K_{\tilde{f}}(x,x)\,dx$ also follows directly from Steps 2–4, since only finitely many elliptic classes contribute.)

Step 2 (finiteness). As in [FK, §18], only finitely many classes $[\gamma]$ have $\tilde{f}(x^{-1}\iota(\gamma)x) \neq 0$ for some x, by discreteness of characteristic polynomials of rational elements. This uses no local hypothesis.

Step 3 (only regular elliptic classes contribute; uses only (ii)). If $\tilde{f}(x^{-1}\iota(\gamma)x) \neq 0$, then $x_u^{-1}\gamma x_u$ lies in the support of $\tilde{f}_u$, so γ is regular and elliptic in $\mathrm{GL}(r,K_u)$. Its characteristic polynomial is then separable and irreducible over K_u, hence over K, so γ is regular and elliptic over K. In [FK] this step needs two places: u'' for regularity, and u' together with a modification of $\tilde{f}_{u'}$ to remove the non-elliptic regular classes. Under (ii) the non-elliptic and singular classes contribute nothing *pointwise*, and no modification is needed.

Step 4 (elliptic terms). For regular elliptic γ the computation in the proof of Proposition 8.1 applies verbatim with G_D replaced by $\mathrm{GL}(r)$. The quotient $Z(\mathbb{A})G_\gamma(K)\backslash G_\gamma(\mathbb{A})$ is compact; $H^Z_\gamma(K)/G_\gamma(K) \cong W_\gamma$ by Skolem–Noether in $M_r(K)$; non-good classes contribute 0 by the character c_γ; and the global orbital integral converges absolutely and

factors into local ones [FK, §§6, 18] **[E]**. The interchange of the finite sum and the integral is the one carried out at the end of [FK, §18]. □

Remark 8.3. In the notation of [FK, §18, Theorem], condition (ii) at u implies the conditions at u' and u'' (Theorem 6.5(c)), and the place u of [FK] (which carries the supercusp form there) is replaced by w_0. In particular u'' need not be distinct from u', and a single place of S suffices for the geometric side. We use $|S| \geq 2$ only to know that $S \neq \emptyset$.

Lemma 8.4 (Cuspidal kernels)**.** ***[P]*** *(using* ***[E]****) Let* $X := \iota(\mathrm{GL}(r,K))\overline{Z}(\mathbb{A}_K)\backslash\overline{G}(\mathbb{A}_K)$*, which is homeomorphic to* $Z(\mathbb{A})\mathrm{GL}(r,K)\backslash\mathrm{GL}(r,\mathbb{A})$ *and has finite volume. Let* $L^2_{\mathrm{cusp}} \subset L^2(\iota(\mathrm{GL}(r,K))\backslash\overline{G}(\mathbb{A}_K),\tilde{\theta})$ *be the cuspidal subspace,* P *the projection onto it, and* ϕ *a genuine test function transforming by* $\tilde{\theta}^{-1}$*.*

(a) ***[P]*** *(using* ***[E]****; §7.4) There is* C_ϕ *such that*

$$\sup_x |(r(\phi)\psi)(x)| \leq C_\phi \|\psi\|_2 \qquad (\psi \in L^2_{\mathrm{cusp}})$$

(Gelfand–Graev–Piatetski-Shapiro, Godement).

(b) $r(\phi)P$ *is Hilbert–Schmidt, with* $\|r(\phi)P\|^2_{HS} \leq C_\phi^2 \operatorname{vol}(X)$*.*

(c) If $\phi = \phi_1 * \phi_2$*, the series* $K^{\mathrm{cusp}}_\phi(x,y) = \sum_i (r(\phi_1)\varphi_i)(x)\overline{(r(\phi_2^*)\varphi_i)(y)}$ *converges absolutely and uniformly on compact sets to a continuous function, and it is the kernel of* $r(\phi)P$*.*

Proof. The map $X \to Z(\mathbb{A})\mathrm{GL}(r,K)\backslash\mathrm{GL}(r,\mathbb{A})$. It is a continuous bijection, since $\mu_n \subset \overline{Z}(\mathbb{A}_K)$. It is open, because p is a local homeomorphism [FK, §2]. Hence it is a homeomorphism, and we give X the transported measure. For genuine $\tilde{\theta}$-equivariant ψ, ψ', the function $\psi\overline{\psi'}$ is invariant under $\overline{Z}(\mathbb{A}_K)$, since $\tilde{\theta}$ is unitary, and under $\iota(\mathrm{GL}(r,K))$, so it is a function on X. Thus the inner product is an integral over X.

(a) This is Lemma 7.8 of §7.4, where the passage from $\mathrm{GL}(r,\mathbb{A})$ to the finite central cover is carried out.

(b) Let $\{\varphi_i\}$ be an orthonormal basis of L^2_{cusp}. For each x and each finite set F of indices, $\sum_{i\in F} |(r(\phi)\varphi_i)(x)|^2 = \sup |(r(\phi)\psi)(x)|^2$, where the supremum is over $\psi = \sum_{i\in F} c_i\varphi_i$ with $\sum |c_i|^2 \leq 1$. By (a) this is at most C_ϕ^2. Integrating over X gives $\sum_i \|r(\phi)\varphi_i\|_2^2 \leq C_\phi^2 \operatorname{vol}(X)$.

(c) By (a) and (b) applied to ϕ_1 and ϕ_2^*, both $\sum_i |(r(\phi_1)\varphi_i)(x)|^2$ and $\sum_i |(r(\phi_2^*)\varphi_i)(y)|^2$ are bounded by constants. So the series converges absolutely, by Cauchy–Schwarz. Moreover $(r(\phi_1)\psi)(x) = \langle \psi, k_x\rangle$ with $k_x := \overline{K_{\phi_1}(x,\cdot)}$, where

$$K_{\phi_1}(x,y) = \sum_{\gamma \in Z(K)\backslash\mathrm{GL}(r,K)} \phi_1(x^{-1}\iota(\gamma)y)$$

is the full kernel of $r(\phi_1)$. For fixed x this is a locally finite sum, compactly supported in y modulo $\iota(\mathrm{GL}(r,K))\overline{Z}(\mathbb{A}_K)$, so $k_x \in L^2$. Hence $\sum_i |(r(\phi_1)\varphi_i)(x)|^2 = \|Pk_x\|_2^2$. The map

$x \mapsto k_x \in L^2$ is continuous, so this sum of continuous nonnegative functions converges to a continuous function. By Dini's theorem the convergence is uniform on compact sets, and likewise in y. Hence the partial sums of K_ϕ^{cusp} converge locally uniformly, and the limit is continuous. It is the kernel of $r(\phi_1)Pr(\phi_2) = r(\phi)P$, by expanding $r(\phi_2)\psi$ in the basis. □

8.3 Comparison

We now show that the two trace formulas agree for matching test functions. The places $w_0 \notin S$ and $u \in S$ and the admissible pairs fixed in Definition 8.5 are used throughout §9.

Definition 8.5. Fix a finite place $w_0 \notin S$, a place $u \in S$, and a finite set $S_D \supseteq S \cup \{w_0\}$ as in §7.3. A pair $(\tilde{f}_D, \tilde{f})$ of factorizable genuine test functions with central characters $\tilde{\theta}^{-1}$ is *admissible* if:

(a) for $v \in S$, $\tilde{f}_{D,v}$ vanishes near $\overline{Z}_v$, $\tilde{f}_v$ is supported in the regular set, and they match (in particular $F(\tilde{\delta}, \tilde{f}_v) = 0$ at good regular non-elliptic $\tilde{\delta}$); moreover $\tilde{f}_u$ is supported on the regular elliptic set. (By Theorem 6.5, every $\tilde{f}_{D,v}$ vanishing near $\overline{Z}_v$ has such a partner, supported on the good regular elliptic set; conversely every such $\tilde{f}_v$ has a partner $\tilde{f}_{D,v}$.)

(b) for $v \notin S$, $\tilde{f}_{D,v} = \tilde{f}_v \circ \psi_v$; for $v \notin S_D$ both are unit elements (Proposition 4.6(c)); at the archimedean places they are K_v-finite;

(c) $\tilde{f}_{w_0}$ is a supercusp form.

Theorem 8.6. ***[P]****For every admissible pair,*

$$\sum_{\overline{\pi}_D} m(\overline{\pi}_D) \operatorname{tr} \overline{\pi}_D(\tilde{f}_D) = \sum_{\overline{\pi} \text{ cuspidal}} m(\overline{\pi}) \operatorname{tr} \overline{\pi}(\tilde{f}),$$

both sides converging absolutely.

Proof. Apply Proposition 8.1 to $\tilde{f}_D$; the identity term vanishes by (a). Apply Proposition 8.2 to $\tilde{f}$, with $u \in S$ and w_0 as given. We compare the geometric sides class by class.

By Corollary 7.7, $\gamma \mapsto j(\gamma)$ is a bijection from $\mathcal{O}_{G_D}$ onto the set of classes in $\mathcal{O}_{\mathrm{GL}}$ whose field embeds in D. It commutes with multiplication by $K^\times$, so it is well defined on classes modulo $Z(K)$. Classes in $\mathcal{O}_{\mathrm{GL}}$ whose field does not embed in D are non-elliptic (and regular) at some $v \in S$, where $F(\cdot, \tilde{f}_v)$ vanishes by (M2), so they contribute 0. For corresponding classes, $F(\iota(\gamma), \tilde{f}_D) = F(\iota(j\gamma), \tilde{f})$ by Theorem 7.6. Finally $c(\gamma) = c(j\gamma)$: the field $E = K(\gamma) \cong K(j\gamma)$ and the measure dt are the same (§7.3), and $W_\gamma = W_{j\gamma}$ because j is a K-algebra isomorphism of E onto $K(j\gamma)$. □

Remark 8.7. The **[E]** inputs of the global sections are: the sup-norm bound for cusp forms on $\mathrm{GL}(r, \mathbb{A})$ [GGPS], [God], [MW, I.2], transferred to the cover in §7.4, and the

Dixmier–Malliavin theorem (A2) [DM], used for the trace formula on the cuspidal spectrum; the trace class property on the compact quotient; absolute convergence and factorization of elliptic orbital integrals [FK, §6]; finiteness of cusp forms of fixed level, type and infinitesimal character [HC68], [MW, I.2]; Arthur's multipliers [Art] (as in [JC, Lemma 6.3]); and the splitting-off of supercuspidal representations (Lemma 9.1).

9 Spectral separation and the global transfer

This section deduces Theorem C from Theorem 8.6. We first use matrix coefficients at w_0 to fix the component there (§9.1). We then separate by archimedean infinitesimal characters and by the local components outside S (§9.2). This uses corner Hecke algebras and linear independence of traces of finitely many simple modules, but no multiplicity-one theorem, and it yields the packet identity of Proposition 9.6 unconditionally. The section depends on Theorem 8.6 and on the local representation theory cited in Lemmas 9.1 and 9.5. It introduces the set $V = S \cup \{w_0\}$, the packets $P_D(\rho)$ and $P(\rho)$, the corner algebras $e\mathcal{H}e$, and the hypotheses (H_D) and (H).

Dependency structure. The passage from the packet identity to the existence of a transfer (§9.3) needs the D-side packet character to be nonzero on matching regular-supported test functions. This is hypothesis (H_D) (Definition 9.9). We prove it when the packet is finite (Proposition 9.10), and we show exactly what a failure would require (Lemma 9.8), but we cannot prove it in general. So Theorem C is unconditional for finite packets and conditional on (H_D) in general. Similarly, the converse is conditional on (H).

9.1 The place w_0

We fix the component at w_0 by a choice of test function, so that no separation is needed there. The supercuspidal σ and the test function $\tilde{f}_\sigma$ fixed here are kept for the rest of the section.

Lemma 9.1. ***[P]*** *(using **[E]**) Let σ be a genuine irreducible supercuspidal representation of $\overline{G}_{w_0}$ with (unitary) central character $\tilde{\theta}_{w_0}$. Then σ is unitarizable, since its matrix coefficients are compactly supported modulo $\overline{Z}_{w_0}$ and hence square integrable modulo the center; we fix an invariant inner product. Let $v \in \sigma$ and $v^\vee \in \sigma^\vee$ with $\langle v, v^\vee\rangle = 1$, and let $d(\sigma)$ be the formal degree (for dg_{w_0}). Put $\tilde{f}_\sigma(\tilde{g}) := d(\sigma)\langle\sigma(\tilde{g}^{-1})v, v^\vee\rangle$. Then $\tilde{f}_\sigma$ is genuine, transforms by $\tilde{\theta}_{w_0}^{-1}$ under $\overline{Z}_{w_0}$, and is a supercusp form. Moreover $\operatorname{tr}\sigma(\tilde{f}_\sigma) = 1$, and $\pi(\tilde{f}_\sigma) = 0$ for every irreducible smooth $\pi \not\cong \sigma$ with central character $\tilde{\theta}_{w_0}$.*

Proof. Genuineness and the central character follow from $\sigma(\zeta\tilde{g}) = \zeta\sigma(\tilde{g})$ and $\sigma(z) = \tilde{\theta}_{w_0}(z)$. Matrix coefficients of supercuspidal representations are compactly supported modulo the center and are supercusp forms [FK, §13] **[E]**. The trace formula $\operatorname{tr}\sigma(\tilde{f}_\sigma) = \langle v, v^\vee\rangle = 1$ is Schur orthogonality for representations that are square-integrable modulo the center. The vanishing for $\pi \not\cong \sigma$ is the fact that a supercuspidal representation splits off the category of smooth representations [Ber, BZ] **[E]**. Its proof uses only that $\overline{G}_{w_0}$ is a

totally disconnected group with $\overline{Z}_{w_0}$ central of finite index in the center modulo μ_n and cocompact in the support of matrix coefficients, so it applies to covers. □

Fix a genuine supercuspidal σ of $\overline{G}_{w_0}$ and take $\tilde{f}_{w_0} = \tilde{f}_\sigma$ and $\tilde{f}_{D,w_0} = \tilde{f}_\sigma \circ \psi_{w_0}$ in every admissible pair. Then Theorem 8.6 becomes

$$\sum_{\overline{\pi}_D:\ \overline{\pi}_{D,w_0} \cong \sigma\circ\psi_{w_0}} m(\overline{\pi}_D)\,\mathrm{tr}\,\overline{\pi}_D^{w_0}(\tilde{f}_D^{w_0}) = \sum_{\overline{\pi}:\ \overline{\pi}_{w_0}\cong\sigma} m(\overline{\pi})\,\mathrm{tr}\,\overline{\pi}^{w_0}(\tilde{f}^{w_0}), \tag{8}$$

where the superscript w_0 denotes the components away from w_0. All $\overline{\pi}$ on the right are cuspidal.

9.2 Separation

We separate (8) first by archimedean infinitesimal characters, to reduce to finite sums, and then by the components outside $S \cup \{w_0\}$. For the second step we use corner Hecke algebras $e\mathcal{H}e$ and linear independence of traces of finitely many simple modules. Packets are *not* assumed finite. Instead, the packet identity is proved as an identity of locally finite sums: for each test function, only finitely many terms are nonzero.

Lemma 9.2. ***[P]*** *(using **[E]**) For each $W_{\infty,\mathbb{C}}$-orbit ν, (8) holds with both sums restricted to representations of archimedean infinitesimal character ν. If all finite components of the test functions are bi-invariant under a fixed open compact subgroup U of the finite adeles (split by κ^*), and the archimedean components lie in $e_{\mathfrak{t}} C_c^\infty e_{\mathfrak{t}}$ for a fixed finite set $\mathfrak{t}$ of K_∞-types, then only finitely many terms on either side are nonzero. Specifically, the nonzero terms come from representations $\overline{\pi}$ with $e_U e_{\mathfrak{t}} \overline{\pi} \neq 0$ and infinitesimal character ν.*

Proof. This is the proof of [JC, Lemma 6.12], with the torus side replaced by the GL side. *Finiteness:* a term is nonzero only if $\overline{\pi}(\tilde{f}) \neq 0$, which forces $e_U e_{\mathfrak{t}} \overline{\pi} \neq 0$. On the D side, the automorphic forms of level U, of K_∞-types in $\mathfrak{t}$ and with given Casimir eigenvalue form a finite-dimensional space, by ellipticity of the Casimir operator on the compact quotient, as in [JC, Lemma 6.12]. On the GL side the contributing $\overline{\pi}$ are cuspidal, and the corresponding spaces of cusp forms are finite-dimensional [HC68], [MW, I.2] **[E]**. *Multipliers:* at the complex places both covers split, $\tilde{f}_{D,v} = \tilde{f}_v \circ \psi_v$, and Arthur's multipliers [JC, Lemma 6.3] act by the same scalar $\hat{\mu}(\nu_\pi)$ on both sides, preserving $e_{\mathfrak{t}} C_c^\infty e_{\mathfrak{t}}$ and admissibility. The Paley–Wiener argument of [JC, Lemma 6.12] then separates the orbits ν, using the absolute convergence of both sides (Theorem 8.6). □

Lemma 9.3. ***[E]****Let A be a $\mathbb{C}$-algebra and $M_1, \dots, M_k$ pairwise non-isomorphic simple A-modules of finite dimension. Then the functions $a \mapsto \mathrm{tr}(a|M_i)$ are linearly independent. (By the density theorem, $A \to \prod_i \mathrm{End}(M_i)$ is surjective [Lan, XVII].)*

Let $V := S \cup \{w_0\}$. For a genuine representation $\overline{\pi}$ of $\overline{G}(\mathbb{A}_K)$ let $\overline{\pi}^V$ be its component away from V, and use the same notation on the D side. Via $\psi^V := \otimes_{v\notin V}\psi_v$, genuine representations of $\overline{G}_D(\mathbb{A}^V)$ and of $\overline{G}(\mathbb{A}^V)$ correspond; we write $\rho \circ \psi^V$ for the transport of ρ.

Lemma 9.4 (Tensor products of simple modules)**.** ***[P]****Let $B_1, \dots, B_m$ be $\mathbb{C}$-algebras (not necessarily unital), and let M_i, M_i' be finite-dimensional simple B_i-modules on which B_i acts nondegenerately (i.e. $B_i M_i = M_i$), so that $B_i \to \mathrm{End}(M_i)$ is surjective by density. Then $M := \bigotimes_i M_i$ is a simple $B := \bigotimes_i B_i$-module, and if $M \cong M' := \bigotimes_i M_i'$ as B-modules then $M_i \cong M_i'$ for every i.*

Proof. $B \to \bigotimes_i \mathrm{End}(M_i) = \mathrm{End}(M)$ is surjective, so M is simple. For the second assertion, compare traces: $\mathrm{tr}(b_1 \otimes \cdots \otimes b_m \,|\, M) = \prod_i \mathrm{tr}(b_i|M_i)$. Fix i. For $k \neq i$ choose $b_k \in B_k$ mapping to a rank-one idempotent of $\mathrm{End}(M_k)$, so $\mathrm{tr}(b_k|M_k) = 1$. Then $\mathrm{tr}(b_i|M_i) = c\,\mathrm{tr}(b_i|M_i')$ for all b_i, with $c = \prod_{k\neq i} \mathrm{tr}(b_k|M_k')^{-1}$. Here c is finite and nonzero, since both sides are nonzero functionals of b_i, by surjectivity. Two nonzero proportional trace functionals of simple finite-dimensional modules come from isomorphic modules (Lemma 9.3), so $M_i \cong M_i'$. □

Why fixed-central-character Hecke algebras. All test functions in this paper transform under the full central subgroup $\overline{Z}$ by $\tilde{\theta}^{-1}$, as the L^2 spaces do, and convolution is over G/Z. The resulting Hecke algebras $C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ are not subalgebras of the ordinary Hecke algebra $C_c^\infty(\overline{G}_v)$. In particular, the idempotent of an open compact subgroup $\kappa_v^*(U_v)$, a function supported on $\mu_n \kappa_v^*(U_v)$, does not belong to them when $\overline{Z}_v$ is non-compact. It acts on them as a convolution multiplier, but it is not their unit. The unit of the corner algebra must instead be the *central-character idempotent* supported on $\overline{Z}_v \kappa_v^*(U_v)$, defined below. The archimedean projector $e_{\mathfrak{t}}$ is likewise a multiplier and not a function.

Lemma 9.5 (Corner algebras)**.** ***[P]*** *(using **[E]**) Let $U^V = \prod_{v\notin V,\, v<\infty} U_v$ be an open compact subgroup, with $U_v = \mathrm{GL}(r, \mathcal{O}_v)$ split by κ_v^* for $v \notin S_D$, and let $\mathfrak{t}$ be a finite set of K_∞-types.*

- *At a finite place $v \notin V$, assume that $\tilde{\theta}_v$ is trivial on $\kappa_v^*(U_v \cap K_v^\times)$; this holds for $v \notin S_D$, and after shrinking U_v in general. Let $\mathcal{H}_v := C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ be the algebra of smooth genuine functions, compactly supported modulo $\overline{Z}_v$, with $f(z\tilde{g}) = \tilde{\theta}_v(z)^{-1} f(\tilde{g})$, under the convolution $(f_1 * f_2)(\tilde{g}) = \int_{\overline{G}_v/\overline{Z}_v} f_1(\tilde{h}) f_2(\tilde{h}^{-1}\tilde{g})\, d\dot{h}$. Let $e_v \in \mathcal{H}_v$ be the central-character idempotent*

$$e_v(z\,\kappa_v^*(u)) = \mathrm{vol}\left(\overline{Z}_v \kappa_v^*(U_v)/\overline{Z}_v\right)^{-1} \tilde{\theta}_v(z)^{-1} \quad (z \in \overline{Z}_v,\ u \in U_v),$$

  and $e_v = 0$ off $\overline{Z}_v \kappa_v^*(U_v)$. It is well defined by the assumption on $\tilde{\theta}_v$, and $e_v * e_v = e_v$. For τ_v with central character $\tilde{\theta}_v$, $\tau_v(e_v)$ is the projection onto $\tau_v^{\kappa_v^*(U_v)}$. Put $A_v := e_v * \mathcal{H}_v * e_v$, a unital algebra with unit e_v.

- *At the archimedean places (where the covers split, and $Z_\infty \cap K_\infty$ acts by $\tilde{\theta}_\infty$ on every $\delta \in \mathfrak{t}$), let $e_{\mathfrak{t}}$ be the measure $\sum_{\delta\in\mathfrak{t}} \dim(\delta)\overline{\chi_\delta(k)}\, dk$ on K_∞. This is a distribution, not a function on G_∞, but convolution by it maps $C_c^\infty(G_\infty)_{K\text{-}fin}$ into itself. Let $A_\infty := e_{\mathfrak{t}} * C_c^\infty(G_\infty, \tilde{\theta}_\infty)_{K\text{-}fin} * e_{\mathfrak{t}}$. This is a subalgebra of $C_c^\infty(G_\infty)$ that in general has no unit; $e_{\mathfrak{t}}$ acts on it as a two-sided multiplier.*

- *Let A be the span of the functions $\phi_\infty \otimes \bigotimes_{v<\infty} \phi_v$ with $\phi_\infty \in A_\infty$, $\phi_v \in A_v$, and $\phi_v = e_v$ for almost all v.*

For an admissible genuine irreducible (unitary) representation τ of $\overline{G}(\mathbb{A}^V)$ with central character $\tilde{\theta}^V$, put $e\tau := \tau(e_{\mathfrak{t}}) \prod_v \tau(e_v)\, \tau$. Then:

(a) $e\tau$ is a finite-dimensional, nondegenerate A-module, and $\operatorname{tr} \tau(\phi) = \operatorname{tr}(\phi | e\tau)$ for $\phi \in A$;

(b) if $e\tau \neq 0$, it is a simple A-module;

(c) if $e\tau \cong e\tau' \neq 0$ as A-modules, then $\tau \cong \tau'$.

Proof. (a) Every $\phi \in A$ satisfies $\phi = e_{\mathfrak{t}} * \phi * e_{\mathfrak{t}}$ and $\phi = e_v * \phi * e_v$, so $\tau(\phi)$ maps τ into $e\tau$ and kills its complement $\ker\left(\tau(e_{\mathfrak{t}}) \prod_v \tau(e_v)\right)$. So the trace of $\tau(\phi)$ is its trace on $e\tau$, which is finite-dimensional by admissibility. Nondegeneracy follows from (b) and its proof.

(b) Write $\tau = \tau_\infty \otimes \bigotimes'_v \tau_v$. By admissibility there is a finite set T of finite places such that $\dim e_v\tau_v = 1$ for $v \notin T$, and then $e\tau = e_{\mathfrak{t}}\tau_\infty \otimes \bigotimes_{v\in T} e_v\tau_v$, tensored with the distinguished lines at $v \notin T$. The single-place facts are [**E**]:

- at a finite place, $\tau_v \mapsto e_v\tau_v$ is a bijection from irreducible representations with $e_v\tau_v \neq 0$ to simple A_v-modules [BZ, §2], [Ber];
- at the archimedean places, by Harish-Chandra, $e_{\mathfrak{t}}\tau_\infty$ is a simple, nondegenerate, finite-dimensional A_∞-module whenever it is nonzero, and it determines the irreducible admissible $(\mathfrak{g}, K_\infty)$-module of τ_∞, hence the unitary representation τ_∞ up to isomorphism [HC], [Wal, §3.4–3.5].

Thus each factor is a simple, nondegenerate, finite-dimensional module over the corresponding algebra. The subalgebra $A_\infty \otimes \bigotimes_{v\in T} A_v \otimes \bigotimes_{v\notin T} \mathbb{C}e_v$ of A acts on $e\tau$ through the tensor product. By Lemma 9.4, $e\tau$ is simple, and nondegenerate, over this subalgebra, hence over A.

(c) Choose a finite set T that works for both τ and τ' as in (b), and enlarge it by any finite set of places. Applying Lemma 9.4 to the subalgebra of (b) gives $e_{\mathfrak{t}}\tau_\infty \cong e_{\mathfrak{t}}\tau'_\infty$ and $e_v\tau_v \cong e_v\tau'_v$ for $v \in T$, hence $\tau_\infty \cong \tau'_\infty$ and $\tau_v \cong \tau'_v$ for $v \in T$. Since T can contain any given finite place, $\tau_v \cong \tau'_v$ for every v, and so $\tau \cong \tau'$. □

Proposition 9.6 (Packet identity)**.** ***[P]****Let ρ be an admissible genuine irreducible representation of $\overline{G}(\mathbb{A}^V)$. Put*

$$P_D(\rho) := \{\overline{\pi}_D :\ \overline{\pi}_D^V \cong \rho \circ \psi^V,\ \overline{\pi}_{D,w_0} \cong \sigma \circ \psi_{w_0}\},$$
$$P(\rho) := \{\overline{\pi} \text{ cuspidal} :\ \overline{\pi}^V \cong \rho,\ \overline{\pi}_{w_0} \cong \sigma\},$$

counted with multiplicities $m(\cdot)$. These sets need not be finite. Let $(\tilde{f}_{D,v}, \tilde{f}_v)_{v\in S}$ be matching, with $\tilde{f}_{D,v}$ vanishing near $\overline{Z}_v$ and $\tilde{f}_v$ supported in the regular set. Then only finitely many terms of each side of

$$\sum_{\overline{\pi}_D \in P_D(\rho)} m(\overline{\pi}_D) \prod_{v\in S} \operatorname{tr} \overline{\pi}_{D,v}(\tilde{f}_{D,v}) = \sum_{\overline{\pi}\in P(\rho)} m(\overline{\pi}) \prod_{v\in S} \operatorname{tr} \overline{\pi}_v(\tilde{f}_v) \tag{9}$$

are nonzero, and the identity holds.

Proof. Reduction to elliptically supported $\tilde{f}_u$. Fix $u \in S$, and let $\tilde{f}'_u$ be the function of Theorem 6.5(a) attached to $\tilde{f}_{D,u}$; it is supported on the regular elliptic set. Both $\tilde{f}_u$ and $\tilde{f}'_u$ match $\tilde{f}_{D,u}$, so their normalized orbital integrals agree at all good regular elements: at elliptic ones by (M1), and at non-elliptic ones both vanish by (M2). By Lemma 6.12, $\operatorname{tr}\overline{\pi}_u(\tilde{f}_u) = \operatorname{tr}\overline{\pi}_u(\tilde{f}'_u)$ for every $\overline{\pi}_u$. So we may assume that $\tilde{f}_u$ is supported on the regular elliptic set, i.e. that the family satisfies Definition 8.5(a).

Uniform finiteness. Choose an open compact $U_S \subset \prod_{v\in S} D_v^\times$ (split) such that each $\tilde{f}_{D,v}$ is bi-$U_{S,v}$-invariant, and a level U'_S for the $\tilde{f}_v$. Choose a level U_{w_0} under which $\tilde{f}_\sigma$ is bi-invariant. Choose U^V and $\mathfrak{t}$ as in Lemma 9.5 with $e\rho \neq 0$, and let ν be the infinitesimal character of ρ_∞. Let Ω_D (resp. Ω) be the set of automorphic $\overline{\pi}_D$ (resp. cuspidal $\overline{\pi}$) with $\overline{\pi}_{w_0} \cong \sigma$ (transported), infinitesimal character ν, nonzero U_S- (resp. U'_S-) fixed vectors at S, and $e\overline{\pi}^V \neq 0$. These conditions do not involve the test functions away from V. By the finiteness part of Lemma 9.2, applied at the level $U_S \times U_{w_0} \times U^V$ (resp. $U'_S \times U_{w_0} \times U^V$) with types $\mathfrak{t}$, both Ω_D and Ω are *finite*. For *every* $\phi \in A$, a term in the identity below (with components ϕ away from V) is nonzero only if the representation lies in Ω_D (resp. Ω). Indeed, $\overline{\pi}(\tilde{f}) \neq 0$ forces each local operator to be nonzero, hence nonzero fixed vectors at every place for the level of the corresponding test function; at w_0 it forces $\overline{\pi}_{w_0} \cong \sigma$ (Lemma 9.1); and at infinity it forces the type condition. So all the identities below are identities of sums over the fixed finite sets Ω_D and Ω, uniformly in $\phi \in A$. This uniformity is what allows us to apply Lemma 9.3, which concerns a fixed finite family of modules.

Separation. Keep the components at S and at w_0 fixed, and let the components away from V run through $A = e\mathcal{H}e$ (transported to the D side by ψ^V). By Lemma 9.2 the ν-part of (8) becomes the identity, for $\phi \in A$,

$$\sum_{\overline{\pi}_D} m(\overline{\pi}_D)\Big(\prod_{v\in S}\operatorname{tr}\overline{\pi}_{D,v}(\tilde{f}_{D,v})\Big)\operatorname{tr}(\phi|e(\overline{\pi}_D^V\circ(\psi^V)^{-1})) = \sum_{\overline{\pi}} m(\overline{\pi})\Big(\prod_{v\in S}\operatorname{tr}\overline{\pi}_v(\tilde{f}_v)\Big)\operatorname{tr}(\phi|e\overline{\pi}^V),$$

in which only representations from the finite sets Ω_D, Ω occur. Automorphic representations in L^2 are unitary and admissible, so Lemma 9.5 applies. Group the terms by the isomorphism class of the simple A-module $e\overline{\pi}^V$ (or $e(\overline{\pi}_D^V \circ (\psi^V)^{-1})$). By Lemma 9.3 the total coefficients of the class of $e\rho$ on the two sides agree. By Lemma 9.5(c), the terms in that class are exactly those with $\overline{\pi}^V \cong \rho$ (resp. $\overline{\pi}_D^V \cong \rho\circ\psi^V$). This is (9). □

9.3 Non-cancellation and the transfer

Proposition 9.6 produces a GL-side packet as soon as the D-side of (9) is nonzero for *some* matching regular-supported family. We now show that this happens if $P_D(\rho)$ is finite. We also determine exactly what a failure would look like (Lemma 9.8). Since we cannot exclude such a failure for infinite packets, Theorem C is stated under a hypothesis (H_D) that is parallel to the hypothesis (H) of the converse.

Lemma 9.7 (Finite case)**.** ***[P]****Let $\tau_1, \dots, \tau_k$ be pairwise inequivalent genuine irreducible representations of $\overline{G}_D(K_S) := \prod_{v\in S}\overline{G}_D(K_v)$ (modulo the identification of the μ_n's) with*

central character $\tilde\theta_S$, and let $m_1,\dots,m_k>0$. Then there are $\tilde f_{D,v}\in C_c^\infty(\overline{G}_D(K_v),\tilde\theta_v)_{\mathrm{gen}}$, $v\in S$, each vanishing near $\overline{Z}_v$, with $\sum_i m_i \operatorname{tr}\tau_i(\otimes_v \tilde f_{D,v})\neq 0$.

Proof. Each τ_i is finite-dimensional, since $\overline{G}_D(K_S)$ is compact modulo its center. Its character Θ_i is locally constant with $\Theta_i(z\tilde g)=\tilde\theta_S(z)\Theta_i(\tilde g)$, and $\operatorname{tr}\tau_i(\tilde f)=\int_{\overline{G}_D(K_S)/\overline{Z}_S}\tilde f\,\Theta_i$. By Schur orthogonality on the compact group $\overline{G}_D(K_S)/\overline{Z}_S$ (with fixed central character), the Θ_i are linearly independent, so $\Theta:=\sum_i m_i\Theta_i\neq 0$. Since Θ is locally constant and the product R of the regular sets is dense and open, $\Theta(\tilde x)\neq 0$ for some $\tilde x=(\tilde x_v)\in R$. Choose small open neighborhoods $N_v\ni\tilde x_v$ inside the regular set, on whose product Θ is constant, and let $\tilde f_{D,v}$ be the function supported on $\overline{Z}_vN_v$ with $\tilde f_{D,v}(z\tilde y)=\tilde\theta_v(z)^{-1}$ for $z\in\overline{Z}_v$, $\tilde y\in N_v$ (well defined if N_v is small). Then $\sum_i m_i\operatorname{tr}\tau_i(\otimes\tilde f_{D,v})=\Theta(\tilde x)\prod_v\operatorname{vol}(N_v)\neq 0$. The finiteness of the set $\{\tau_i\}$ is essential: shrinking the N_v raises the level, but no new terms appear. □

The example $\delta_e=\sum_\tau(\dim\tau)\Theta_\tau$ on a compact group shows that the finiteness assumption in Lemma 9.7 cannot simply be dropped. The next lemma determines exactly which coefficient functions cancel at a fixed level. When S has one element, the cancelling functions are the multiples of the Plancherel weight, as in this example. Since $|S|\geq 2$ under our assumptions, the actual condition is weaker: the coefficient function must be a sum of terms, each of which has a Plancherel factor at one place of S.

Lemma 9.8 (When cancellation occurs)**.** ***[P]****For $v\in S$ let $U_v\subset D_v^\times$ be an open compact subgroup over which the cover splits by κ_v, with $\tilde\theta_v\circ\kappa_v$ trivial on $U_v\cap K_v^\times$. Let e_{U_v} be the central-character idempotent of $\overline{Z}_v\kappa_v(U_v)$, as in Lemma 9.5, and let $\mathcal{H}_v:=e_{U_v}*C_c^\infty(\overline{G}_D(K_v),\tilde\theta_v)_{\mathrm{gen}}*e_{U_v}$, which is finite-dimensional. Let $\mathcal{R}_v\subset\mathcal{H}_v$ be the subspace of functions vanishing near $\overline{Z}_v$, and let $\mathcal{T}_v$ be the finite set of genuine irreducible τ_v with central character $\tilde\theta_v$ and $\tau_v^{U_v}\neq 0$. For a function a on $\mathcal{T}:=\prod_{v\in S}\mathcal{T}_v$, put $\Lambda_a(\otimes_v f_v):=\sum_{\tau\in\mathcal{T}}a(\tau)\prod_v\operatorname{tr}\tau_v(f_v)$. Then Λ_a vanishes on $\bigotimes_v\mathcal{R}_v$ if and only if*

$$a\in\sum_{v\in S}d_v\otimes\mathbb{C}^{\prod_{w\in S\smallsetminus\{v\}}\mathcal{T}_w},\qquad d_v(\tau_v):=\dim\tau_v.$$

Equivalently, a maps to 0 in $\bigotimes_{v\in S}\left(\mathbb{C}^{\mathcal{T}_v}/\mathbb{C}d_v\right)$. If S had one element, this would say that a is proportional to $\tau\mapsto\dim\tau$, i.e. to the Plancherel weight. For $|S|\geq 2$ it does not *say that a is proportional to the product weight $\prod_v d_v$. For example, with two places and $d_1=d_2=(1,1)$, the positive matrix $a=\left(\begin{smallmatrix}2&3\\4&5\end{smallmatrix}\right)=d_1\otimes(0,1)+(2,4)\otimes d_2$ cancels.*

Proof. Since the cover splits over U_v and $\tilde\theta_v\circ\kappa_v$ is trivial on $U_v\cap K_v^\times$, every element of $\mathcal{H}_v$ is determined on $\overline{Z}_v\kappa_v(U_v)$ by its value at 1. So $\mathcal{R}_v=\ker(\mathrm{ev}_{1,v})$, with $\mathrm{ev}_{1,v}(f)=f(1)$; indeed, a bi-$U_v$-invariant function vanishing at 1 vanishes on the open set $\overline{Z}_v\kappa_v(U_v)$, and the singular elements of $D_v^\times$ are central. The group $\overline{G}_D(K_v)/\overline{Z}_v$ is compact, so by Peter–Weyl (with fixed central character) $\mathcal{H}_v\cong\bigoplus_{\tau_v\in\mathcal{T}_v}\mathrm{End}(\tau_v^{U_v})$ is semisimple, the functionals $\operatorname{tr}\tau_v$ ($\tau_v\in\mathcal{T}_v$) form a basis of the trace functionals on $\mathcal{H}_v$, and Fourier inversion gives

$$\mathrm{ev}_{1,v}=\operatorname{vol}(\overline{G}_D(K_v)/\overline{Z}_v)^{-1}\sum_{\tau_v\in\mathcal{T}_v}(\dim\tau_v)\operatorname{tr}\tau_v\quad\text{on }\mathcal{H}_v.$$

For finite-dimensional spaces, $\bigotimes_v \ker(\mathrm{ev}_{1,v}) = \bigcap_v \ker(\mathrm{ev}_{1,v} \otimes \mathrm{id})$, so its annihilator in $(\bigotimes_v \mathcal{H}_v)^*$ is $\sum_v \mathbb{C}\,\mathrm{ev}_{1,v} \otimes \bigotimes_{w\neq v} \mathcal{H}_w^*$. Let $E_v : \mathcal{H}_v \to Z(\mathcal{H}_v)$ be the central projection (with kernel $[\mathcal{H}_v, \mathcal{H}_v]$). Then $\lambda \mapsto \lambda \circ (\otimes_v E_v)$ is a projection of $(\bigotimes \mathcal{H}_v)^*$ onto the span of products of trace functionals; it fixes Λ_a and the $\mathrm{ev}_{1,v}$. Applying it to an expression $\Lambda_a = \sum_v \mathrm{ev}_{1,v} \otimes \mu_v$, we may assume that each μ_v is a combination of products of the $\mathrm{tr}\,\tau_w$. Expanding in the basis $\{\prod_v \mathrm{tr}\,\tau_v\}$ gives the stated condition, and conversely. □

Definition 9.9. We say that a packet $P_D(\rho)$ satisfies (H_D) if there are $\tilde{f}_{D,v} \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$ $(v \in S)$, each vanishing near $\overline{Z}_v$, with

$$\sum_{\overline{\pi}'_D \in P_D(\rho)} m(\overline{\pi}'_D) \prod_{v\in S} \mathrm{tr}\,\overline{\pi}'_{D,v}(\tilde{f}_{D,v}) \neq 0$$

(a sum with finitely many nonzero terms).

By Lemma 9.8, (H_D) fails if and only if, at every level $U = \prod_{v\in S} U_v$, the function a_U on $\prod_v \mathcal{T}_v$ is of the form in that lemma. Here $a_U(\tau)$ is the total multiplicity of the members of $P_D(\rho)$ whose S-component is τ. So a failure requires that, at every level, the coefficient function of the packet at S be a sum of terms each having a Plancherel factor at some place of S. This is a strong but not obviously contradictory condition when $P_D(\rho)$ is infinite.

Proposition 9.10. ***[P]****If $P_D(\rho)$ is finite and nonempty, then (H_D) holds.*

Proof. The S-components of the members of $P_D(\rho)$ are pairwise inequivalent. Two members with isomorphic components at S have isomorphic components everywhere, and so are the same automorphic representation, counted once with its multiplicity. Apply Lemma 9.7. □

Theorem 9.11 (Theorem C)**.** ***[P]*** *under (H_D) Let $\overline{\pi}_D$ be an irreducible genuine automorphic representation of $\overline{G}_D(\mathbb{A}_K)$ with central character $\tilde{\theta}$, such that $\overline{\pi}_{D,w_0}$ is supercuspidal for some finite place $w_0 \notin S$. Let $\rho = \overline{\pi}_D^V \circ (\psi^V)^{-1}$ and $\sigma = \overline{\pi}_{D,w_0} \circ \psi_{w_0}^{-1}$, and assume that $P_D(\rho)$ satisfies (H_D); this holds, for example, if $P_D(\rho)$ is finite (Proposition 9.10). Then there is a genuine cuspidal automorphic representation $\overline{\pi}$ of $\overline{G}(\mathbb{A}_K)$ with central character $\tilde{\theta}$ such that $\overline{\pi}_v \cong \overline{\pi}_{D,v} \circ \psi_v^{-1}$ for every $v \notin S$, and the packet identity (9) holds.*

Proof. By (H_D) there are $\tilde{f}_{D,v}$ $(v \in S)$ vanishing near $\overline{Z}_v$ for which the left side of (9) is nonzero. By Theorem 6.5(a) they have matching $\tilde{f}_v$ supported on the good regular elliptic set. By Proposition 9.6 the right side is nonzero, so $P(\rho) \neq \emptyset$. Any $\overline{\pi} \in P(\rho)$ is cuspidal, with $\overline{\pi}^V \cong \rho$ and $\overline{\pi}_{w_0} \cong \sigma$, which is the assertion. □

Theorem 9.12 (Converse)**.** ***[P]*** *under hypothesis (H) Let $\overline{\pi}$ be a genuine cuspidal automorphic representation of $\overline{G}(\mathbb{A}_K)$ with central character $\tilde{\theta}$, with $\overline{\pi}_{w_0}$ supercuspidal for some finite $w_0 \notin S$. Let $\rho = \overline{\pi}^V$, $\sigma = \overline{\pi}_{w_0}$. Assume*

(H) there are $\tilde{f}_v$ $(v \in S)$, supported in the regular set of $\overline{G}_v$ with vanishing orbital integrals at good regular non-elliptic elements, such that $\sum_{\overline{\pi}' \in P(\rho)} m(\overline{\pi}') \prod_{v\in S} \mathrm{tr}\,\overline{\pi}'_v(\tilde{f}_v) \neq 0$.

Then there is $\overline{\pi}_D$ with $\overline{\pi}_{D,v} \cong \overline{\pi}_v \circ \psi_v$ for all $v \notin S$.

Proof. By Theorem 6.5(b), the $\tilde{f}_v$ in (H) have matching $\tilde{f}_{D,v}$ vanishing near $\overline{Z}_v$. Proposition 9.6 applies to this family, and its right side is nonzero by (H). Hence $P_D(\rho) \neq \emptyset$, and any $\overline{\pi}_D \in P_D(\rho)$ has the required components. □

Remark 9.13. (1) Hypothesis (H) holds if $P(\rho) = \{\overline{\pi}\}$ and, for each $v \in S$, $\Theta_{\overline{\pi}_v}$ does not vanish identically on the good regular elliptic set: bump functions as in Lemma 9.7, supported near a good regular elliptic point, detect a single product of characters. The second condition holds when $\overline{\pi}_v$ is supercuspidal. Indeed, $\operatorname{tr} \overline{\pi}_v(\tilde{f}_{\overline{\pi}_v}) = 1$ for a matrix coefficient $\tilde{f}_{\overline{\pi}_v}$, and its orbital integrals vanish at regular non-elliptic elements [FK, §13(i)], so only the elliptic terms of the Weyl integration formula contribute.

(2) (H_D) and (H) play symmetric roles. Each would follow from finiteness of the relevant packet. On the D side this is Proposition 9.10. On the GL side one needs, in addition, that the elliptic characters of the members of a finite packet are linearly independent, which is not automatic. Proving finiteness of near-equivalence packets for $\overline{G}_D(\mathbb{A}_K)$, for instance by bounding the level at S of the members of $P_D(\rho)$ in terms of ρ, would make Theorem C unconditional. This is goal (iv) of §1.3. For $n = 1$, strong multiplicity one for $D^\times$ [Bad] gives $|P_D(\rho)| \leq 1$.

10 What is left after Part I

This section lists the steps that need new input, and exactly where each would enter. They correspond to the goals in §1.3, where their status in Parts II and III is recorded: (D1) is proved in Part II; Part III proves (D2) at tame places where the corrected Flicker–Kazhdan correspondence is available (in particular when $(n, N_r) = 1$), as explained in Remark 10.1. The section depends only on the statements of §§5–9, and it introduces the labels (D1)–(D4), which Parts II–IV use.

(D1) **[C]** *Matching of arbitrary functions.* (a) For $\tilde{f}_D$ not vanishing near $\overline{Z}_v$: needed to remove the regular-support condition at $v \in S$ in Theorems 8.6–9.11. Inputs: the germ expansion of [FK, §8] at central good elements, the restriction of germs to the elliptic set, and a genuine function with the orbital integrals of a Steinberg pseudo-coefficient near the center (Proposition 6.6 and the Remark after it). (b) For $\tilde{f}$ with vanishing non-elliptic orbital integrals but not regular support (e.g. supercusp forms at $v \in S$): needed to put w_0 in S and to drop the regular-support condition in (H). Input: a characterization of orbital integrals of genuine functions on $\overline{G}_v$ (the analog of [DKV, A.3]), via [FK, §8], [KP2, §1].

(D2) **[C]** *Local character relations.* For square-integrable genuine $\overline{\pi}_v$ there should be $\overline{\pi}_{D,v}$ with $\Theta_{\overline{\pi}_{D,v}} = (-1)^{r-1}\Theta_{\overline{\pi}_v} \circ \jmath$ on good regular elements, and conversely. Needed to turn (9) into a correspondence and to remove (H). Inputs: isolating single representations in (9), using multiplicity one and rigidity for $\overline{G}(\mathbb{A}_K)$ ([FK, §§27–28] as corrected by [Z]); the elliptic orthogonality relations [FK, §§24–25]; and a canonical normalization of $dg_{D,v}$ relative to dg_v.

(D3) **[C]** *Finiteness of near-equivalence packets.* Needed to remove (H_D) from Theorem 9.11, via Proposition 9.10. For the converse one needs in addition linear independence of the elliptic characters of the members of a finite GL-side packet (Remark 9.13).

(D4) **[C]** *Composing with [JC].* The lifts of [JC, Theorem 6.13] are unramified almost everywhere, with no supercuspidal component imposed. Applying Theorem 9.11 requires one supercuspidal $\overline{\pi}_{D,w_0}$. That means either extending [JC] to allow an arbitrary test function at one extra place inert in L (the local transfer [JC, Lemma 4.5] holds there), or removing the supercuspidal condition, by [Li4] or by (D1)(b).

Remark 10.1 ((D2) at places prime to n)**.** The character relation (D2) follows at a tame place $v \in S$ in the published range of the corrected square-integrable Flicker–Kazhdan correspondence, not merely from the unitarity of the division-side transform. Put $F = K_v$, and assume $p \nmid n$. A sufficient published hypothesis for the external correspondence is $(n, N_r) = 1$, where N_r is the least common multiple of the composite positive integers at most r (and $N_r = 1$ if none occur) [FK, §26; Z, Proposition 3.6 and Definition 3.7].

Part I, Proposition 4.3(b), identifies the restricted ambient cover, after relabeling μ_n, with the parameter-c_v Kazhdan–Patterson extension. Its diagonal cocycle is $\prod_{i<j}(a_i, b_j)_n(\det a, \det b)_n^{c_v}$, which is Zou's formula [Z, (2.1)] for one-dimensional blocks. The scalar cocycle exponent is $e_r = \binom{r}{2} + c_v r^2$; it is divisible by n because $2e_r = r(r - 1 + 2rc_v)$, n is odd, and $n \mid r - 1 + 2rc_v$. Consequently the scalar-section and central-character condition in Zou's correspondence agree with those of this paper; in particular the associated ordinary scalar character is trivial on μ_n.

For odd n, the star map is intrinsically $x^* = \tilde{x}^n$ [FK, §4]. Any change of extension isomorphism by a determinant character χ with values in μ_n disappears on star images. The canonical torus map of Theorem 5.9 satisfies $\jmath(zt_D^*) = z(jt)_{\mathrm{GL}}^*$ for $z \in \overline{Z}_v$. The discriminants on the two inner forms agree on matching elliptic elements. For a regular elliptic t the root sum in [Z, (3.2)] has exactly n equal contributions (the roots ζt, $\zeta \in \mu_n$), yielding the coefficient

$$b_D = \frac{n}{d_F(n)|n|_F^{r/2}} = |n|_F^{-(r-1)/2}, \qquad d_F(m) = m|m|_F^{-1/2}.$$

Here $d_r = \gcd(n, 2rc_v + r - 1) = n$; at a tame place $b_D = 1$ and the discriminant ratio $\Delta(t)/\Delta(t^n)$ is 1. Thus Part III, Corollary 7.8 (Draft 9), composes its tame irreducible-character theorem (Theorem 7.6), ordinary local Jacquet–Langlands [DKV], and the corrected Flicker–Kazhdan–Zou correspondence to obtain (D2), with the sign $(-1)^{r-1}$, on matching good regular elliptic elements. This is **[P]** within the explicitly stated external-correspondence range (using the cited **[E]** inputs), and is *not* an unconditional assertion for every pair (n, r) with $p \nmid n$. At wild places, irreducible-character integrality of L_D remains open.

11 Conclusion: what the local theorem gives on its own

This section records what the local results of §§4–6 establish independently of the global argument. It proves nothing new and introduces no notation.

At a place v where D is ramified, the cover $\overline{G}_D(K_v)$ of [JC] and the Kazhdan–Patterson cover $\overline{G}(K_v)$ with parameter c_v are subgroups of one cover of $\mathrm{GL}(r, L_v)$. This identifies their centers and central characters without any choice. For every maximal subfield $E \subset D_v$, the covered tori on the two sides have identical commutator pairings and good elements, and Theorem 5.9 gives a *canonical* identification of their good parts. Consequently there is a canonical bijection between good regular elliptic conjugacy classes (Corollary 5.12). Matching of genuine test functions is therefore well defined, with no choices beyond Haar measures. Theorem 6.5 then shows that matching is possible in both directions for functions supported on the regular set, for arbitrary (not necessarily cyclic) maximal subfields, by a self-contained argument (§6.6).

These statements are the local prerequisites for *any* trace-formula comparison between $\overline{G}_D$ and $\overline{G}$. They do not depend on the trace formula of [FK], on finiteness of packets, or on multiplicity one. Proposition 6.6 also shows exactly where the regular-support condition has to be removed (germs at the center); Part II removes it. Globally, the packet identity of Proposition 9.6 is unconditional. What remains conditional is only the final non-cancellation step, isolated as (H_D) and (H).

12 Summary of results

The results of Part I, with status labels (§3.1) and pointers; notation as in Assumption 1.1 and §3, with $v \in S$ and $F = K_v$.

1. **One ambient cover** (Lemma 4.2, Proposition 4.3; **[P]** using **[E]**). The cover $\overline{G}_D(K_v)$ of [JC] and the Kazhdan–Patterson cover $\overline{G}_v$ of $\mathrm{GL}(r, K_v)$ are subgroups of one cover $\overline{\mathcal{G}}_v$ of $\mathrm{GL}(r, L_v)$, with the same central subgroup $\overline{Z}_v$. As an extension, $\overline{G}_v$ is the parameter-c_v cover after relabeling μ_n, and the parameter is forced (Remark 5.3). At $v \notin S$ they are identified canonically by conjugation (Proposition 4.6; **[P]**).

2. **Theorem A: covered elliptic tori** (Proposition 5.1, Proposition 5.5, Theorem 5.9, Corollaries 5.12 and 5.13; **[P]**). For every maximal subfield $E \subset D_v$ the two covered tori have the commutator pairing $((Nx)^{1+2c_v}x^{-1}, y)_E$ and the good elements $p^{-1}(F^\times E^{\times n})$, of index $|n|_F^{-(r-1)}$. Every isomorphism between them over $\iota_{\mathrm{GL}} \circ j \circ \iota_D^{-1}$ that is the identity on $\overline{Z}_v$ has the same restriction $\jmath_{E,j}$ to the good elements. This gives a canonical bijection $\jmath$ of good regular elliptic classes and of genuine representations of covered tori.

3. **Theorem B: matching on the regular set** (Theorem 6.5, §6.6; **[P]**). Matching (Definition 6.3) carries the sign $(-1)^{r-1}$. Every genuine $\tilde{f}_D$ vanishing near $\overline{Z}_v$ has a

matching $\tilde{f}$ supported on the good regular elliptic set, and conversely every regular-supported $\tilde{f}$ with vanishing non-elliptic orbital integrals has a matching $\tilde{f}_D$. Near the center the method fails (Proposition 6.6; [**P**]).

4. **Rational elements and trace formulas** (Theorem 7.6, Corollary 7.7, Propositions 8.1 and 8.2, Theorem 8.6; [**P**] using the [**E**] inputs of Remark 8.7). The elliptic terms of the two trace formulas agree term by term; on the GL side we use a variant of [FK, §18] with the supercusp form at a place $w_0 \notin S$.

5. **Packet identity** (Proposition 9.6; [**P**], unconditional). The D-side and cuspidal GL-side near-equivalence packets with the same components outside $S \cup \{w_0\}$ have equal traces at S against matching regular-supported test functions; both sums are locally finite.

6. **Theorem C: global transfer** (Theorem 9.11; [**P**] under (H_D)). A genuine automorphic $\overline{\pi}_D$ with a supercuspidal component at some finite $w_0 \notin S$ transfers to a genuine cuspidal $\overline{\pi}$ with the same components outside S, if its packet satisfies (H_D); this holds when the packet is finite (Proposition 9.10; [**P**]). The converse holds under (H) (Theorem 9.12), and Lemma 9.8 ([**P**]) shows exactly how (H_D) could fail.

7. **What is left** (§10; [**C**]). (D1) matching of arbitrary functions, (D2) local character relations, (D3) finiteness of packets, (D4) composition with [JC]. Part II proves (D1), and Part III proves (D2) at tame places in the range of the corrected Flicker–Kazhdan correspondence (Remark 10.1); the rest is open.

Running example (Example 5.7): the covered elliptic tori are non-abelian at $\mathfrak{p}_3$ (index 9, dimension 3) and abelian at $\mathfrak{p}_7$; the sign in Theorem B is -1.

Acknowledgments

The three authors are David Joyner, Claude (Anthropic) and ChatGPT (OpenAI). This paper was drafted by David Joyner and Claude; ChatGPT refereed the drafts and supplied independent proofs and corrections. A separate revision log records the drafting and referee history.

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